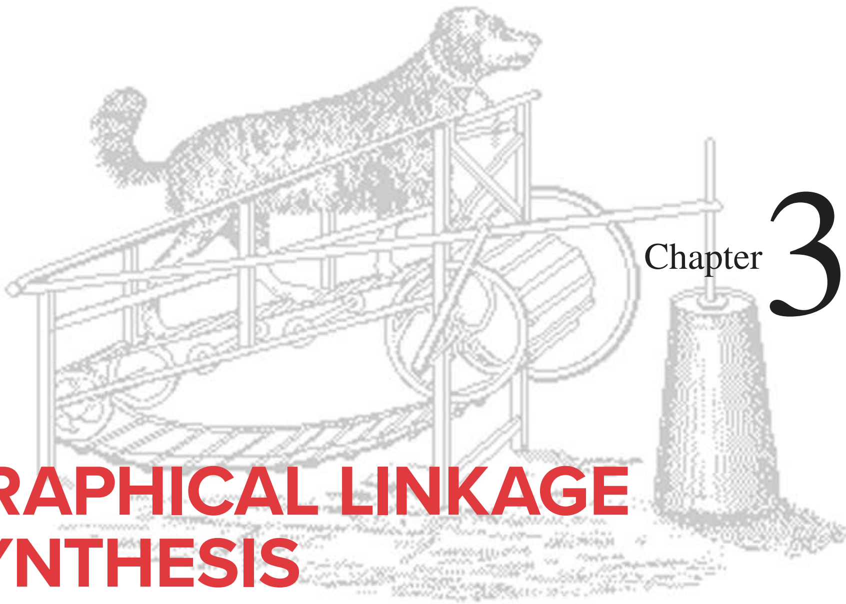


Chapter 3

GRAPHICAL LINKAGE SYNTHESIS

Genius is 1% inspiration
and 99% perspiration
THOMAS A. EDISON

3.0 INTRODUCTION

Most engineering design practice involves a combination of synthesis and analysis. Most engineering courses deal primarily with analysis techniques for various situations. However, one cannot analyze anything until it has been synthesized into existence. Many machine design problems require the creation of a device with particular motion characteristics. Perhaps you need to move a tool from position *A* to position *B* in a particular time interval. Perhaps you need to trace out a particular path in space to insert a part into an assembly. The possibilities are endless, but a common denominator is often the need for a linkage to generate the desired motions. So, we will now explore some simple synthesis techniques to enable you to create potential linkage design solutions for some typical kinematic applications.

3.1 SYNTHESIS

QUALITATIVE SYNTHESIS means *the creation of potential solutions in the absence of a well-defined algorithm that configures or predicts the solution*. Since most real design problems will have many more unknown variables than you will have equations to describe the system's behavior, you cannot simply solve the equations to get a solution.

Nevertheless you must work in this fuzzy context to create a potential solution and to also judge its **quality**. You can then analyze the proposed solution to determine its viability, and iterate between synthesis and analysis, as outlined in the **design process**, until you are satisfied with the result. Several tools and techniques exist to assist you in this process. The traditional tool is the drafting board, on which you lay out, to scale, multiple orthographic views of the design, and investigate its motions by drawing arcs, showing multiple positions, and using transparent, movable overlays. *Computer-aided drafting* (CAD) systems can speed this process to some degree, but you will probably find that the quickest way to get a sense of the quality of your linkage design is to model it, to scale, in cardboard, foam board, or drafting Mylar® and see the motions directly.

Other tools are available in the form of computer programs such as LINKAGES, DYNACAM, and MATRIX (included with this text), some of which do synthesis, but these are mainly analysis tools. They can analyze a trial mechanism solution so rapidly that their dynamic graphical output gives almost instantaneous visual feedback on the quality of the design. Commercially available programs such as *Solidworks, Pro-Engineer,* and *Working Model* also allow rapid analysis of a proposed mechanical design. The process then becomes one of **qualitative design by successive analysis,** which is really *an iteration between synthesis and analysis*. Very many trial solutions can be examined in a short time using these *computer-aided engineering* (CAE) tools. We will develop the mathematical solutions used in these programs in subsequent chapters in order to provide the proper foundation for understanding their operation. But if you want to try these programs to reinforce some of the concepts in these early chapters, you may do so. Appendix A describes these programs, and they each contain a manual for their use. Reference will be made to program features that are germane to topics in each chapter, as they are introduced. Data files for input to these computer programs are also provided as downloads for example problems and figures in these chapters. The data filenames are noted near the figure or example. The student is encouraged to open these sample files in the programs in order to observe more dynamic examples than the printed page can provide. These examples can be run by merely accepting the defaults provided for all inputs.

TYPE SYNTHESIS refers to *the definition of the proper type of mechanism best suited to the problem* and is a form of qualitative synthesis.* This is perhaps the most difficult task for the student as it requires some experience and knowledge of the various types of mechanisms that exist and which also may be feasible from a performance and manufacturing standpoint. As an example, assume that the task is to design a device to track the straight-line motion of a part on a conveyor belt and spray it with a chemical coating as it passes by. This has to be done at high, constant speed, with good accuracy and repeatability, and it must be reliable. Moreover, the solution must be inexpensive. Unless you have had the opportunity to see a wide variety of mechanical equipment, you might not be aware that this task could conceivably be accomplished by any of the following devices:

- *A straight-line linkage*
- *A cam and follower*
- *An air cylinder*
- *A hydraulic cylinder*
- *A robot*
- *A solenoid*

Each of these solutions, while possible, may not be optimal or even practical. Greater detail needs to be known about the problem to make that judgment, and that detail will come from the research phase of the design process. The straight-line linkage may prove

* A good discussion of type synthesis and an extensive bibliography on the topic can be found in Olson, D. G., et al. (1985). "A Systematic Procedure for Type Synthesis of Mechanisms with Literature Review." *Mechanism and Machine Theory*, 20(4), pp. 285-295.

to be too large and to have undesirable accelerations; the cam and follower will be expensive, though accurate and repeatable. The air cylinder itself is inexpensive but is noisy and unreliable. The hydraulic cylinder is more expensive, as is the robot. The solenoid, while cheap, has high impact loads and high impact velocity. So, you can see that the choice of device type can have a significant effect on the quality of the design. A poor choice at the type synthesis stage can create insoluble problems later on. The design might have to be scrapped after completion, at great expense. **Design is essentially an exercise in trade-offs**. Each proposed type of solution in this example has good and bad points. Seldom will there be a clear-cut, obvious solution to a real engineering design problem. It will be your job as a design engineer to balance these conflicting features and find a solution that gives the best trade-off of functionality against cost, reliability, and all other factors of interest. Remember, *an engineer can do, with one dollar, what any fool can do for ten dollars*. Cost is always an important constraint in engineering design.

QUANTITATIVE SYNTHESIS, OR ANALYTICAL SYNTHESIS, means the generation of one or more solutions of a particular type that you know to be suitable to the problem, and more importantly, one for which there is a synthesis algorithm defined. As the name suggests, this type of solution can be quantified, as some set of equations exists that will give a numerical answer. Whether that answer is a good or suitable one is still a matter for the judgment of the designer and requires analysis and iteration to optimize the design. Often the available equations are fewer than the number of potential variables, in which case you must assume some reasonable values for enough unknowns to reduce the remaining set to the number of available equations. Thus some qualitative judgment enters into the synthesis in this case as well. Except for very simple cases, a CAE tool is needed to do quantitative synthesis. Examples of such tools are the programs LINKAGES by R. L. Norton that solves the three-position multibar linkage synthesis problem and LINCAGES,* by Erdman and Gustafson.[1], that solves the four-position fourbar linkage synthesis problem. Program LINKAGES, provided with this text, does both three-position **analytical synthesis** as defined in Chapter 5, and general linkage **design by successive analysis**. The fast computation of these programs allows one to analyze the performance of many trial mechanism designs in a short time and promotes rapid iteration to a better solution.

* Available from Prof. A. Erdman, U. Minn., 111 Church St. SE, Minneapolis, MN 55455 612-625-8580

DIMENSIONAL SYNTHESIS of a linkage *is the determination of the proportions (lengths) of the links necessary to accomplish the desired motions* and can be a form of quantitative synthesis if an algorithm is defined for the particular problem, but can also be a form of qualitative synthesis if there are more variables than equations. The latter situation is more common for linkages. (Dimensional synthesis of cams is quantitative.) Dimensional synthesis assumes that, through *type synthesis*, you have already determined that a linkage (or a cam) is the most appropriate solution to the problem. This chapter discusses **graphical dimensional (position) synthesis** of linkages in detail. Chapter 5 presents methods of **analytical linkage synthesis**, and Chapter 8 presents **cam synthesis**.

3.2 FUNCTION, PATH, AND MOTION GENERATION

FUNCTION GENERATION is defined as *the correlation of an input motion with an output motion in a mechanism*. A function generator is conceptually a "black box" that delivers some predictable output in response to a known input. Historically, before the advent of electronic computers, mechanical function generators found wide application in artillery rangefinders and shipboard gun aiming systems, and many other tasks. They are, in fact, **mechanical analog computers**

microcomputers for control systems coupled with the availability of compact servo and stepper motors has reduced the demand for these mechanical function generator linkage devices. Many such applications can now be served more economically and efficiently with electromechanical devices.* Moreover, the computer-controlled electromechanical function generator is programmable, allowing rapid modification of the function generated as demands change. For this reason, while presenting some simple examples in this chapter and a general, analytical design method in Chapter 5, we will not emphasize mechanical linkage function generators in this text. Note, however, that the cam-follower system, discussed extensively in Chapter 8, is in fact a form of mechanical function generator, and it is typically capable of higher force and power levels per dollar than electromechanical systems.

PATH GENERATION is defined as *the control of a* ***point*** *in the plane such that it follows some prescribed path.* This is typically accomplished with at least four bars, wherein a point on the coupler traces the desired path. Specific examples are presented in the section on coupler curves below. Note that no attempt is made in path generation to control the orientation of the link that contains the point of interest. However, it is common for the timing of the arrival of the point at particular locations along the path to be defined. This case is called *path generation with prescribed timing* and is analogous to function generation in that a particular output function is specified. Analytical path and function generation will be dealt with in Chapter 5.

MOTION GENERATION is defined as *the control of a* ***line*** *in the plane such that it assumes some prescribed set of sequential positions*. Here orientation of the link containing the line is important. This is a more general problem than path generation, and in fact, path generation is a subset of motion generation. An example of a motion generation problem is the control of the "bucket" on a bulldozer. The bucket must assume a set of positions to dig, pick up, and dump the excavated earth. Conceptually, the motion of a line, painted on the side of the bucket, must be made to assume the desired positions. A linkage is the usual solution.

PLANAR MECHANISMS VERSUS SPATIAL MECHANISMS The above discussion of controlled movement has assumed that the motions desired are planar (2-D). We live in a three-dimensional world, however, and our mechanisms must function in that world. **Spatial mechanisms** *are* 3-D *devices*. Their design and analysis are much more complex than those of **planar mechanisms**, which *are* 2-D *devices*. The study of spatial mechanisms is beyond the scope of this introductory text. Some references for further study are in the bibliography to this chapter. However, the study of planar mechanisms is not as practically limiting as it might first appear since many devices in three dimensions are constructed of multiple sets of 2-D devices coupled together. An example is any folding chair. It will have some sort of linkage in the left side plane that allows folding. There will be an identical linkage on the right side of the chair. These two *XY* planar linkages will be connected by some structure along the *Z* direction, which ties the two planar linkages into a 3-D assembly. Many real mechanisms are arranged in this way, as **duplicate planar linkages**, displaced in the *Z* direction in parallel planes and rigidly connected. When you open the hood of a car, take note of the hood hinge mechanism. It will be duplicated on each side of the car. The hood and the car body tie the two planar linkages together into a 3-D assembly. Look and you will see many other such examples of assemblies of planar linkages into 3-D configurations. So, the 2-D techniques of synthesis and analysis presented here will prove to be of practical value in designing in 3-D as well.

* It is worth noting that the day is long past when a mechanical engineer can be content to remain ignorant of electronics and electromechanics. Virtually all modern machines are controlled by electronic devices. Mechanical engineers must understand their operation.

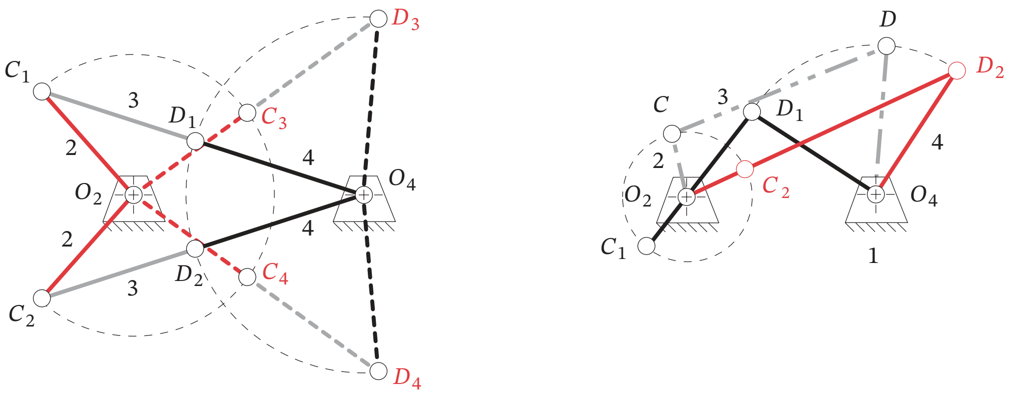
3

3.3 LIMITING CONDITIONS

The manual, graphical, dimensional synthesis techniques presented in this chapter and the computerizable, analytical synthesis techniques presented in Chapter 5 are reasonably rapid means to obtain a trial solution to a motion control problem. Once a potential solution is found, it must be evaluated for its quality. There are many criteria that may be applied. In later chapters, we will explore the analysis of these mechanisms in detail. However, one does not want to expend a great deal of time analyzing, in great detail, a design that can be shown to be inadequate by some simple and quick evaluations.

TOGGLE POSITIONS One important test can be applied within the synthesis procedures described below. You need to check that the linkage can in fact reach all of the specified design positions without encountering a limit position. Linkage synthesis procedures often only provide that the particular positions specified will be obtained. They say nothing about the linkage's behavior between those positions. Figure 3-1a shows a non-Grashof fourbar linkage at its limits of motion called **toggle positions**. *The toggle positions are determined by the **colinearity** of two of the moving links.* C_1D_1 and C_2D_2 (solid lines) are the toggle positions reached when driven from link 2. C_3D_3 and C_4D_4 (dashed lines) are the toggle positions reached when driven from link 4. A fourbar triple-rocker mechanism will have four, and a Grashof double-rocker two, of these toggle positions in which the linkage assumes a triangular configuration. When in a triangular (toggle) position, it will not allow further input motion in one direction from one of its rocker links (either of link 2 from positions C_1D_1 and C_2D_2 or link 4 from positions C_3D_3 and C_4D_4). A different link will then have to be driven to get it out of toggle.

STATIONARY POSITIONS A Grashof fourbar crank-rocker linkage will also assume two stationary positions as shown in Figure 3-1b, when the shortest link (crank O_2C) is colinear with the coupler CD (link 3), either *extended colinear* ($O_2C_2D_2$) or *overlapping colinear* ($O_2C_1D_1$). It cannot be *back driven* from the rocker O_4D (link 4) through these colinear positions (which then act as toggles), but when the crank O_2C (link 2) is driven, it will carry through both stationary positions because it is Grashof. Note that the stationary positions define the limits of motion of the driven rocker (link 4), at which its angular



(*a*) Non-Grashof triple-rocker toggle positions (*b*) Grashof crank-rocker stationary configurations

FIGURE 3-1

Linkage limit positions

3

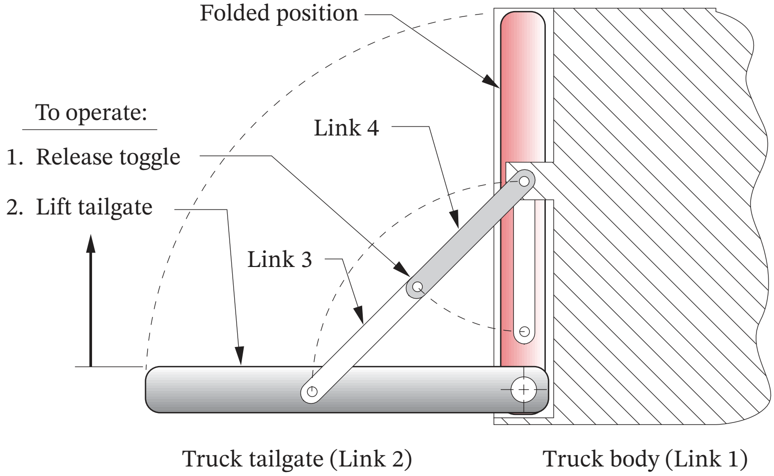


FIGURE 3-2

Deltoid toggle linkage used to control truck tailgate motion

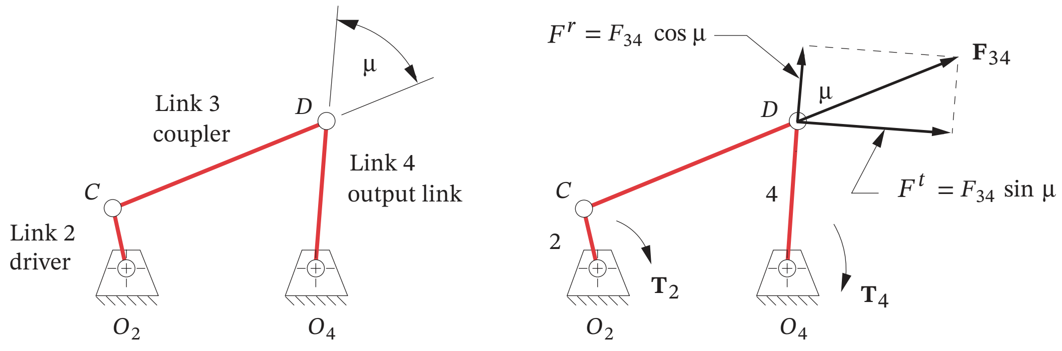
velocity will go through zero. Use program LINKAGES to read the data files F03-01A.4br and F03-1b.4br and animate these examples.

After synthesizing a **double- or triple-rocker** solution to a multiposition (motion generation) problem, you **must** check for the presence of toggle positions *between* your design positions. *An easy way to do this is with a model of the linkage.* A CAE tool such as LINKAGES or *Working Model* will also check for this problem. It is important to realize that a toggle condition is only undesirable if it is preventing your linkage from getting from one desired position to the other. In other circumstances the toggle is very useful. It can provide a self-locking feature when a linkage is moved slightly beyond the toggle position and against a fixed stop. Any attempt to reverse the motion of the linkage then causes it merely to jam harder against the stop. It must be manually pulled "over center," out of toggle, before the linkage will move. You have encountered many examples of this application, as in card table or ironing board leg linkages and also pickup truck or station wagon tailgate linkages. An example of such a toggle linkage is shown in Figure 3-2. It happens to be a special-case Grashof linkage in the deltoid configuration (see also Figure 2-17d), which provides a locking toggle position when open, and folds on top of itself when closed, to save space. We will analyze the toggle condition in greater detail in a later chapter.

TRANSMISSION ANGLE Another useful test that can be very quickly applied to a linkage design to judge its quality is the measurement of its transmission angle. This can be done analytically, graphically on the drawing board, or via a model for a rough approximation. (Extend the links beyond the pivot to measure the angle.) The **transmission angle** μ is shown in Figure 3-3a and is defined as *the angle between the output link and the coupler.** It is usually taken as the *absolute value of the acute angle of the pair of angles at the intersection of the two links* and *varies continuously from some minimum to some maximum value as the linkage goes through its range of motion.* It is a measure of the quality of force and velocity transmission at the joint. Note in Figure 3-2 that the linkage cannot be moved from the open position shown by any force applied to the tailgate, link 2, since the transmission angle between links 3 and 4 is zero at that position. But a force

* The transmission angle as defined by Alt[2] has limited application. It only predicts the quality of force or torque transmission if the input and output links are pivoted to ground. If the output force is taken from a floating link (coupler), then the transmission angle is of no value. A different index of merit called the joint force index (JFI) is presented in Chapter 11 which discusses force analysis in linkages. (See Section 11.12) The JFI is useful for situations in which the output link is floating as well as for giving the same kind of information when the output is taken from a link rotating against the ground. However, the JFI requires that a complete force analysis of the linkage be done, whereas the transmission angle is determined from linkage geometry alone.

3



(*a*) Linkage transmission angle μ (*b*) Static forces at a linkage joint

FIGURE 3-3

Transmission angle in the fourbar linkage

applied to link 4 as the input link will move it. The transmission angle is now between links 3 and 2 and is 45 degrees.

Figure 3-3b shows a torque $\mathbf{T}_2$ applied to link 2. Even before any motion occurs, this causes a static, colinear force $\mathbf{F}_3$ to b applied by link 3 to link 4 at point *D*. Its radial and tangential components F^r_{34} and F^t_{34} are resolved parallel and perpendicular to link 4, respectively. Ideally, we would like all of the force $\mathbf{F}_{34}$ to go into producing output torque $\mathbf{T}_4$ on link 4. However, only the tangential component creates torque on link 4. The radial component F^r_{34} provides only tension or compression in link 4. This radial component only increases pivot friction and does not contribute to the output torque. Therefore, the optimum value for the **transmission angle** is 90°. When μ is less than 45° the radial component will be larger than the tangential component. Most machine designers try to keep the **minimum transmission angle above about 40**° to promote smooth running and good force transmission. However, if in your particular design there will be little or no external force or torque applied to link 4, you may be able to get away with even lower values of μ.* The transmission angle provides one means to judge the quality of a newly synthesized linkage. If it is unsatisfactory, you can iterate through the synthesis procedure to improve the design. We will investigate the transmission angle in greater detail in later chapters.

3.4 POSITION SYNTHESIS *View the lecture video* (47:57)†

Position synthesis of a linkage is *the determination of the proportions (lengths) of the links necessary to accomplish the desired motions*. This section assumes that, through *type synthesis*, you have determined that a linkage is the most appropriate solution to the problem. Many techniques exist to accomplish this task of **position synthesis of a fourbar linkage**. The simplest and quickest methods are graphical. These work well for up to three design positions. Beyond that number, a numerical, analytical synthesis approach as described in Chapter 5, using a computer, is usually necessary.

* Alt[2] who defined the transmission angle, recommended keeping $\mu_{min} > 40°$. But it can be argued that at higher speeds, the momentum of the moving elements and/or the addition of a flywheel will carry a mechanism through locations of poor transmission angle. The most common example is the back-driven slider-crank (as used in internal combustion engines) which has $\mu = 0$ twice per revolution. Also, the transmission angle is only critical in a fourbar linkage when the rocker is the output link on which the working load impinges. If the working load is taken by the coupler rather than by the rocker, then minimum transmission angles less than 40° may be viable. A more definitive way to qualify a mechanism's dynamic function is to compute the variation in its required driving torque. Driving torque and flywheels are addressed in Chapter 11. A joint force index (JFI) can also be calculated. (See footnote on previous page.)

† http://www.designofmachinery.com/DOM/Position_Synthesis.mp4

Note that the principles used in these graphical synthesis techniques are simply those of **euclidean geometry**. The rules for bisection of lines and angles, properties of parallel and perpendicular lines, and definitions of arcs, etc., are all that is needed to generate these linkages. **Compass**, **protractor**, and **rule** are the only tools needed for graphical linkage synthesis. Refer to any introductory (high school) text on geometry if your geometric theorems are rusty.

Two-Position Synthesis

Two-position synthesis subdivides into two categories: **rocker output** (pure rotation) and **coupler output** (complex motion). Rocker output is most suitable for situations in which a Grashof crank-rocker is desired and is, in fact, a trivial case of *function generation* in which the output function is defined as two discrete angular positions of the rocker. Coupler output is more general and is a simple case of *motion generation* in which two positions of a line in the plane are defined as the output. This solution will frequently lead to a triple-rocker. However, the fourbar triple-rocker can be motor driven by the addition of a **dyad** (twobar chain), which makes the final result a **Watt sixbar** containing a **Grashof fourbar subchain**. We will now explore the synthesis of each of these types of solution for the two-position problem.

EXAMPLE 3-1

Rocker Output - Two Positions with Angular Displacement. (Function Generation)

Problem: Design a fourbar Grashof crank-rocker to give 45° of rocker rotation with equal time forward and back, from a constant speed motor input.

Solution: (See Figure 3-4†.)

1 Draw the output link O_4B in both extreme positions, B_1 and B_2 in any convenient location, such that the desired angle of motion θ_4 is subtended.

2 Draw the chord B_1B_2 and extend it in either direction.

3 Select a convenient point O_2 on line B_1B_2 extended.

4 Bisect line segment B_1B_2 , and draw a circle of that radius about O_2.

5 Label the two intersections of the circle and B_1B_2 extended, A_1 and A_2.

6 Measure the length of the coupler as A_1 to B_1 or A_2 to B_2.

7 Measure ground length 1, crank length 2, and rocker length 4.

8 Find the Grashof condition. If non-Grashof, redo steps 3 to 8 with O_2 farther from O_4.

9 Make a model of the linkage and check its function and transmission angles.

10 You can input the file F03-04.4br to program LINKAGES to see this example come alive.

† This figure is provided as animated AVI and Working Model files. Its filename is the same as the figure number.

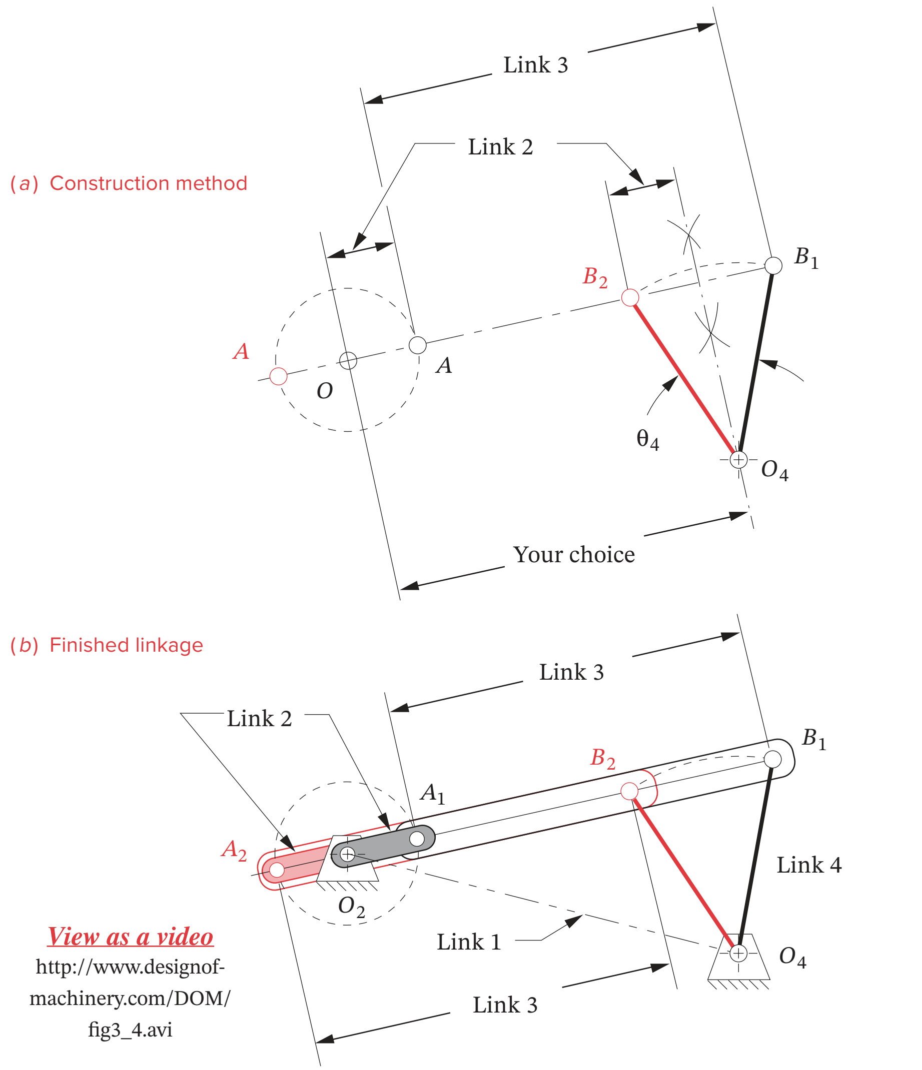


FIGURE 3-4

Two-position function synthesis with rocker output (non-quick-return)

Note several things about this synthesis process. We started with the output end of the system, as it was the only aspect defined in the problem statement. We had to make many quite arbitrary decisions and assumptions to proceed because there were many more variables than we could have provided "equations" for. We are frequently forced to make "free choices" of "a convenient angle or length." These free choices are actually definitions of design parameters. A poor choice will lead to a poor design. Thus these are **qualitative synthesis** approaches and require an iterative process, even for this simple example. The first solution you reach will probably not be satisfactory, and several attempts (iterations) should be expected to be necessary. As you gain more experience in designing kinematic solutions, you will be able to make better choices for these design parameters with fewer iterations. **The value of making a simple model of your design cannot be overstressed**! You will get the *most insight* into your design's quality for the *least effort* by making, articulating, and studying the model. These general observations will hold for most of the linkage synthesis examples presented.

EXAMPLE 3-2

Rocker Output - Two Positions with Complex Displacement. (Motion Generation)

Problem: Design a fourbar linkage to move link CD from position C_1D_1 to C_2D_2.

Solution: (See Figure 3-5*.)

1 Draw the link CD in its two desired positions, C_1D_1 and C_2D_2, in the plane as shown.

2 Draw construction lines from point C_1 to C_2 and from point D_1 to D_2.

3 Bisect line C_1C_2 and line D_1D_2 and extend their perpendicular bisectors to intersect at O_4. Their intersection is the **rotopole**.

4 Select a convenient radius and draw an arc about the rotopole to intersect both lines O_4C_1 and O_4C_2. Label the intersections B_1 and B_2.

5 Do steps 2 to 8 of Example 3-1 to complete the linkage.

6 Make a model of the linkage and articulate it to check its function and its transmission angles.

* This figure is provided as animated AVI and Working Model files. Its filename is the same as the figure number.

Note that Example 3-2 reduces to the method of Example 3-1 once the **rotopole** is found. Thus a link represented by a line in complex motion can be reduced to the simpler problem of pure rotation and moved to any two positions in the plane as the rocker on a fourbar linkage. The next example moves the same link through the same two positions as the coupler of a fourbar linkage.

EXAMPLE 3-3

Coupler Output - Two Positions with Complex Displacement. (Motion Generation)

Problem: Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 (with moving pivots at C and D).

Solution: (See Figure 3-6.)

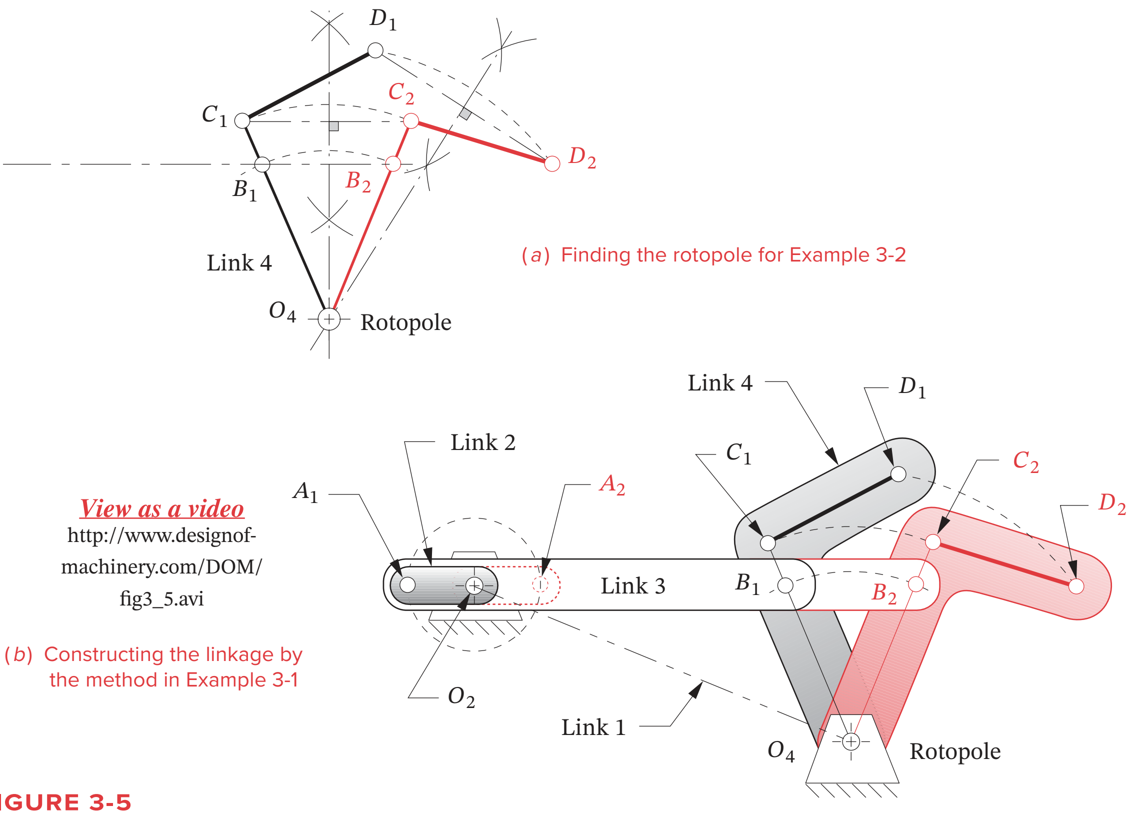
1 Draw the link CD in its two desired positions, C_1D_1 and C_2D_2, in the plane as shown.

2 Draw construction lines from point C_1 to C_2 and from point D_1 to D_2.

3 Bisect line C_1C_2 and line D_1D_2 and extend the perpendicular bisectors in convenient directions. The rotopole will **not** be used in this solution.

4 Select any convenient point on each bisector as the fixed pivots O_2 and O_4, respectively.

3



View as a video
http://www.designof-machinery.com/DOM/fig3_5.avi

(b) Constructing the linkage by the method in Example 3-1

FIGURE 3-5

Two-position motion synthesis with rocker output (non-quick-return)

5 Connect O_2 with C_1 and call it link 2. Connect O_4 with D_1 and call it link 4.

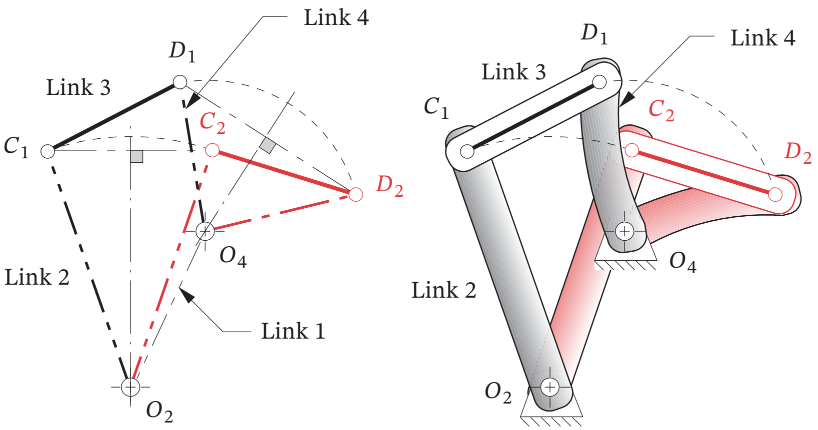
6 Line C_1D_1 is link 3. Line O_2O_4 is link 1.

7 Check the Grashof condition, and repeat steps 4 to 7 if unsatisfied. Note that any Grashof condition is potentially acceptable in this case.

8 Construct a model and check its function to be sure it can get from the initial to final position without encountering any limit (toggle) positions.

9 Check the transmission angles.

Input file F03-06.4br to program LINKAGES to see Example 3-3. Note that this example had nearly the same problem statement as Example 3-2, but the solution is quite different. Thus a link can also be moved to any two positions in the plane as the coupler of a four-bar linkage, rather than as the rocker. However, to limit its motions to those two coupler positions as extrema, two additional links are necessary. These additional links can be designed by the method shown in Example 3-4 and Figure 3-7.



(*a*) Two-position synthesis (*b*) Finished non-Grashof fourbar

FIGURE 3-6

Two-position motion synthesis with coupler output

EXAMPLE 3-4

Adding a Dyad (Twobar Chain) to Control Motion in Example 3-3.

Problem: Design a **dyad** to control and limit the extremes of motion of the linkage in Example 3-3 to its two design positions.

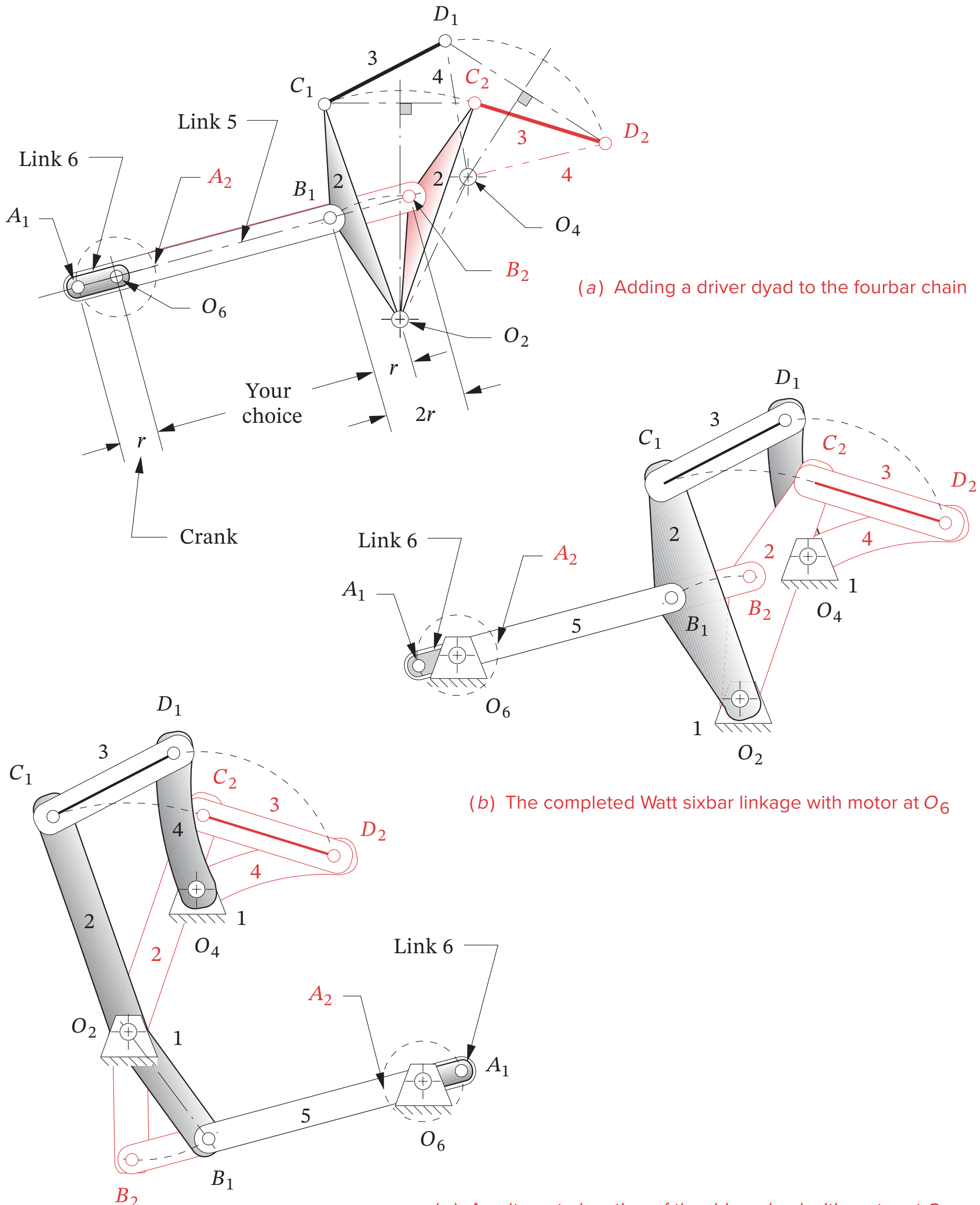
Solution: (See Figure 3-7a.)

1 Select a convenient point on link 2 of the linkage designed in Example 3-3. Note that it need not be on the line O_2C_1. Label this point B_1.

2 Draw an arc about center O_2 through B_1 to intersect the corresponding line O_2B_2 in the second position of link 2. Label this point B_2. The chord B_1B_2 provides us with the same problem as in Example 3-1.

3 Do steps 2 to 9 of Example 3-1 to complete the linkage, except add links 5 and 6 and center O_6 rather than links 2 and 3 and center O_2. Link 6 will be the driver crank. The fourbar subchain of links O_6, A_1, B_1, O_2 must be a Grashof crank-rocker.

Note that we have used the approach of Example 3-1 to add a **dyad** to serve as a *driver stage* for our existing fourbar. This results in a **sixbar Watt mechanism** whose first stage is Grashof as shown in Figure 3-7b. Thus we can drive this with a motor on link 6. Note also that we can locate the motor center O_6 anywhere in the plane by judicious choice of point B_1 on link 2. If we had put B_1 below center O_2, the motor would be to the right of links 2, 3, and 4 as shown in Figure 3-7c. There is *an infinity of driver dyads* possible that will drive any double-rocker assemblage of links. Input the files F03-07b.6br and F03-07c.6br to program LINKAGES to see Example 3-4 in motion for these two solutions.



(*a*) Adding a driver dyad to the fourbar chain

(*b*) The completed Watt sixbar linkage with motor at O_6

(*c*) An alternate location of the driver dyad with motor at O_6

FIGURE 3-7

Driving a non-Grashof linkage with a dyad (non-quick-return)

Three-Position Synthesis with Specified Moving Pivots

Three-position synthesis allows the definition of three positions of a line in the plane and will create a fourbar linkage configuration to move it to each of those positions. This is a **motion generation** problem. The synthesis technique is a logical extension of the method used in Example 3-3 for two-position synthesis with coupler output. The resulting linkage may be of any Grashof condition and will usually require the addition of a dyad to control and limit its motion to the positions of interest. Compass, protractor, and rule are the only tools needed in this graphical method.

EXAMPLE 3-5

Coupler Output - 3 Positions with Complex Displacement. (Motion Generation)

Problem: Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3. Moving pivots are at C and D. Find the fixed pivot locations.

Solution: (See Figure 3-8.)

1 Draw link CD in its three design positions C_1D_1, C_2D_2, C_3D_3 in the plane as shown.

2 Draw construction lines from point C_1 to C_2 and from point C_2 to C_3.

3 Bisect line C_1C_2 and line C_2C_3 and extend their perpendicular bisectors until they intersect. Label their intersection O_2.

4 Repeat steps 2 and 3 for lines D_1D_2 and D_2D_3. Label the intersection O_4.

5 Connect O_2 with C_1 and call it link 2. Connect O_4 with D_1 and call it link 4.

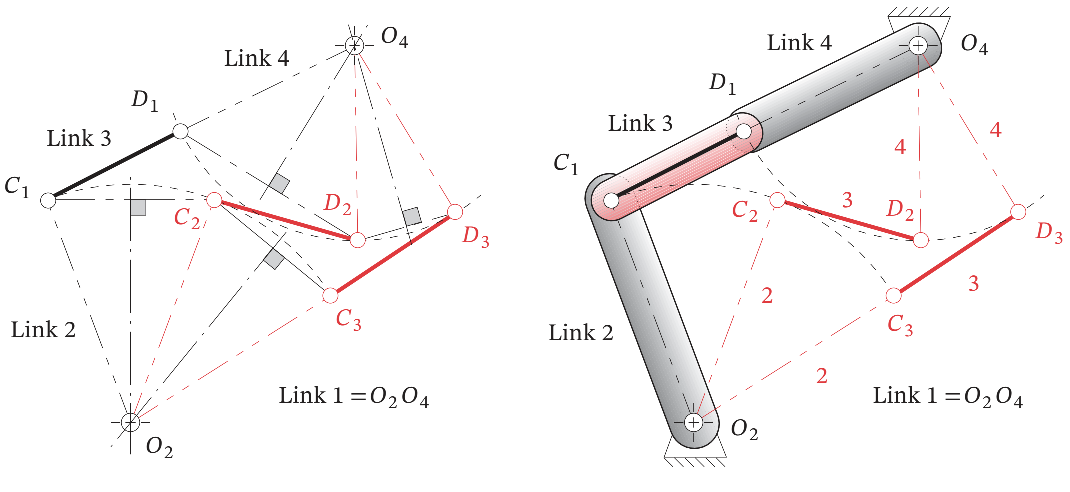
6 Line C_1D_1 is link 3. Line O_2O_4 is link 1.

7 Check the Grashof condition. Any Grashof condition is potentially acceptable in this case.

8 Construct a model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions.

9 Construct a driver dyad according to the method in Example 3-4 using an extension of link 3 to attach the dyad.

Note that while a solution is usually obtainable for this case, it is possible that you may not be able to move the linkage continuously from one position to the next without disassembling the links and reassembling them to get them past a limiting position. That will obviously be unsatisfactory. In the particular solution presented in Figure 3-8, note that links 3 and 4 are in toggle at position one, and links 2 and 3 are in toggle at position three. In this case we will have to drive link 3 with a driver dyad, since any attempt to drive either link 2 or link 4 will fail at the toggle positions. No amount of torque applied to link 2 at position C_1 will move link 4 away from point D_1, and driving link 4 will not move link 2 away from position C_3. Input the file F03-08.4br to program LINKAGES to see Example 3-5.



(*a*) Construction method (*b*) Finished non-Grashof fourbar

FIGURE 3-8

Three-position motion synthesis

Three-Position Synthesis with Alternate Moving Pivots

Another potential problem is the possibility of an undesirable location of the fixed pivots O_2 and O_4 with respect to your packaging constraints. For example, if the fixed pivot for a windshield wiper linkage design ends up in the middle of the windshield, you may want to redesign it. Example 3-6 shows a way to obtain an alternate configuration for the same three-position motion as in Example 3-5. And, the method to be shown in Example 3-8 allows you to specify the location of the fixed pivots in advance and then find the locations of the moving pivots on link 3 that are compatible with those fixed pivots.

EXAMPLE 3-6

Coupler Output - Three Positions with Complex Displacement - Alternate Attachment Points for Moving Pivots. (Motion Generation)

Problem: Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3. Use different moving pivots than CD. Find the fixed pivot locations.

Solution: (See Figure 3-9.)

1 Draw the link CD in its three desired positions C_1D_1, C_2D_2, C_3D_3, in the plane as done in Example 3-5.

2 Define new attachment points E_1 and F_1 that have a fixed relationship between C_1D_1 and E_1F_1 within the link. Now use E_1F_1 to define the three positions of the link.

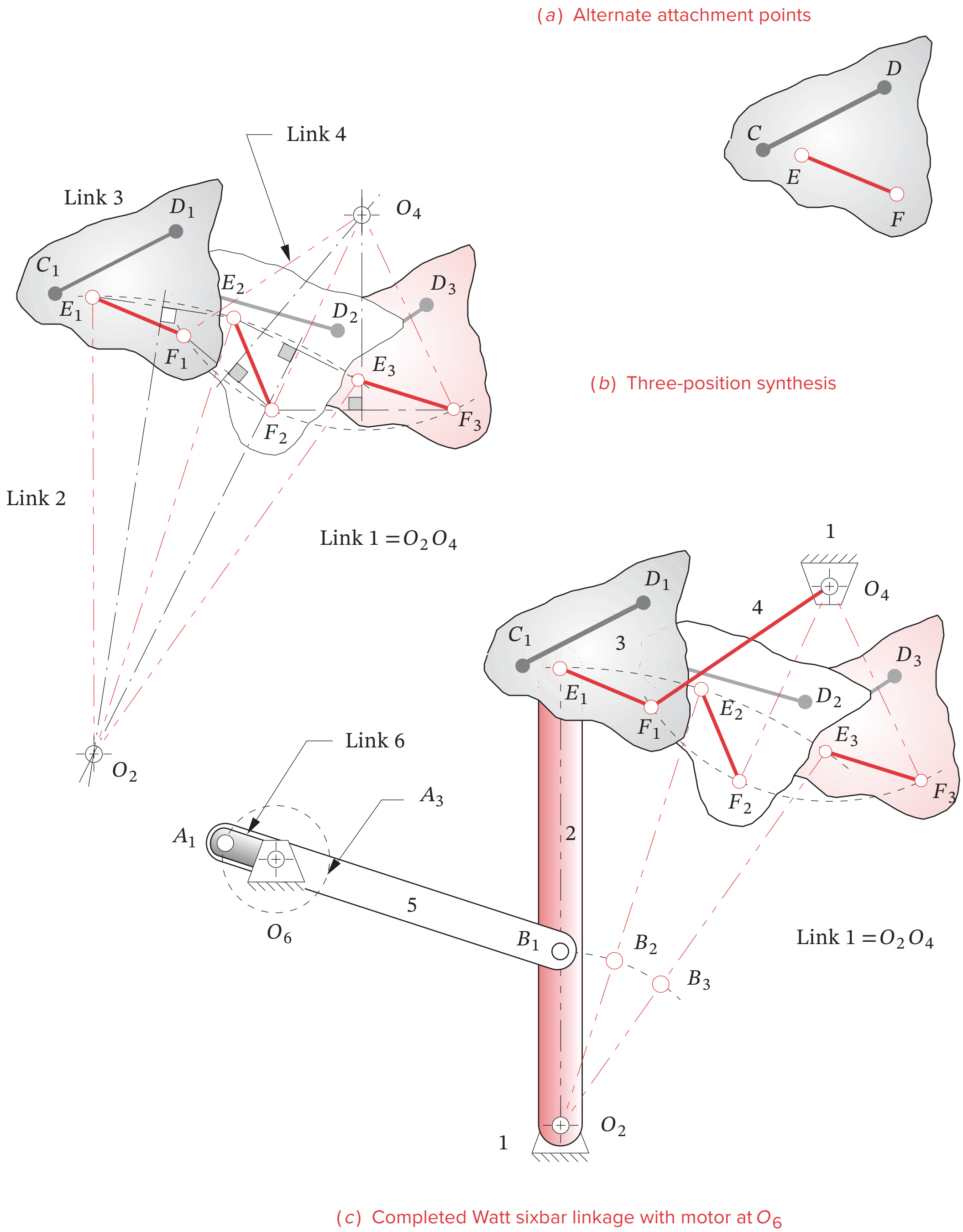


FIGURE 3-9

Three-position synthesis with alternate moving pivots

3 Draw construction lines from point E_1 to E_2 and from point E_2 to E_3.

4 Bisect line E_1E_2 and line E_2E_3 and extend the perpendicular bisectors until they intersect. Label the intersection O_2.

5 Repeat steps 2 and 3 for lines F_1F_2 and F_2F_3. Label the intersection O_4 .

6 Connect O_2 with E_1 and call it link 2. Connect O_4 with F_1 and call it link 4.

7 Line E_1F_1 is link 3. Line O_2O_4 is link 1.

8 Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case.

9 Construct a model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions. If not, change locations of points E and F and repeat steps 3 to 9.

10 Construct a driver dyad acting on link 2 according to the method in Example 3-4.

Note that the shift of the attachment points on link 3 from CD to EF has resulted in a shift of the locations of fixed pivots O_2 and O_4 as well. Thus they may now be in more favorable locations than they were in Example 3-5. It is important to understand that any two points on link 3, such as E and F, can serve to fully define that link as a rigid body, and that there is an infinity of such sets of points to choose from. While points C and D have some particular location in the plane that is defined by the linkage's function, points E and F can be anywhere on link 3, thus creating an infinity of solutions to this problem.

The solution in Figure 3-9 is different from that of Figure 3-8 in several respects. It avoids the toggle positions and thus can be driven by a dyad acting on one of the rockers, as shown in Figure 3-9c, and the transmission angles are better. However, the toggle positions of Figure 3-8 might actually be of value if a self-locking feature were desired. *Recognize that both of these solutions are to the same problem,* and the solution in Figure 3-8 is just a special case of that in Figure 3-9. Both solutions may be useful. Line CD moves through the same three positions with both designs. There is an infinity of other solutions to this problem waiting to be found as well. Input the file F03-09c.6br to program LINKAGES to see Example 3-6.

Three-Position Synthesis with Specified Fixed Pivots

Even though one can probably find an acceptable solution to the three-position problem by the methods described in the two preceding examples, it can be seen that the designer will have little direct control over the location of the fixed pivots since they are one of the results of the synthesis process. The fixed pivots need to be located where the ground plane of the package exists and is accessible. It would be preferable if we could define the fixed pivot locations, as well as the three positions of the moving link, and then synthesize the appropriate attachment points, E and F, to the moving link to satisfy these more realistic constraints. The principle of **inversion** can be applied to this problem. Examples 3-5 and 3-6 showed how to find the required fixed pivots for three chosen positions of the moving pivots. Inverting this problem allows specification of the fixed pivot locations and determination of the required moving pivots for those locations. The first step is to find the three positions of the ground plane that correspond to the three desired coupler positions. This is done by **inverting the linkage*** as shown in Figure 3-10 and Example 3-7.

* This method and example were supplied by Mr. Homer D. Eckhardt, Consulting Engineer, Lincoln, MA.

3

EXAMPLE 3-7

Three-Position Synthesis with Specified Fixed Pivots – Inverting the Three-Position Motion Synthesis Problem

Problem: Invert a fourbar linkage which moves the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3. Use specified fixed pivots O_2 and O_4.

Solution: First find the inverted positions of the ground link corresponding to the three coupler positions specified. (See Figure 3-10.)

1 Draw the link CD in its three desired positions C_1D_1, C_2D_2, C_3D_3, in the plane, as was done in Example 3-5 and as shown in Figure 3-10a.

2 Draw the ground link O_2O_4 in its desired position in the plane with respect to the first coupler position C_1D_1 as shown in Figure 3-10a.

3 Draw construction arcs from point C_2 to O_2 and from point D_2 to O_2 whose radii define the sides of triangle $C_2O_2D_2$. This defines the relationship of the fixed pivot O_2 to the coupler line CD in the second coupler position as shown in Figure 3-10b.

4 Draw construction arcs from point C_2 to O_4 and from point D_2 to O_4 to define the triangle $C_2O_4D_2$. This defines the relationship of the fixed pivot O_4 to the coupler line CD in the second coupler position as shown in Figure 3-10b.

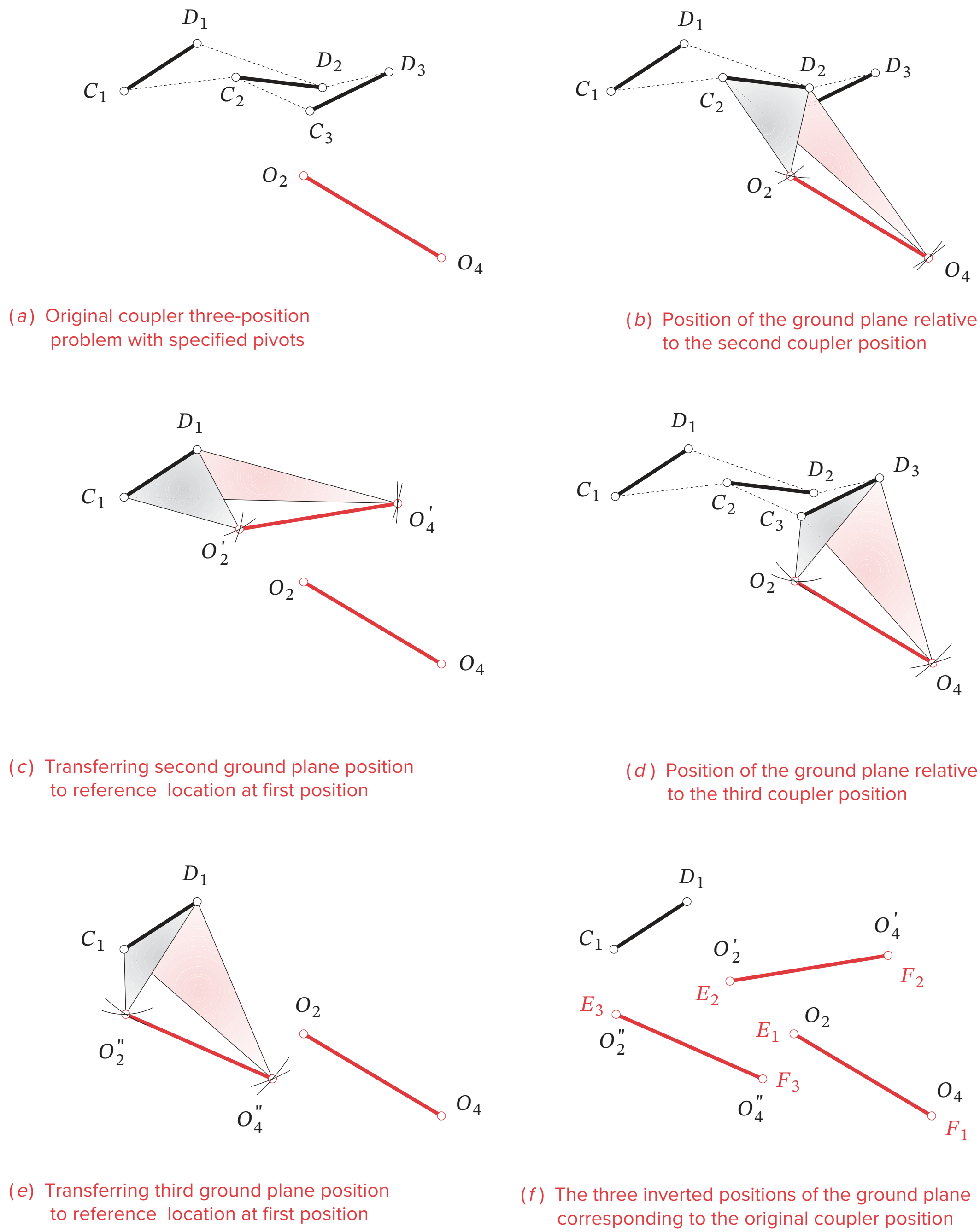
5 Now transfer this relationship back to the first coupler position C_1D_1 so that the ground plane position $O_2'O_4'$ bears the same relationship to C_1D_1 as O_2O_4 bore to the second coupler position C_2D_2. In effect, you are sliding C_2 along the dotted line C_2-C_1 and D_2 along the dotted line D_2-D_1. By doing this, we have pretended that the ground plane moved from O_2O_4 to $O_2'O_4'$ instead of the coupler moving from C_1D_1 to C_2D_2. We have *inverted* the problem.

6 Repeat the process for the third coupler position as shown in Figure 3-10d and transfer the third relative ground link position to the first, or reference, position as shown in Figure 3-10e.

7 The three inverted positions of the ground plane that correspond to the three desired coupler positions are labeled O_2O_4, $O_2'O_4'$, and $O_2''O_4''$ and have also been renamed E_1F_1, E_2F_2, and E_3F_3 as shown in Figure 3-10f. These correspond to the three coupler positions shown in Figure 3-10a. Note that the original three lines C_1D_1, C_2D_2, and C_3D_3 are not now needed for the linkage synthesis.

We can use these three new lines E_1F_1, E_2F_2, and E_3F_3 to find the attachment points GH (moving pivots) on link 3 that will allow the desired fixed pivots O_2 and O_4 to be used for the three specified output positions. In effect we will now consider the ground link O_2O_4 to be a coupler moving through the inverse of the original three positions, find the "ground pivots" GH needed for that inverted motion, and put them on the real coupler instead. The inversion process done in Example 3-7 and Figure 3-10 has swapped the roles of coupler and ground plane. The remaining task is identical to that done in Example 3-5 and Figure 3-8. The result of the synthesis then must be reinverted to obtain the solution.

3



(*a*) Original coupler three-position problem with specified pivots

(*b*) Position of the ground plane relative to the second coupler position

(*c*) Transferring second ground plane position to reference location at first position

(*d*) Position of the ground plane relative to the third coupler position

(*e*) Transferring third ground plane position to reference location at first position

(*f*) The three inverted positions of the ground plane corresponding to the original coupler position

FIGURE 3-10

Inverting the three-position motion synthesis problem

EXAMPLE 3-8

Finding the Moving Pivots for Three Positions and Specified Fixed Pivots

Problem: Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3. Use specified fixed pivots O_2 and O_4. Find the required moving pivot locations on the coupler by inversion.

Solution: Using the inverted ground link positions E_1F_1, E_2F_2, and E_3F_3 found in Example 3-7, find the fixed pivots for that inverted motion, then reinvert the resulting linkage to create the moving pivots for the three positions of coupler CD that use the selected fixed pivots O_2 and O_4 as shown in Figure 3-10a (see also Figure 3-11*).

1 Start with the inverted three positions in the plane as shown in Figures 3-10f and 3-11a. Lines E_1F_1, E_2F_2, and E_3F_3 define the three positions of the inverted link to be moved.

2 Draw construction lines from point E_1 to E_2 and from point E_2 to E_3.

3 Bisect line E_1E_2 and line E_2E_3 and extend the perpendicular bisectors until they intersect. Label the intersection G.

4 Repeat steps 2 and 3 for lines F_1F_2 and F_2F_3. Label the intersection H.

5 Connect G with E_1 and call it link 2. Connect H with F_1 and call it link 4. See Figure 3-11b.

6 In this inverted linkage, line E_1F_1 is the coupler, link 3. Line GH is the "ground" link 1.

7 We must now reinvert the linkage to return to the original arrangement. Line E_1F_1 is really the ground link O_2O_4, and GH is really the coupler. Figure 3-11c shows the reinversion of the linkage in which points G and H are now the moving pivots on the coupler and E_1F_1 has resumed its real identity as ground link O_2O_4. (See Figure 3-10e)

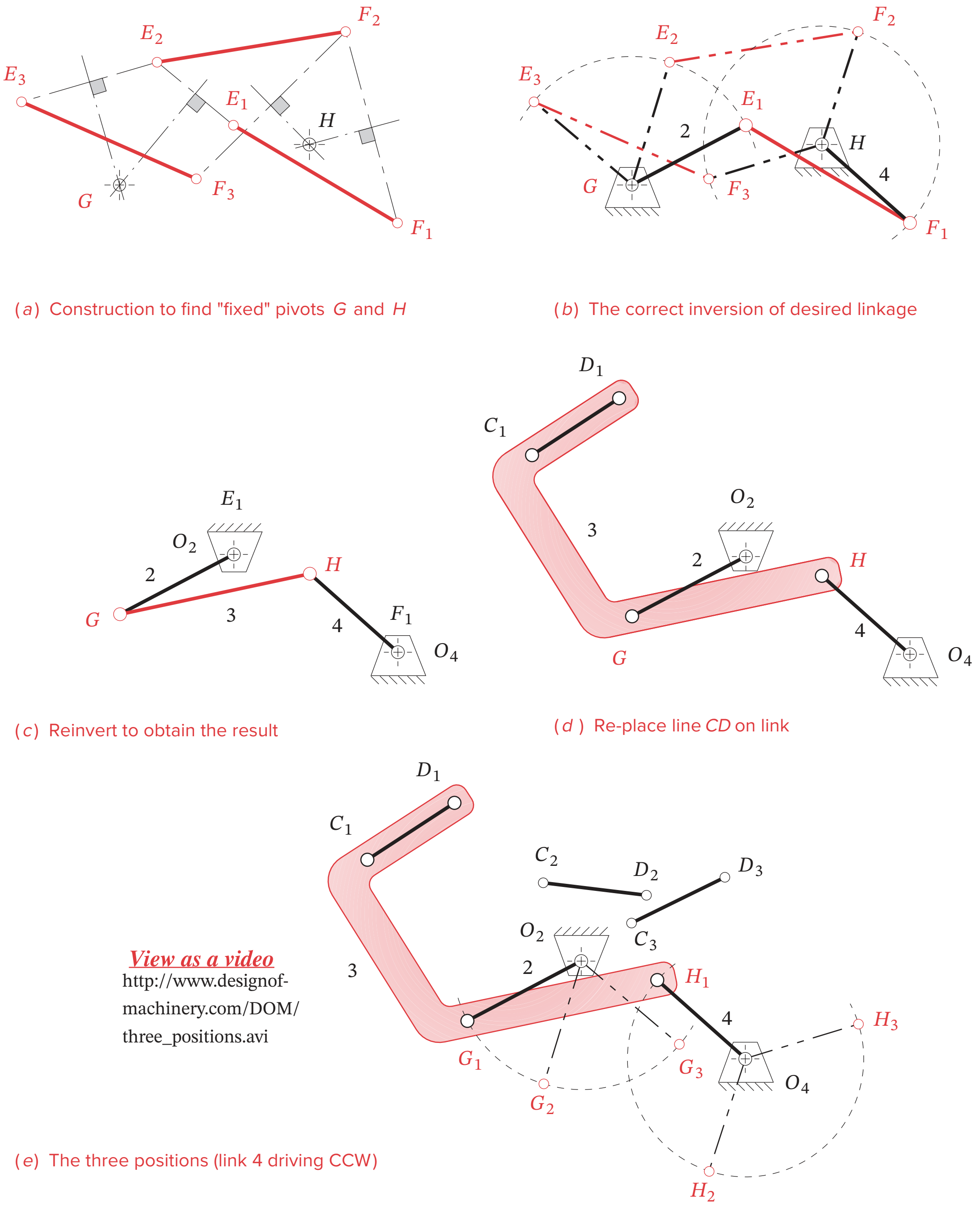
8 Figure 3-11d reintroduces the original line C_1D_1 in its correct relationship to line O_2O_4 at the initial position as shown in the original problem statement in Figure 3-10a. This forms the required coupler plane and defines a minimal shape of link 3.

9 The angular motions required to reach the second and third positions of line CD shown in Figure 3-11e are the same as those defined in Figure 3-11b for the linkage inversion. The angle F_1HF_2 in Figure 3-11b is the same as angle $H_1O_4H_2$ in Figure 3-11e and F_2HF_3 is the same as angle $H_2O_4H_3$. The angular excursions of link 2 retain the same relationship between Figure 3-11b and e as well. The angular motions of links 2 and 4 are the same for both inversions as the link excursions are relative to one another.

10 Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case provided that the linkage has mobility among all three positions. This solution is a non-Grashof linkage.

11 Construct a model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions. In this case links 3 and 4 reach a toggle position between points H_1 and H_2. This means that this linkage cannot be driven from link 2 as it will hang up at that toggle position. It must be driven from link 4.

* This figure is provided as animated AVI and Working Model files. Its filename is the same as the figure number.



(a) Construction to find "fixed" pivots G and H

(b) The correct inversion of desired linkage

(c) Reinvert to obtain the result

(d) Re-place line CD on link

View as a video
http://www.designof-machinery.com/DOM/three_positions.avi

(e) The three positions (link 4 driving CCW)

FIGURE 3-11

Constructing the linkage for three positions with specified fixed pivots by inversion

By inverting the original problem, we have reduced it to a more tractable form that allows a direct solution by the general method of three-position synthesis from Examples 3-5 and 3-6.

Position Synthesis for More Than Three Positions

It should be obvious that the more constraints we impose on these synthesis problems, the more complicated the task becomes to find a solution. When we define more than three positions of the output link, the difficulty increases substantially.

FOUR-POSITION SYNTHESIS does not lend itself as well to manual graphical solutions, though Hall[3] does present one approach. Probably the best approach is that used by Sandor and Erdman[4] and others, which is a quantitative synthesis method and requires a computer to execute it. Briefly, a set of simultaneous vector equations is written to represent the desired four positions of the entire linkage. These are then solved after some free choices of variable values are made by the designer. The computer program LINCAGES[1] by Erdman and Gustafson and the program KINSYN[5] by Kaufman, both provide a convenient and user-friendly computer graphics-based means to make the necessary design choices to solve the four-position problem. See Chapter 5 for further discussion.

3.5 QUICK-RETURN MECHANISMS *View the lecture video (55:10)*†

† http://www.designofmachinery.com/DOM/Quick_Return_Linkages.mp4

Many machine design applications have a need for a difference in average velocity between their "forward" and "return" strokes. Typically some external work is being done by the linkage on the forward stroke, and the return stroke needs to be accomplished as rapidly as possible so that a maximum of time will be available for the working stroke. Many arrangements of links will provide this feature. The only problem is to synthesize the right one!

Fourbar Quick-Return

The linkage synthesized in Example 3-1 is perhaps the simplest example of a fourbar linkage design problem (see Figure 3-4, and program LINKAGES disk file F03-04.4br). It is a crank-rocker that gives two positions of the rocker with equal time for the forward stroke and the return stroke. This is called a *non-quick-return* linkage, and it is a special case of the more general **quick-return** case. The reason for its non-quick-return state is the positioning of the crank center O_2 on the chord B_1B_2 extended. This results in equal angles of 180 degrees being swept out by the crank as it drives the rocker from one extreme (toggle position) to the other. If the crank is rotating at constant angular velocity, as it will tend to when motor driven, then each 180 degree sweep, forward and back, will take the same time interval. Try this with your model from Example 3-1 by rotating the crank at uniform velocity and observing the rocker motion and velocity.

If the crank center O_2 is located off the chord B_1B_2 extended, as shown in Figure 3-1b and Figure 3-12, then unequal angles will be swept by the crank between the toggle positions (defined as colinearity of crank and coupler). Unequal angles will give unequal time, when the crank rotates at constant velocity. These angles are labeled α and β in Figure 3-12. Their ratio α/β is called the **time ratio** (T_R) and defines the degree of quick return of the linkage. Note that the term **quick return** is arbitrarily used to describe this kind of linkage. If the crank is rotated in the opposite direction, it will be a **quick-forward** mechanism. Given a completed linkage, it is a trivial task to estimate the time ratio by measuring or calculating the angles α and β. It is a more difficult task to design the linkage for a chosen time ratio. Hall[6] provides a graphical method to synthesize a quick-return Grashof fourbar. To do so we need to compute the values of α an β that will give the specified time ratio. We can write two equations involving α and β and solve them simultaneously.

$$T_R = \frac{\alpha}{\beta} \qquad \alpha + \beta = 360 \qquad \therefore \quad T_R = \frac{\alpha}{360 - \alpha} \tag{3.1}$$

We also must define a construction angle,

$$\delta = |180 - \alpha| = |180 - \beta| \tag{3.2}$$

which will be used to synthesize the linkage.

EXAMPLE 3-9

Fourbar Crank-Rocker Quick-Return Linkage for Specified Time Ratio

Problem: Redesign Example 3-1 to provide a time ratio of 1:1.25 with 45° output rocker motion.

Solution: (See Figure 3-12.)

1 Draw the output link O_4B in both extreme positions, in any convenient location, such that the desired angle of motion, θ_4, is subtended.

2 Calculate α, β, and δ using equations 3.1 and 3.2. In this example, $\alpha = 160°$, $\beta = 200°$, $\delta = 20°$.

3 Draw a construction line through point B_1 at any convenient angle.

4 Draw a construction line through point B_2 at angle δ from the first line.

5 Label the intersection of the two construction lines O_2.

6 The line O_2O_4 now defines the ground link.

7 Find lengths of crank and coupler by measuring O_2B_1 and O_2B_2 and simultaneously solving:

$$\text{Coupler} + \text{crank} = O_2B_1$$

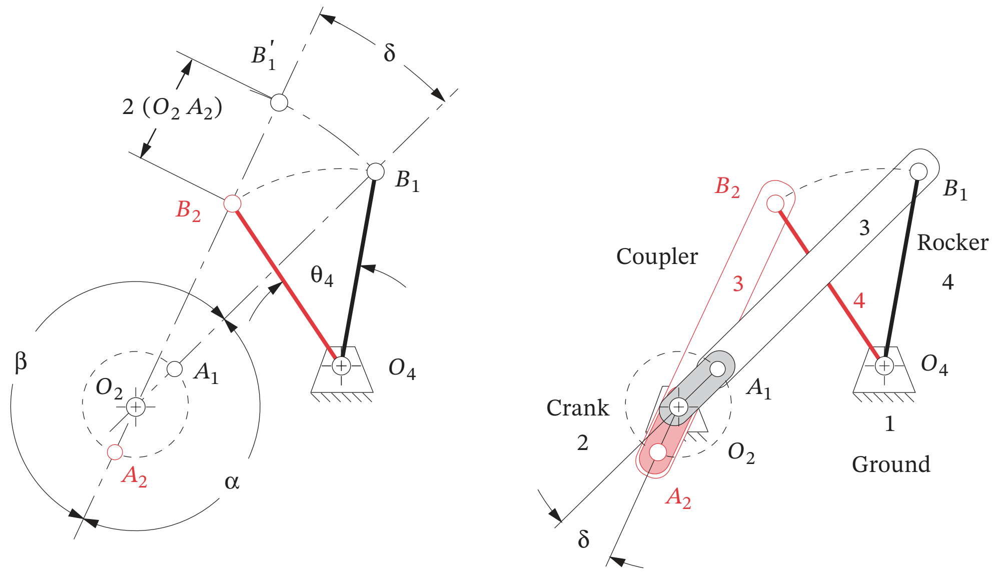
$$\text{Coupler} - \text{crank} = O_2B_2$$

or you can construct the crank length by swinging an arc centered at O_2 from B_1 to cut line O_2B_2 extended. Label that intersection B_1'. The line B_2B_1' is twice the crank length. Bisect this line segment to measure crank length O_2A_1.

8 Calculate the Grashof condition. If non-Grashof, repeat steps 3 to 8 with O_2 farther from O_4.

9 Make a model of the linkage and articulate it to check its function.

10 Check the transmission angles.



(*a*) Construction of a quick-return Grashof crank-rocker

(*b*) The finished linkage in its two toggle positions

FIGURE 3-12

Quick-return Grashof fourbar crank-rocker linkage

This method works well for time ratios down to about 1:1.5. Beyond that value the transmission angles become poor, and a more complex linkage is needed. Input the file F03-12.4br to program LINKAGES to see Example 3-9.

Sixbar Quick-Return

Larger time ratios, up to about 1:2, can be obtained by designing a sixbar linkage. The strategy here is to first design a fourbar drag link mechanism that has the desired time ratio between its driver crank and its driven or “dragged” crank, and then add a dyad (twobar) output stage, driven by the dragged crank. This dyad can be arranged to have either a rocker or a translating slider as the output link. First the drag link fourbar will be synthesized; then the dyad will be added.*

EXAMPLE 3-10

Sixbar Drag Link Quick-Return Linkage for Specified Time Ratio.

Problem: Provide a time ratio of 1:1.4 with 90° rocker motion.

Solution: (See Figure 3-13.)

1 Calculate α and β using equations 3.1. For this example, $\alpha = 150°$ and $\beta = 210°$.

2 Draw a line of centers XX at any convenient location.

3 Choose a crank pivot location O_2 on line XX. Draw an axis YY perpendicular to XX through O_2.

4 Draw a circle of convenient radius O_2A about center O_2.

5 Lay out angle α with vertex at O_2, symmetrical about quadrant one.

6 Label points A_1 and A_2 at the intersections of the lines subtending angle α and the circle of radius O_2A .

7 Set the compass to a convenient radius AC long enough to cut XX in two places on either side of O_2 when swung from both A_1 and A_2. Label the intersections C_1 and C_2.

8 The line O_2A_1 is the driver crank, link 2, and line A_1C_1 is the coupler, link 3.

9 The distance C_1C_2 is twice the driven (dragged) crank length. Bisect it to find fixed pivot O_4.

10 The line O_2O_4 now defines the ground link. Line O_4C_1 is the driven crank, link 4.

11 Calculate the Grashof condition. If non-Grashof, repeat steps 7 to 11 using a smaller radius in step 7.

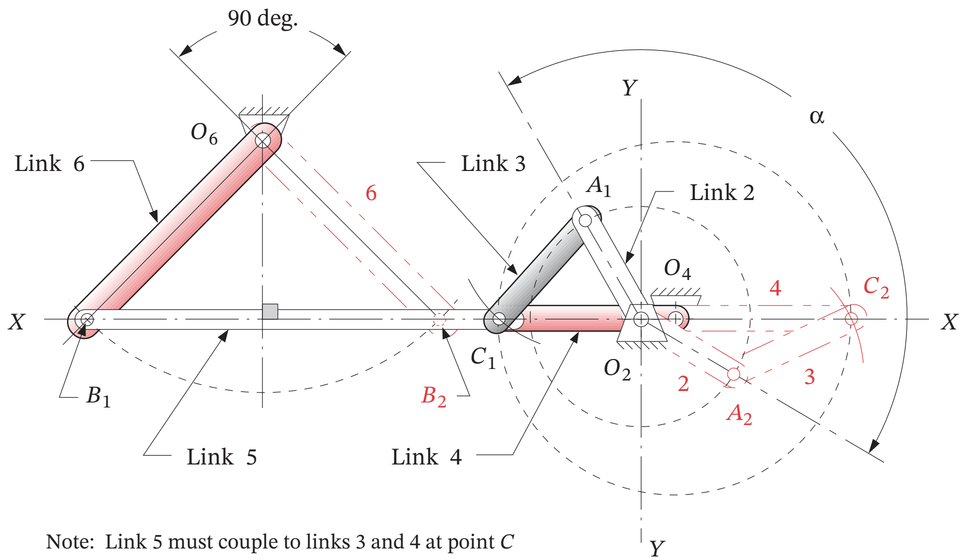
12 Invert the method of Example 3-1 to create the output dyad using XX as the chord and O_4C_1 as the driving crank. The points B_1 and B_2 will lie on line XX and be spaced apart a distance $2O_4C_1$. The pivot O_6 will lie on the perpendicular bisector of B_1B_2, at a distance from line XX which subtends the specified output rocker angle.

13 Check the transmission angles.

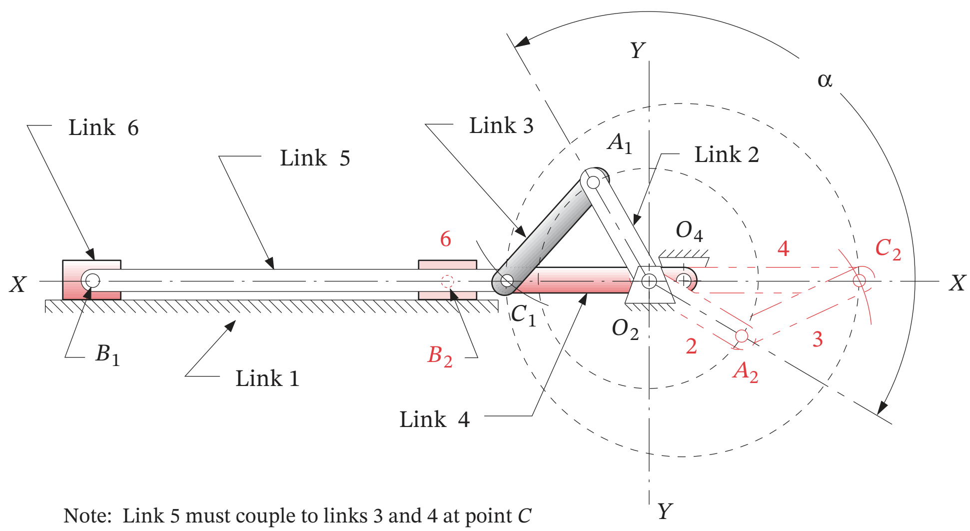
This linkage provides a quick return when a constant-speed motor is attached to link 2. Link 2 will go through angle α while link 4 (which is dragging the output dyad along) goes through the first 180 degrees, from position C_1 to C_2. Then, while link 2 completes its cycle through β degrees, the output stage will complete another 180 degrees from C_2 to C_1. Since angle β is greater than α, the forward stroke takes longer. Note that the chordal stroke of the output dyad is twice the crank length O_4C_1. This is independent of the angular displacement of the output link which can be tailored by moving the pivot O_6 closer to or farther from the line XX.

The transmission angle at the joint between link 5 and link 6 will be optimized if the fixed pivot O_6 is placed on the perpendicular bisector of the chord B_1B_2 as shown in Figure 3-13a. If a translating output is desired, the slider (link 6) will be located on line XX and will oscillate between B_1 and B_2 as shown in Figure 3-13b. The arbitrarily chosen size of this or any other linkage can be scaled up or down, simply by multiplying all link lengths by the same scale factor. Thus a design made to arbitrary size can be fit to any package. Input the file F03-13a.6br to program LINKAGES to see Example 3-10 in action.

CRANK-SHAPER QUICK RETURN A commonly used mechanism capable of large time ratios is shown in Figure 3-14. It is often used in metal shaper machines to provide a slow cutting stroke and a quick-return stroke when the tool is doing no work. It is the inversion #2 of the crank-slider mechanism as was shown in Figure 2-15b. It is very easy to synthesize this linkage by simply moving the rocker pivot O_4 along the vertical centerline



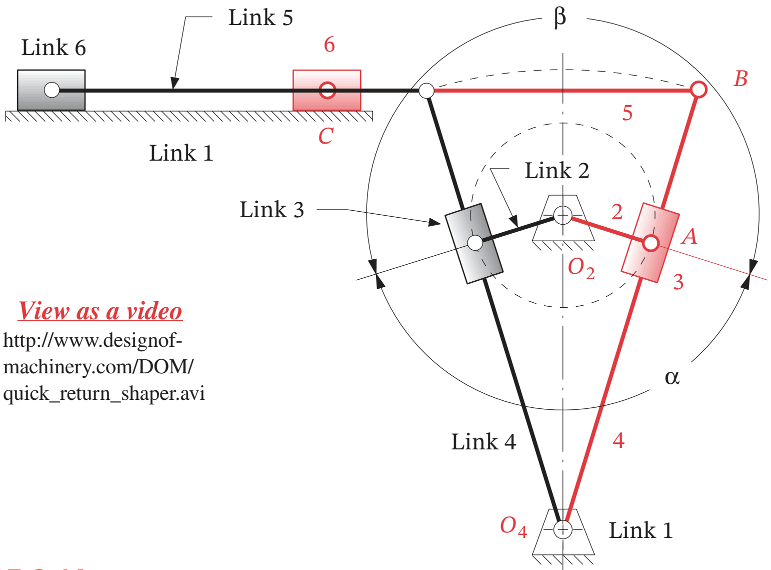
(*a*) Rocker output sixbar drag link quick-return mechanism



(*b*) Slider output sixbar drag link quick-return mechanism

FIGURE 3-13

Synthesizing a sixbar drag link quick-return mechanism



View as a video
http://www.designofmachinery.com/DOM/quick_return_shaper.avi

FIGURE 3-14

Crank-shaper quick-return mechanism

O_2O_4 while keeping the two extreme positions of link 4 tangent to the circle of the crank, until the desired time ratio (α / β) is achieved. Note that the angular displacement of link 4 is then defined as well. Link 2 is the input and link 6 is the output.

Depending on the relative lengths of the links, this mechanism is known as a **Whitworth** or **crank-shaper** mechanism. If the ground link is the shortest, then it will behave as a double-crank linkage, or *Whitworth mechanism*, with both pivoted links making full revolutions as shown in Figure 2-13b. If the driving crank is the shortest link, then it will behave as a crank-rocker linkage, or *crank-shaper mechanism*, as shown in Figure 3-14.* They are the same inversion as the slider block is in complex motion in each case.

3.6 COUPLER CURVES *View the lecture video (59:57)*§

A **coupler** is the most interesting link in any linkage. It is in complex motion, and thus points on the coupler can have path motions of high degree.† In general, the more links, the higher the degree of curve generated, where **degree** here means *the highest power of any term in its equation*. A curve (function) can have *up to* as many intersections (roots) with any straight line as the degree of the function. The *fourbar crank-slider* has, in general, fourth-degree coupler curves; the *pin-jointed fourbar*, up to sixth degree.†† The geared fivebar, the sixbar, and more complicated assemblies all have still higher-degree curves. Wunderlich[7b] derived an expression for the highest degree m possible for a coupler curve of a mechanism of n links connected with only revolute joints.

$$m = 2 \cdot 3^{(n/2-1)} \tag{3.3}$$

This gives, respectively, degrees of 6, 18, and 54 for the fourbar, sixbar, and eightbar linkage coupler curves. Specific points on their couplers may have degenerate curves of

3

* This figure is provided as animated AVI and Working Model files. Its filename is the same as the figure number.

† In 1876, Kempe[7a] proved his theory that a linkage with only revolute (pin) and prismatic (slider) joints can be found that will trace any algebraic curve of any order or complexity. But the linkage for a particular curve may be excessively complex, may be unable to traverse the curve without encountering limit (toggle) positions, and may even need to be disassembled and reassembled to reach all points on the curve. See the discussion of circuit and branch defects in Section 4.12. Nevertheless this theory points to the potential for interesting motions from the coupler curve.

§ http://www.designofmachinery.com/DOM/Coupler_Curves.mp4

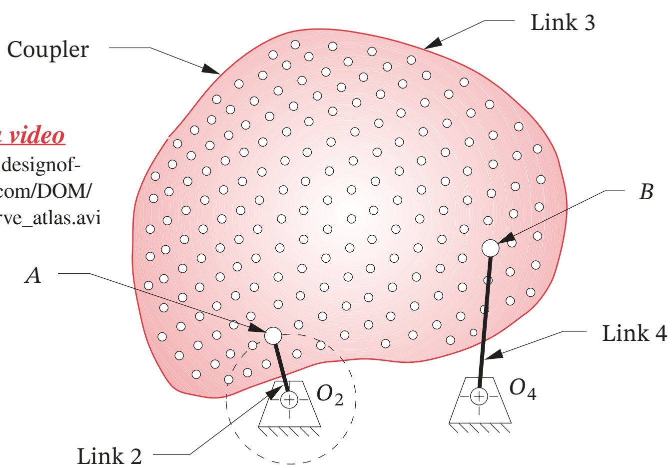
†† The algebraic equation of the coupler curve is sometimes referred to as a "tricircular sextic" referring respectively to its circularity of 3 (it can contain 3 loops) and its degree of 6. See Chapter 5 for its equation.

lower degree as, for example, the pin joints between any crank or rocker and the coupler that describes second-degree curves (circles). The parallelogram fourbar linkage has degenerate coupler curves, all of which are circles.

All linkages that possess one or more "floating" coupler links will generate coupler curves. It is interesting to note that these will be closed curves even for non-Grashof linkages. The coupler (or any link) can be extended infinitely in the plane. Figure 3-15† shows a fourbar linkage with its coupler extended to include a large number of points, each of which describes a different coupler curve. Note that these points may be anywhere on the coupler, including along line *AB*. There is, of course, an infinity of points on the coupler, each of which generates a different curve.

Coupler curves can be used to generate quite useful path motions for machine design problems. They are capable of *approximating straight lines* and *large circle arcs* with remote centers. Recognize that the coupler curve is a solution to the path generation problem described in Section 3.2. It is not by itself a solution to the motion generation problem, since the attitude or orientation of a line on the coupler is not predicted by the information contained in the path. Nevertheless it is a very useful device, and it can be converted to a parallel motion generator by adding two links as described in the next section. As we shall see, approximate straight-line motions, dwell motions, and more complicated symphonies of timed motions are available from even the simple fourbar linkage and its infinite variety of often surprising coupler curve motions.

CUSPS AND CRUNODES come in a variety of shapes which can be crudely categorized as shown in Figure 3-16. There is an infinite range of variation between these generalized shapes. Interesting features of some coupler curves are the **cusp** and **crunode**. A **cusp** is *a sharp point on the curve which has the useful property of instantaneous zero velocity*. Note that *the acceleration at the cusp is not zero*. The simplest example of a curve with a cusp is the cycloid curve which is generated by a point on the rim of a wheel rotating on a flat surface. When the point touches the surface, it has the same (zero) velocity as all points on the stationary surface, provided there is pure rolling and no slip between



View a video
http://www.designofmachinery.com/DOM/coupler_curve_atlas.avi

FIGURE 3-15
The fourbar coupler extended to include a large number of coupler points

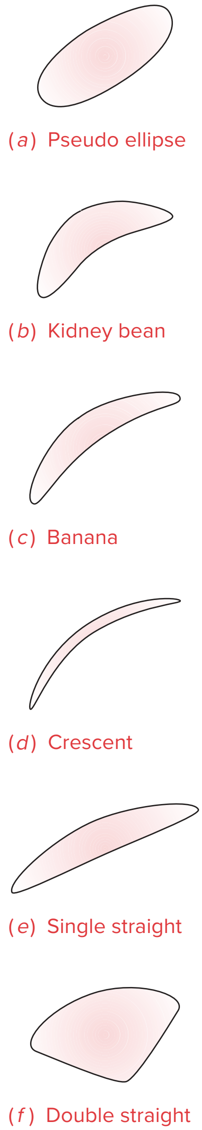


FIGURE 3-16 Part 1
A "Cursory Catalog" of coupler curve shapes

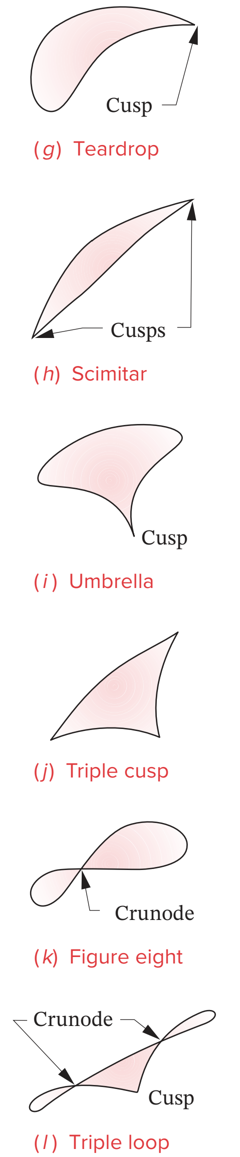


FIGURE 3-16 Part 2

A "Cursory Catalog" of coupler curve shapes

the elements. Anything attached to a cusp point will come smoothly to a stop along one path and then accelerate smoothly away from that point on a different path. The cusp's feature of zero velocity has value in such applications as transporting, stamping, and feeding processes. A **crunode** *is a double point that occurs where the coupler curve crosses itself creating multiple loops*. The two slopes (tangents) at a crunode give the point two different velocities, neither of which is zero in contrast to the cusp. In general, a fourbar coupler curve can have up to three real double points,† which may be a combination of cusps and crunodes as can be seen in Figure 3-16.

The Hrones and Nelson (H&N) atlas of fourbar coupler curves[8a] is a useful reference which can provide the designer with a starting point for further design and analysis. It contains about 7000 coupler curves and defines the linkage geometry for each of its Grashof crank-rocker linkages. Figure 3-17a* reproduces a page from this book and the entire atlas is reproduced as PDF files in the books downloadable files. The H&N atlas is logically arranged, with all linkages defined by their link ratios, based on a unit length crank. The coupler is shown as a matrix of fifty coupler points for each linkage geometry, arranged ten to a page. Thus each linkage geometry occupies five pages. Each page contains a schematic "key" in the upper right corner which defines the link ratios.

Figure 3-17b shows a "fleshed out" linkage drawn on top of the H&N atlas page to illustrate its relationship to the atlas information. The double circles in Figure 3-17a define the fixed pivots. The crank is always of unit length. The ratios of the other link lengths to the crank are given on each page. The actual link lengths can be scaled up or down to suit your package constraints and this will affect the size but not the shape of the coupler curve. Any one of the ten coupler points shown can be used by incorporating it into a triangular coupler link. The location of the chosen coupler point can be scaled from the atlas and is defined within the coupler by the position vector **R** whose constant angle ϕ is measured with respect to the line of centers of the coupler. The H&N coupler curves are shown as dashed lines. Each dash station represents **five degrees** of crank rotation. So, for an assumed constant crank velocity, the dash spacing is proportional to path velocity. The changes in velocity and the quick-return nature of the coupler path motion can be clearly seen from the dash spacing.

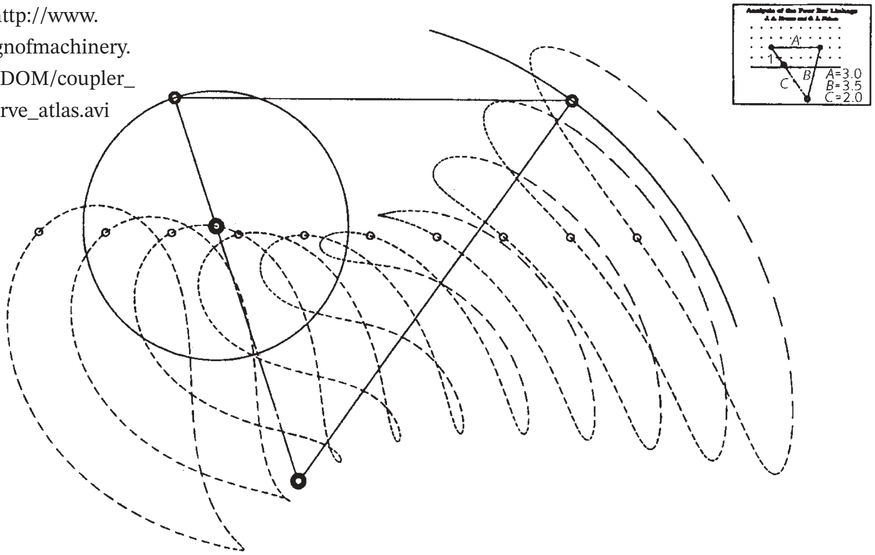
One can peruse this linkage atlas resource and find an approximate solution to virtually any path generation problem. Then one can take the tentative solution from the atlas to a CAE resource such as the LINKAGES program and further refine the design, based on the complete analysis of positions, velocities, and accelerations provided by the program. The only data needed for the LINKAGES program are the four link lengths and the location of the chosen coupler point with respect to the line of centers of the coupler link as shown in Figure 3-17. These parameters can be changed within program LINKAGES to alter and refine the design. Input the file F03-17b.4br to program LINKAGES to animate the linkage shown in that figure. Also see the video "Coupler Curves" for more information.

An example of an application of a fourbar linkage to a practical problem is shown in Figure 3-18* which is a movie camera (or projector) film advance mechanism. Point O_2 is

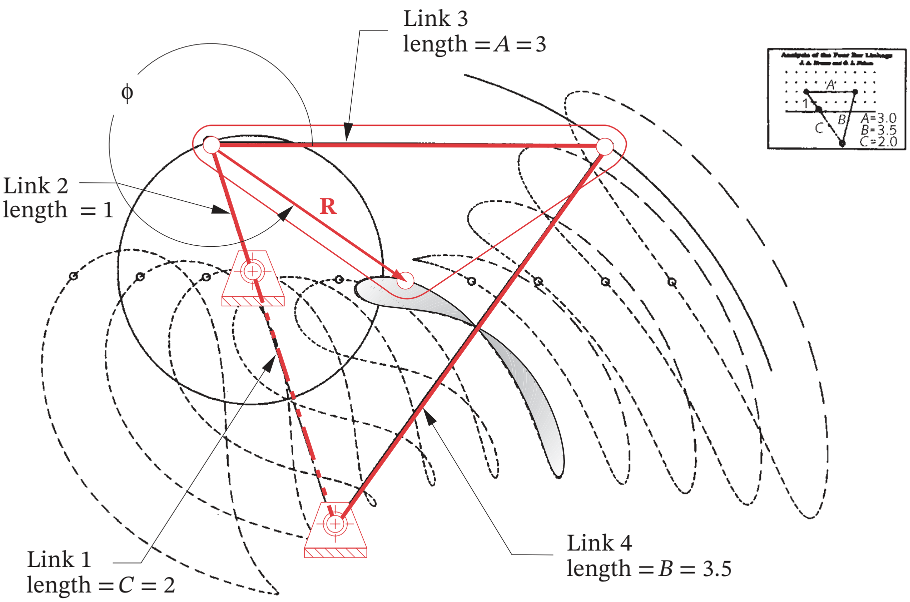
* These figures are provided as animated AVI and Working Model files. Its filename is the same as the figure number.

† Actually, the fourbar coupler curve has 9 double points of which 6 are usually imaginary. However, Fichter and Hunt[8b] point out that some unique configurations of the fourbar linkage (i.e., rhombus parallelograms and those close to this configuration) can have up to 6 real double points which they denote as comprising 3 "proper" and 3 "improper" real double points. For non-special-case Grashof fourbar linkages with minimum transmission angles suitable for engineering applications, only the 3 "proper" double points will appear.

View as a video
http://www.designofmachinery.com/DOM/coupler_curve_atlas.avi



(*a*) A page from the Hrones and Nelson atlas of fourbar coupler curves
Hrones, J. A., and G. L. Nelson (1951). Analysis of the Fourbar Linkage MIT Technology Press, Cambridge, MA. Reprinted with permission.



(*b*) Creating the linkage from the information in the atlas

FIGURE 3-17*
Selecting a coupler curve and constructing the linkage from the Hrones and Nelson atlas

3

* The Hrones and Nelson atlas is long out of print, but a reproduction is included as downloadable PDF files with this book. A video, ”Coupler Curves” is also provided that describes the curve’s properties and shows how to extract the information from the atlas and use it to design a practical mechanism.

Also, a similar volume to the H&N book called the *Atlas of Linkage Design and Analysis Vol. 1: The Four Bar Linkage* is available from Saltire Software, 9725 SW Gemini Drive, Beaverton, OR 97005, (800) 659-1874.

There is also a web site at http://www.cedarville.edu/cf/engineering/kinematics/ccapdf/fccca.htm created by Prof. Thomas J. Thompson of Cedarville College, which provides an interactive coupler curve atlas that allows the link dimensions to be changed and generates the coupler curves on screen.[21]

Program LINKAGES, included with this text, also allows rapid investigation of coupler curve shapes. For any defined linkage geometry, the program draws the coupler curve. By shift-clicking the mouse pointer on the coupler point and dragging it around, you will see the coupler curve shape instantly update for each new coupler point location. When you release the mouse button, the new linkage geometry is preserved for that curve.

View as a video
http://www.designofmachinery.com/DOM/camera.avi

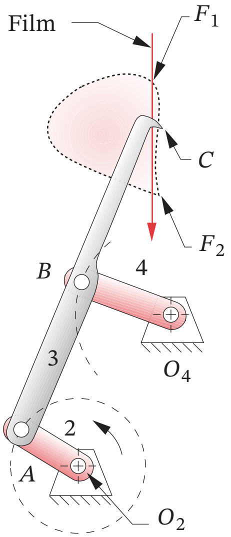


FIGURE 3-18

Movie camera film-advance mechanism. (Input the file F03-18.4br to program LINKAGES to animate this linkage.

the crank pivot which is motor driven at constant speed. Point O_4 is the rocker pivot, and points A and B are the moving pivots. Points A, B, and C define the coupler where C is the coupler point of interest. A movie is really a series of still pictures, each "frame" of which is projected for a small fraction of a second on the screen. Between each picture, the film must be moved very quickly from one frame to the next while the shutter is closed to blank the screen. The whole cycle takes only 1/24 of a second. The human eye's response time is too slow to notice the flicker associated with this discontinuous stream of still pictures, so it appears to us to be a continuum of changing images.

The linkage shown in Figure 3-18* is cleverly designed to provide the required motion. A hook is cut into the coupler of this fourbar Grashof crank-rocker at point C which generates the coupler curve shown. The hook will enter one of the sprocket holes in the film as it passes point F_1. Notice that the direction of motion of the hook at that point is nearly perpendicular to the film, so it enters the sprocket hole cleanly. It then turns abruptly downward and follows a crudely approximate straight line as it rapidly pulls the film downward to the next frame. The film is separately guided in a straight track called the "gate." The shutter (driven by another linkage from the same driveshaft at O_2) is closed during this interval of film motion, blanking the screen. At point F_2 there is a cusp on the coupler curve which causes the hook to decelerate smoothly to zero velocity in the vertical direction, and then as smoothly accelerate up and out of the sprocket hole. The abrupt transition of direction at the cusp allows the hook to back out of the hole without jarring the film, which would make the image jump on the screen as the shutter opens. The rest of the coupler curve motion is essentially "wasting time" as it proceeds up the back side, to be ready to enter the film again to repeat the process. Input the file F03-18.4br to program LINKAGES to animate the linkage shown in that figure.

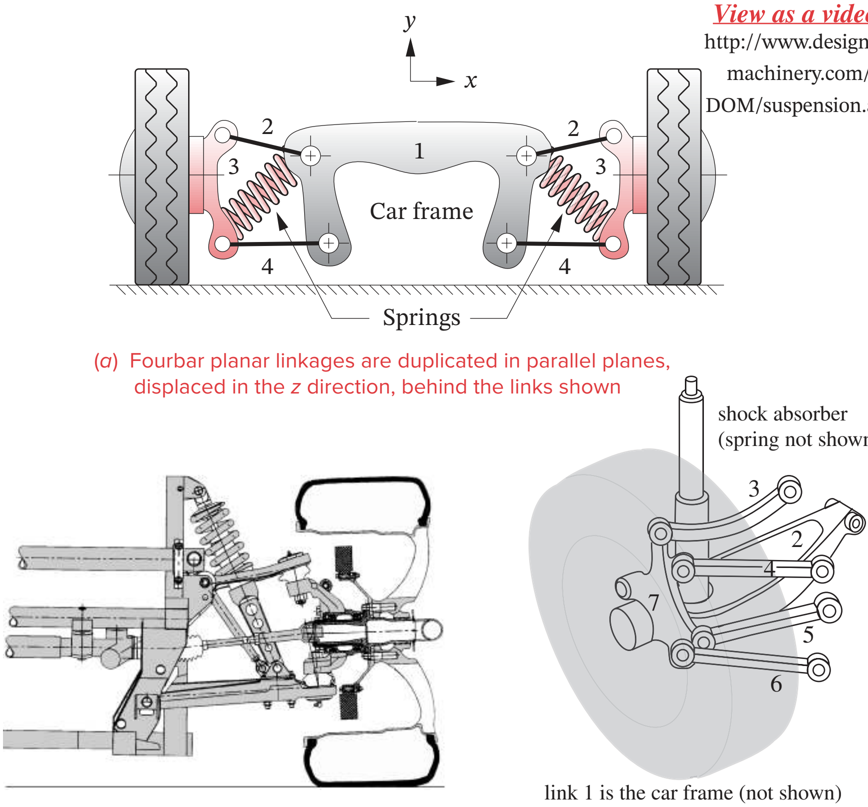
Some advantages of using this type of device for this application are that it is very simple and inexpensive (only four links, one of which is the frame of the camera), is extremely reliable, has low friction if good bearings are used at the pivots, and can be reliably timed with the other events in the overall camera mechanism through common shafting from a single motor. There are a myriad of other examples of fourbar coupler curves used in machines and mechanisms of all kinds.

One other example of a very different application is that of the automobile suspension (Figure 3-19). Typically, the up and down motions of the car's wheels are controlled by some combination of planar fourbar linkages, arranged in duplicate to provide three-dimensional control as described in Section 3.2. Only a few manufacturers currently use a true spatial linkage in which the links are not arranged in parallel planes. In all cases the wheel assembly is attached to the coupler of the linkage assembly, and its motion is along a set of coupler curves. The orientation of the wheel is also of concern in this case, so this is not strictly a path generation problem. By designing the linkage to control the paths of multiple points on the wheel (tire contact patch, wheel center, etc.—all of which are points on the same coupler link extended), motion generation is achieved as the coupler has complex motion. Figure 3-19a* and b* shows parallel planar fourbar linkages suspending the wheels. The coupler curve of the wheel center is nearly a straight line over the small vertical displacement required. This is desirable as the idea is to keep the tire perpendicular to the ground for best traction under all cornering and attitude changes of the car body. This is an application in which a non-Grashof linkage is perfectly acceptable, as full rotation of the wheel in this plane might have some undesirable results and surprise the driver. Limit stops are of course provided to prevent such behavior, so even a Grashof linkage could be

* These figures are provided as animated AVI and Working Model files. Its filename is the same as the figure number.

3

View as a video
http://www.designofmachinery.com/DOM/suspension.avi



(*a*) Fourbar planar linkages are duplicated in parallel planes, displaced in the *z* direction, behind the links shown

(*b*) Fourbar linkage used to control wheel motion

(*c*) Multilink spatial linkage used to control rear wheel motion

FIGURE 3-19

Linkages used in automotive chassis suspensions

used. The springs support the weight of the vehicle and provide a fifth, variable-length "force link" that stabilizes the mechanism as was described in Section 2.15. The function of the fourbar linkage is solely to guide and control the wheel motions. Figure 3-19c shows a true spatial linkage of seven links (including frame and wheel) and nine joints (some of which are ball-and-socket joints) used to control the motion of the rear wheel. These links do not move in parallel planes but rather control the three-dimensional motion of the coupler which carries the wheel assembly.

Symmetrical-Linkage Coupler Curves *View the lecture video (05:48)*§

When a fourbar linkage's geometry is such that the coupler and rocker are the same length pin-to-pin, all coupler points that lie on a circle centered on the coupler-rocker joint with radius equal to the coupler length will generate symmetrical coupler curves. Figure 3-20 shows such a linkage, its symmetrical coupler curve, and the locus of all points that will give symmetrical curves. Using the notation of that figure, the criterion for coupler curve symmetry can be stated as:

§ http://www.designofmachinery.com/DOM/Symmetrical_Coupler_Curves.mp4

3

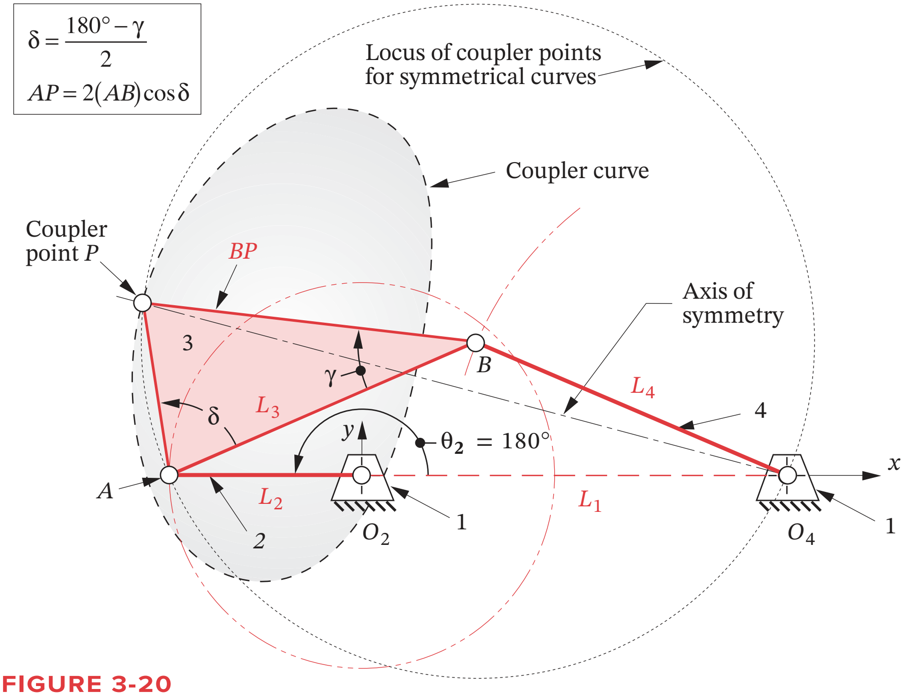


FIGURE 3-20

A fourbar linkage with a symmetrical coupler curve

$$AB = O_4B = BP \tag{3.4}$$

A linkage for which equation 3.4 is true is referred to as a **symmetrical fourbar linkage**. The axis of symmetry of the coupler curve is the line O_4P drawn when the crank O_2A and the ground link O_2O_4 are colinear-extended (i.e., $\theta_2 = 180°$). Symmetrical coupler curves prove to be quite useful as we shall see in the next several sections. Some give good approximations to circular arcs and others give very good approximations to straight lines (over a portion of the coupler curve).

In the general case, nine parameters are needed to define the geometry of a **nonsymmetrical fourbar linkage** with one coupler point.* We can reduce this to five as follows. Three parameters can be eliminated by fixing the location and orientation of the ground link. The four link lengths can be reduced to three parameters by normalizing three link lengths to the fourth. The shortest link (the crank, if a Grashof crank-rocker linkage) is usually taken as the reference link, and three link ratios are formed as L_1/L_2, L_3/L_2, L_4/L_2, where L_1 = ground, L_2 = crank, L_3 = coupler, and L_4 = rocker length as shown in Figure 3-20. Two parameters are needed to locate the coupler point: the distance from a convenient reference point on the coupler (either B or A in Figure 3-20) to the coupler point P, and the angle that the line BP (or AP) makes with the line of centers of the coupler AB (either δ or γ). Thus, with a defined ground link, five parameters that will define the geometry of a nonsymmetrical fourbar linkage (using point B as the reference in link 3 and the labels of Figure 3-20) are: L_1/L_2, L_3/L_2, L_4/L_2, BP/L_2, and γ. Note that multiplying

* The nine independent parameters of a fourbar linkage are: four link lengths, two coordinates of the coupler point with respect to the coupler link, and three parameters that define the location and orientation of the fixed link in the global coordinate system.

these parameters by a scaling factor will change the size of the linkage and its coupler curve but will not change the coupler curve's shape.

A **symmetrical fourbar linkage** with a defined ground link needs only *three parameters* to define its geometry because three of the five nonsymmetrical parameters are now equal per equation 3.4: $L_3/L_2 = L_4/L_2 = BP/L_2$. Three possible parameters to define the geometry of a symmetrical fourbar linkage in combination with equation 3.4 are then: L_1/L_2, L_3/L_2, and γ. Having only three parameters to deal with rather than five greatly simplifies an analysis of the behavior of the coupler curve shape when the linkage geometry is varied. Other relationships for the isosceles-triangle coupler are shown in Figure 3-20. Length AP and angle δ are needed for input of the linkage geometry to program LINKAGES.

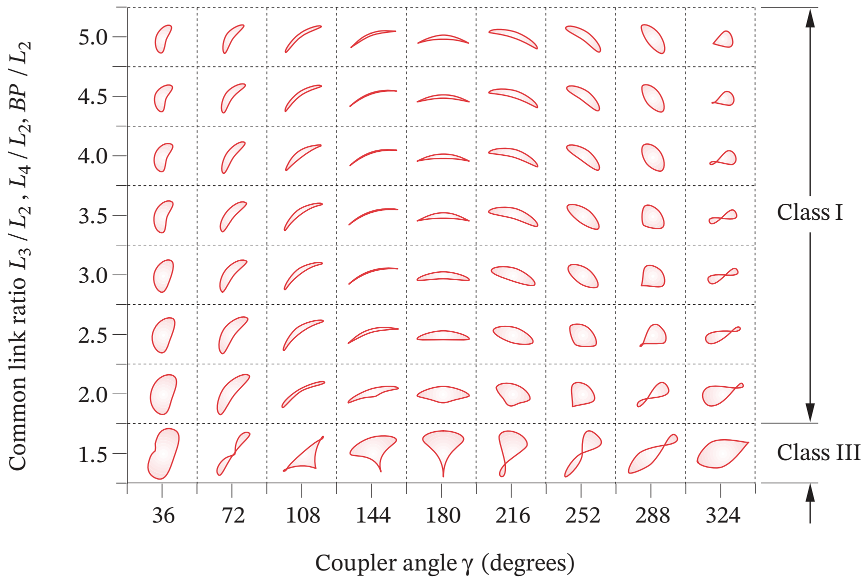
Kota[9] did an extensive study of the characteristics of coupler curves of symmetrical fourbar linkages and mapped coupler curve shape as a function of the three linkage parameters defined above. He defined a three-dimensional design space to map the coupler curve shape. Figure 3-21 shows two orthogonal plane sections taken through this design space for particular values of link ratios,† and Figure 3-22 shows a schematic of the design space. Though the two cross sections of Figure 3-21 show only a small fraction of the information in the 3-D design space of Figure 3-22, they nevertheless give a sense of the way that variation of the three linkage parameters affects the coupler curve shape. Used in combination with a linkage design tool such as program LINKAGES, these design charts can help guide the designer in choosing suitable values for the linkage parameters to achieve a desired path motion.

† Adapted from materials provided by Professor Sridhar Kota, University of Michigan.

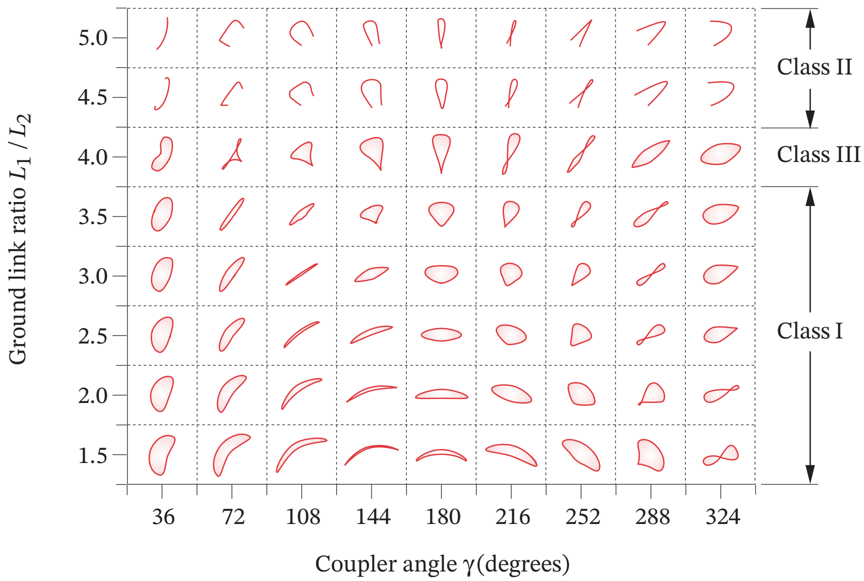
GEARED FIVEBAR COUPLER CURVES (Figure 3-23) are more complex than the fourbar variety. Because there are three additional, independent design variables in a geared fivebar compared to the fourbar (an additional link ratio, the gear ratio, and the phase angle between the gears), the coupler curves can be of higher degree than those of the fourbar. This means that the curves can be more convoluted, having more cusps and crunodes (loops). In fact, if the gear ratio used is noninteger, the input link will have to make a number of revolutions equal to the factor necessary to make the ratio an integer before the coupler curve pattern will repeat. The *Zhang, Norton, Hammond (ZNH) Atlas of Geared FiveBar Mechanisms (GFBM)*[10] shows typical coupler curves for these linkages limited to symmetrical geometry (e.g., link 2 = link 5 and link 3 = link 4) and gear ratios of ±1 and ±2. A page from the ZNH atlas is reproduced in Figure 3-23. Additional pages are in Appendix E, and the entire atlas is downloadable. Each page shows the family of coupler curves obtained by variation of the phase angle for a particular set of link ratios and gear ratio. A key in the upper right corner of each page defines the ratios: α = link 3 / link 2, β = link 1 / link 2, λ = gear 5 / gear 2. Symmetry defines links 4 and 5 as noted above. The phase angle ϕ is noted on the axes drawn at each coupler curve and can be seen to have a significant effect on the resulting coupler curve shape.

This reference atlas is intended to be used as a starting point for a geared fivebar linkage design. The link ratios, gear ratio, and phase angle can be input to the program FIVEBAR and then varied to observe the effects on coupler curve shape, velocities, and accelerations. Asymmetry of links can be introduced, and a coupler point location other than the pin joint between links 3 and 4 defined within the LINKAGES program as well. Note that program LINKAGES expects the gear ratio to be in the form gear 2 / gear 5 which is the inverse of the ratio λ in the ZNH atlas.

3



(*a*) Variation of coupler curve shape with common link ratio and coupler angle for a ground link ratio $L_1 / L_2 = 2.0$



(*b*) Variation of coupler curve shape with ground link ratio and coupler angle for a common link ratio $L_3 / L_2 = L_4 / L_2 = BP / L_2 = 2.5$

FIGURE 3-21

Coupler curve shapes of symmetrical fourbar linkages *Adapted from reference [9]*

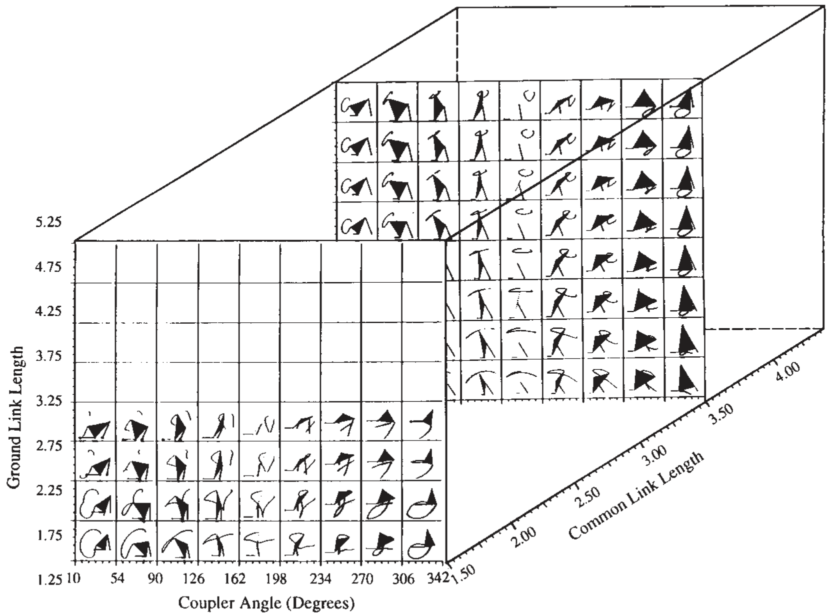


FIGURE 3-22

A three-dimensional map of coupler shapes of symmetrical fourbar linkages[9]

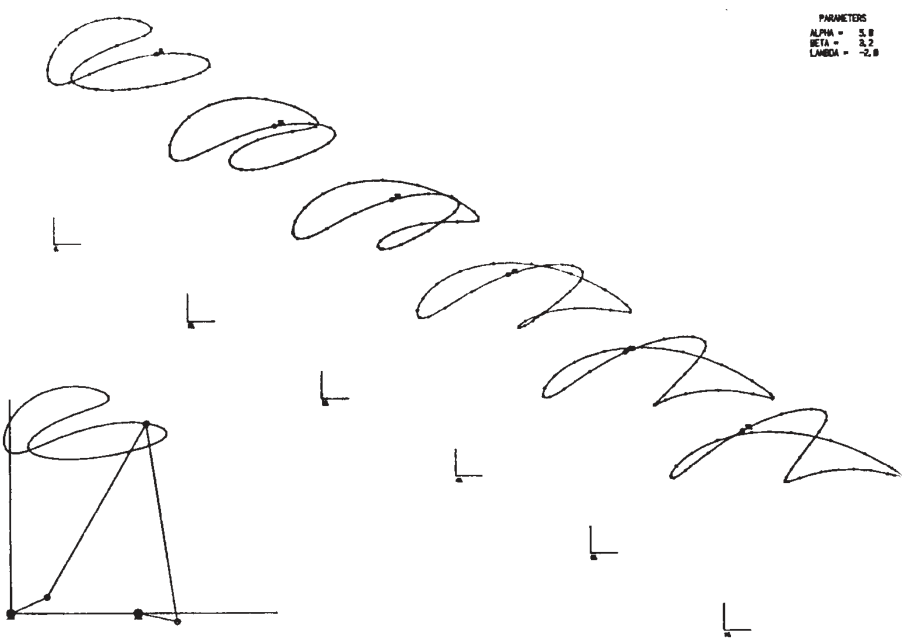


FIGURE 3-23

A page from the Zhang-Norton-Hammond atlas of geared fivebar coupler curves

3

3.7 COGNATES *View the lecture video (18:12)**

It sometimes happens that a good solution to a linkage synthesis problem will be found that satisfies path generation constraints but has the fixed pivots in inappropriate locations for attachment to the available ground plane or frame. In such cases, the use of a **cognate** to the linkage may be helpful. The term **cognate** was used by Hartenberg and Denavit[11] to describe *a linkage, of different geometry, which generates the same coupler curve*. Samuel Roberts (1875)[23] and Chebyschev (1878) independently discovered the theorem which now bears their names:

Roberts-Chebyschev Theorem

Three different planar, pin-jointed fourbar linkages will trace identical coupler curves.

Hartenberg and Denavit[11] presented extensions of this theorem to the crank-slider and the sixbar linkages:

Two different planar crank-slider linkages will trace identical coupler curves.†

The coupler-point curve of a planar fourbar linkage is also described by the joint of a dyad of an appropriate sixbar linkage.

Figure 3-24a shows a fourbar linkage for which we want to find the two cognates. The first step is to release the fixed pivots O_A and O_B. While holding the coupler stationary, rotate links 2 and 4 into colinearity with the line of centers (A_1B_1) of link 3 as shown in Figure 3-24b. We can now construct lines parallel to all sides of the links in the original linkage to create the **Cayley diagram**[24] in Figure 3-24c. This schematic arrangement defines the lengths and shapes of links 5 through 10 which belong to the cognates. All three fourbars share the original coupler point P and will thus generate the same path motion on their coupler curves.

In order to find the correct location of the fixed pivot O_C from the Cayley diagram, the ends of links 2 and 4 are returned to the original locations of the fixed pivots O_A and O_B as shown in Figure 3-25a. The other links will follow this motion, maintaining the parallelogram relationships between links, and fixed pivot O_C will then be in its proper location on the ground plane. This configuration is called a **Roberts diagram**—three fourbar linkage cognates which share the same coupler curve.

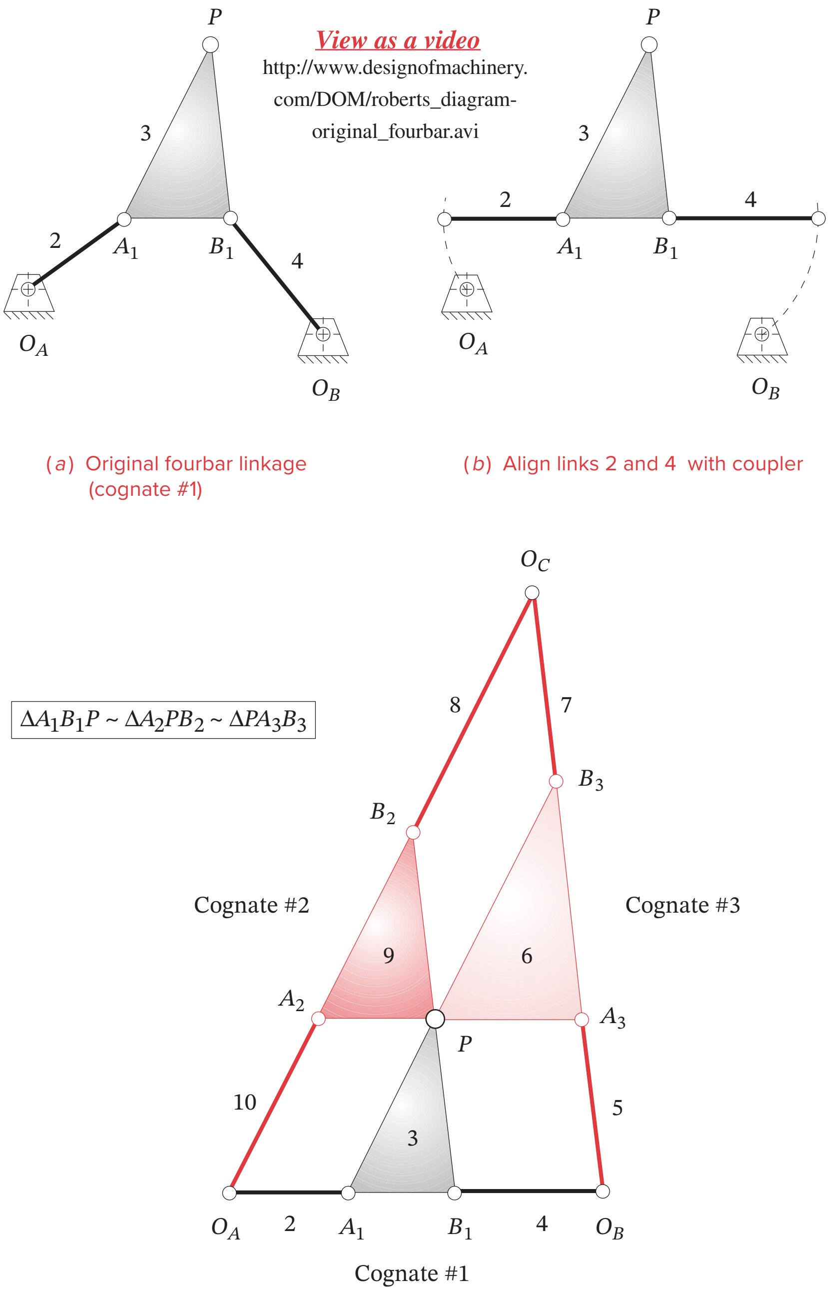
The Roberts diagram can be drawn directly from the original linkage without resort to the Cayley diagram by noting that the parallelograms which form the other cognates are also present in the Roberts diagram and the three couplers are similar triangles. It is also possible to locate fixed pivot O_C directly from the original linkage as shown in Figure 3-25a. Construct a similar triangle to that of the coupler, placing its base (AB) between O_A and O_B. Its vertex will be at O_C.

The ten-link Roberts configuration (Cayley's nine plus the ground) can now be articulated up to any toggle positions, and point P will describe the original coupler path which is the same for all three cognates. Point O_C will not move when the Roberts linkage is articulated, proving that it is a ground pivot. The cognates can be separated as shown in Figure 3-25b and any one of the three linkages used to generate the same coupler curve. Corresponding links in the cognates will have the same angular velocity as the original mechanism as defined in Figure 3-25.

* http://www.designofmachinery.com/DOM/Cognates_of_Linkages.mp4

† Dijksman and Smals [25] state that an inverted crank-slider linkage does not possess any cognates.

View as a video
http://www.designofmachinery.com/DOM/roberts_diagram-original_fourbar.avi



(*a*) Original fourbar linkage (cognate #1)

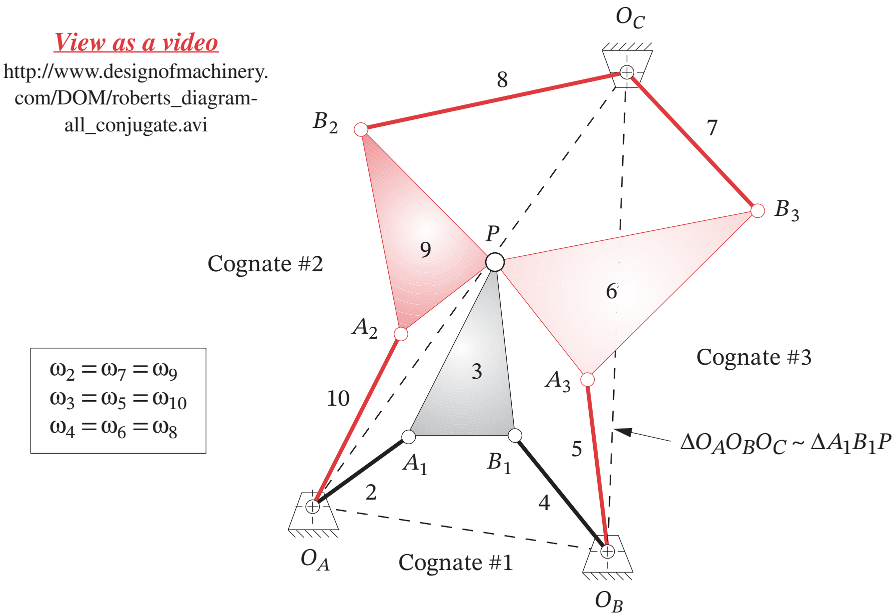
(*b*) Align links 2 and 4 with coupler

(*c*) Construct lines parallel to all sides of the original fourbar linkage to create cognates

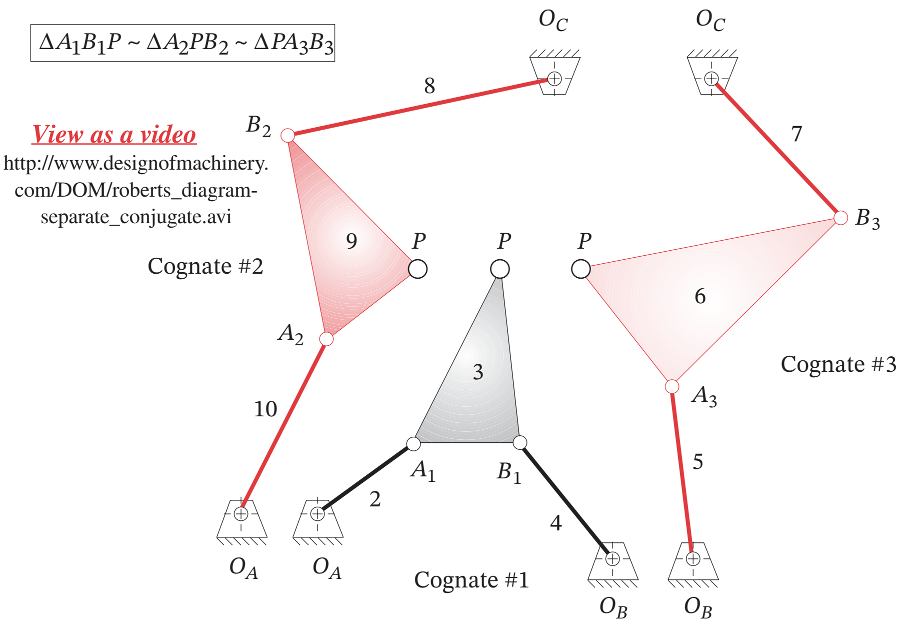
FIGURE 3-24

Cayley diagram to find cognates of a fourbar linkage

3



(*a*) Return links 2 and 4 to their fixed pivots O_A and O_B. Point O_C will assume its proper location.



(*b*) Separate the three cognates. Point *P* has the same path motion in each cognate.

FIGURE 3-25

Roberts diagram of three fourbar cognates

Nolle[12] reports on work by Luck[13] (in German) that defines the character of all fourbar cognates and their transmission angles. If the original linkage is a Grashof crank-rocker, then one cognate will be also, and the other will be a Grashof double-rocker. The minimum transmission angle of the crank-rocker cognate will be the same as that of the original crank-rocker. If the original linkage is a Grashof double-crank (drag link), then both cognates will be also and their minimum transmission angles will be the same in pairs that are driven from the same fixed pivot. If the original linkage is a non-Grashof triple-rocker, then both cognates are also triple-rockers.

These findings indicate that cognates of Grashof linkages do not offer improved transmission angles over the original linkage. Their main advantages are the different fixed pivot location and different velocities and accelerations of other points in the linkage. While the coupler path is the same for all cognates, its velocities and accelerations will not generally be the same since each cognate's overall geometry is different.

When the coupler point lies on the line of centers of link 3, the Cayley diagram degenerates to a group of colinear lines. A different approach is needed to determine the geometry of the cognates. Hartenberg and Denavit[11] give the following set of steps to find the cognates in this case. The notation refers to Figure 3-26.

1 Fixed pivot O_C lies on the line of centers O_AO_B extended and divides it in the same ratio as point P divides AB (i.e., $O_C / O_A = PA / AB$).

2 Line O_AA_2 is parallel to A_1P and A_2P is parallel to O_AA_1, locating A_2.

3 Line O_BA_3 is parallel to B_1P and A_3P is parallel to O_BB_1, locating A_3.

4 Joint B_2 divides line A_2P in the same ratio as point P divides AB. This defines the first cognate $O_AA_2B_2O_C$.

5 Joint B_3 divides line A_3P in the same ratio as point P divides AB. This defines the second cognate $O_BA_3B_3O_C$.

The three linkages can then be separated and each will independently generate the same coupler curve. The example chosen for Figure 3-26 is unusual in that the two cognates of the original linkage are identical, mirror-image twins. These are special linkages and will be discussed further in the next section.

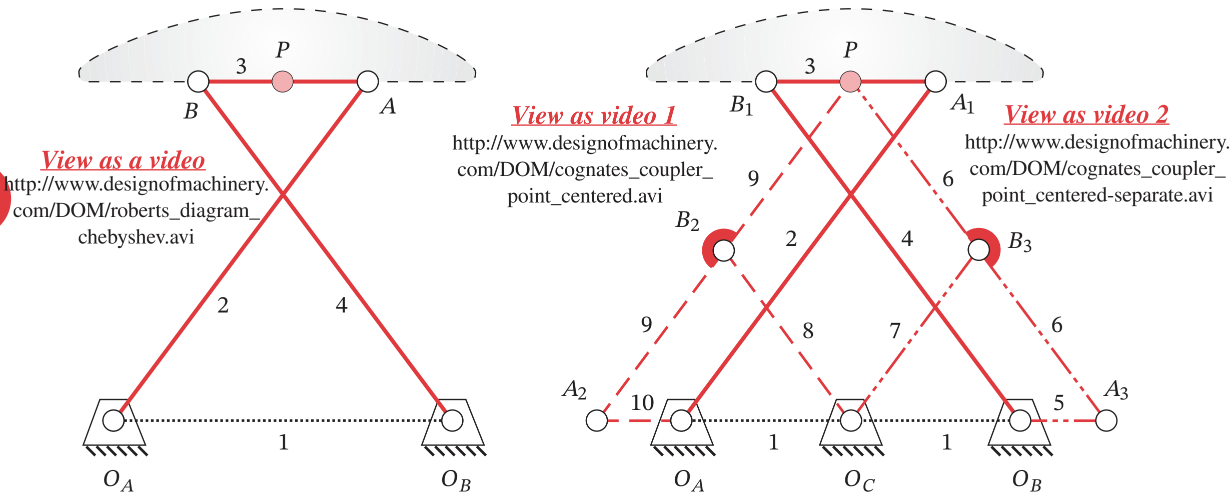
Program LINKAGES will automatically calculate the two cognates for any linkage configuration input to it. The velocities and accelerations of each cognate can then be calculated and compared. The program also draws the Cayley diagram for the set of cognates. Input the file F03-24.4br to program LINKAGES to display the Cayley diagram of Figure 3-24. Input the files COGNATE1.4br, COGNATE2.4br, and COGNATE3.4br to animate and view the motion of each cognate shown in Figure 3-25. Their coupler curves (at least those portions that each cognate can reach) will be seen to be identical.

Parallel Motion *View the lecture video* (21:50)*

It is quite common to want the output link of a mechanism to follow a particular path without any rotation of the link as it moves along the path. Once an appropriate path motion in the form of a coupler curve and its fourbar linkage have been found, a cognate of that linkage provides a convenient means to replicate the coupler path motion and provide curvilinear translation (i.e., no rotation) of a new output link that follows the coupler path.

* http://www.designofmachinery.com/DOM/Parallel_Motion.mp4

3



(*a*) A fourbar linkage and its coupler curve

(*b*) Cognates of the fourbar linkage

FIGURE 3-26

Finding cognates of a fourbar linkage when its coupler point lies on the line of centers of the coupler

This is referred to as **parallel motion.** Its design is best described with an example, the result of which will be a Watt I sixbar linkage† that incorporates the original fourbar and parts of one of its cognates. The method shown is as described in Soni.[14]

†Another common method used to obtain parallel motion is to duplicate the same linkage (i.e., the identical cognate), connect them with a parallelogram loop, and remove two redundant links. This results in an eight-link mechanism. See Figure P3-7 for an example of such a mechanism. The method shown here using a different cognate results in a simpler linkage, but either approach will accomplish the desired goal.

EXAMPLE 3-11

Parallel Motion from a Fourbar Linkage Coupler Curve.

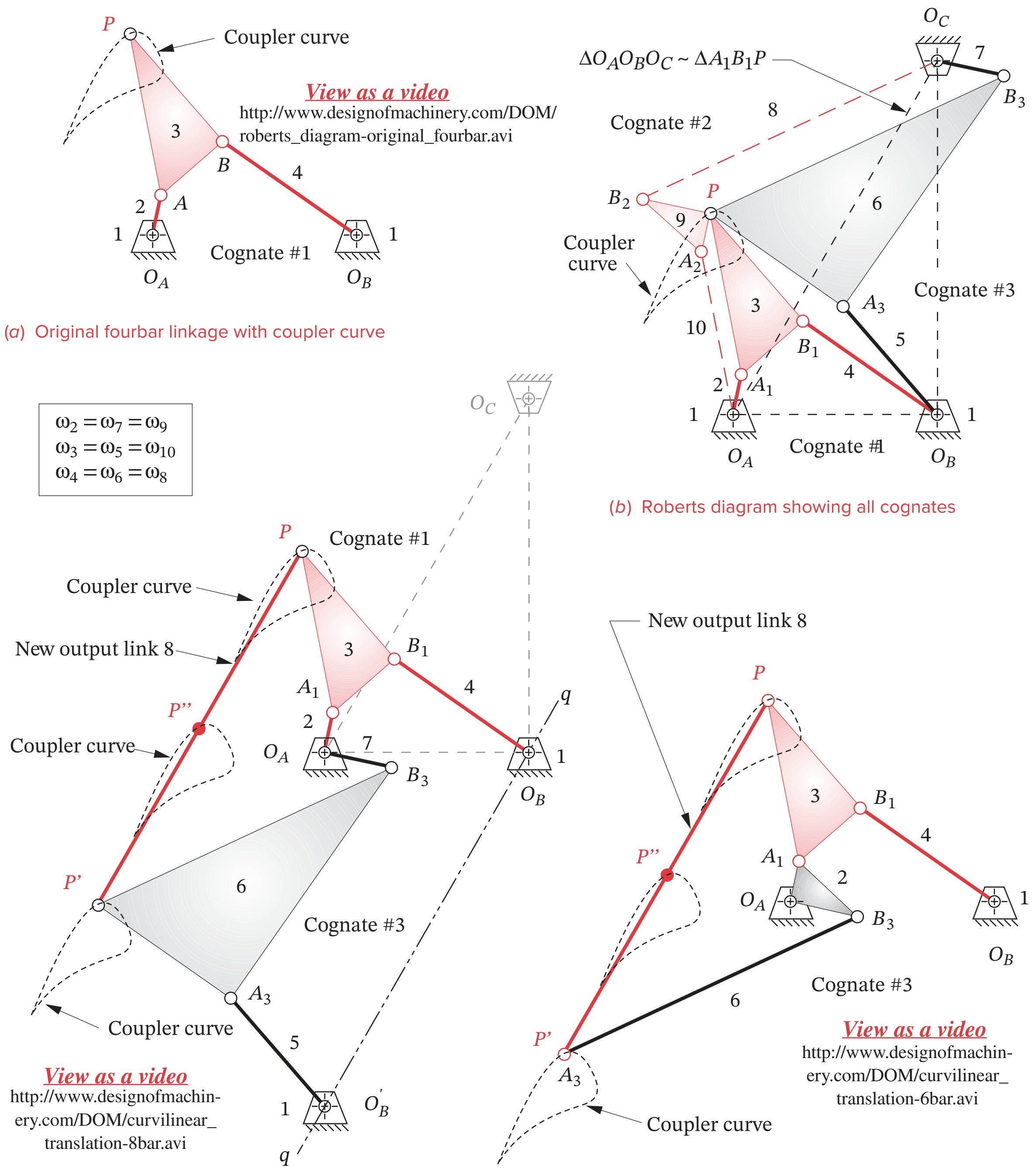
Problem: Design a sixbar linkage for parallel motion over a fourbar linkage coupler path.

Solution: (See Figure 3-27.)

1 Figure 3-27a shows the chosen Grashof crank-rocker fourbar linkage and its coupler curve. The first step is to create the Roberts diagram and find its cognates as shown in Figure 3-27b. The Roberts linkage can be found directly, without resort to the Cayley diagram, as described above. The fixed center O_C is found by drawing a triangle similar to the coupler triangle A_1B_1P with base O_AO_B.

2 One of a crank-rocker linkage's cognates will also be a crank-rocker (here cognate #3) and the other is a Grashof double-rocker (here cognate #2). Discard the double-rocker, keeping the links numbered 2, 3, 4, 5, 6, and 7 in Figure 3-27b. Note that links 2 and 7 are the two cranks, and both have the same angular velocity. The strategy is to coalesce these two cranks on a common center (O_A) and then combine them into a single link.

3 Draw the line qq parallel to line O_AO_C and through point O_B as shown in Figure 3-27c.



(a) Original fourbar linkage with coupler curve

(b) Roberts diagram showing all cognates

(c) Cognate #3 shifted with O_C moving to O_A

(d) Redundant link 5 omitted and links 2 & 7 combined leaving a Watt I sixbar

FIGURE 3-27

Method to construct a Watt-I sixbar that replicates a coupler path with curvilinear translation (parallel motion)

3

4 Without allowing links 5, 6, and 7 to rotate, slide them as an assembly along lines O_AO_C and qq until the free end of link 7 is at point O_A. The free end of link 5 will then be at point O_B' and point P on link 6 will be at P'.

5 Add a new link of length O_AO_C between P and P'. This is the *new output link* 8, and all points on it describe the original coupler curve as depicted at points P, P', and P'' in Figure 3-27c.

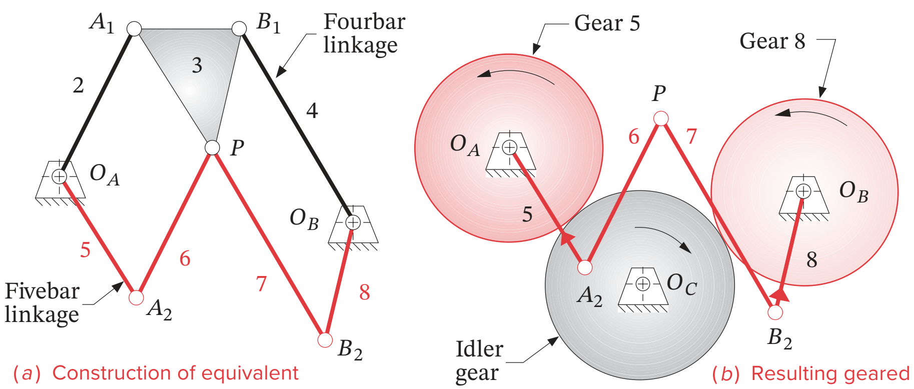
6 The mechanism in Figure 3-27c has 8 links, 10 revolute joints, and one *DOF*. When driven by **either** crank 2 or 7, all points on link 8 will duplicate the coupler curve of point P.

7 This is an *overclosed linkage* with redundant links. Because links 2 and 7 have the same angular velocity, they can be joined into one link as shown in Figure 3-27d. Then link 5 can be removed and link 6 reduced to a binary link supported and constrained as part of the loop 2, 6, 8, 3. The resulting mechanism is a Watt-I sixbar (see Figure 2-16.) with the links numbered 1, 2, 3, 4, 6, and 8. Link 8 is in *curvilinear translation* and follows the coupler path of the original point P.*

* Another example of a parallel motion sixbar linkage is the Chebyschev straight-line linkage of Figure P2-5a. It is a combination of two of the cognates shown in Figure 3-26, assembled by the method described in Example 3-11 and shown in Figure 3-27.

Geared Fivebar Cognates of the Fourbar

Chebyschev also discovered that any fourbar coupler curve can be duplicated with a **geared fivebar mechanism whose gear ratio is plus one,** meaning that the gears turn with the same speed and direction. The geared fivebar's link lengths will be different from those of the fourbar but can be determined directly from the fourbar. Figure 3-28a shows the construction method, as described by Hall[15], to obtain the geared fivebar which will give the same coupler curve as a fourbar. The original fourbar is $O_AA_1B_1O_B$ (links 1, 2, 3, 4). The fivebar is $O_AA_2PB_2O_B$ (links 1, 5, 6, 7, 8). The two linkages share only the coupler point P and fixed pivots O_A and O_B. The fivebar is constructed by simply drawing link 6 parallel to link 2, link 7 parallel to link 4, link 5 parallel to A_1P, and link 8 parallel to B_1P.



(*a*) Construction of equivalent fivebar linkage

View as a video

http://www.designofmachinery.com/DOM/geared_5bar.avi

(*b*) Resulting geared fivebar linkage

View as a video

http://www.designofmachinery.com/DOM/geared_5bar-separate.avi

FIGURE 3-28

A geared fivebar linkage cognate of a fourbar linkage

A three-gear set is needed to couple links 5 and 8 with a ratio of plus one (gear 5 and gear 8 have the same diameter and have the same direction of rotation, due to the idler gear), as shown in Figure 3-28b. Link 5 is attached to gear 5, as is link 8 to gear 8. This construction technique may be applied to each of the three fourbar cognates, yielding three geared fivebars (which may or may not be Grashof). The three fivebar cognates can actually be seen in the Roberts diagram. Note that in the example shown, a non-Grashof triple-rocker fourbar yields a Grashof fivebar, which can be motor driven. This conversion to a GFBM linkage could be an advantage when the "right" coupler curve has been found on a non-Grashof fourbar linkage, but continuous output through the fourbar's toggle positions is needed. Thus we can see that there are at least seven linkages which will generate the same coupler curve, three fourbars, three GFBMs and one or more sixbars.

Program LINKAGES calculates the equivalent geared fivebar configuration for any fourbar linkage and displays the result. The file F03-28a.4br can be opened in LINKAGES to animate the linkage shown in Figure 3-28a. Then also open the file F03-28b.5br in program LINKAGES to see the motion of the equivalent geared fivebar linkage. Note that the original fourbar linkage is a triple-rocker, so it cannot reach all portions of the coupler curve when driven from one rocker. But its geared fivebar equivalent linkage can make a full revolution and traverses the entire coupler path. Program LINKAGES will create the equivalent GFBM of any fourbar linkage.

3.8 STRAIGHT-LINE MECHANISMS *View the lecture video (9:21)*[§]

A very common application of coupler curves is the generation of approximate straight lines. Straight-line linkages have been known and used since the time of James Watt in the 18th century. Many kinematicians, such as Watt, Chebyschev, Peaucellier, Kempe, Evans, and Hoeken (as well as others) over a century ago, developed or discovered either approximate or exact straight-line linkages, and their names are associated with those devices to this day. Figure 3-29 shows a collection of the better-known ones, most of which are also provided as animated files.

The first recorded application of a coupler curve to a motion problem is that of **Watt's straight-line linkage,** patented in 1784, and shown in Figure 3-29a. Watt devised several straight-line linkages to guide the long-stroke piston of his steam engine at a time when metal-cutting machinery that could create a long, straight guideway did not yet exist.[*] Figure 3-29b shows the Watt linkage used to guide the steam engine piston.[†] This triple-rocker linkage is still used in automobile suspension systems to guide the rear axle up and down in a straight line as well as in many other applications.

Richard Roberts (1789-1864) (not to be confused with Samuel Roberts of the cognates) discovered the **Roberts straight-line linkage** shown in Figure 3-29c. This is a triple-rocker. Other values for AP and BP are possible, but the ones shown give the most accurate straight line with a deviation from straight of only 0.04% (0.0004 dec%) of the length of link 2 over the range of $49° < \theta_2 < 69°$.

Chebyschev (1821-1894) also devised many **straight-line linkages**. His well-known Grashof double-rocker is shown in Figure 3-29d.[**]

§ http://www.designofmachinery.com/DOM/Straight_Line_Linkages.mp4

* In Watt's time, straight-line motion was dubbed "parallel motion" though we use that term somewhat differently now. James Watt is reported to have told his son, "*Though I am not over anxious after fame, yet I am more proud of the parallel motion than of any other mechanical invention I have made.*" Quoted in Muirhead, J. P. (1854). *The Origin and Progress of the Mechanical Inventions of James Watt*, Vol. 3, London, p. 89.

† Note also in Figure 3-29b (and in Figure P2-10) that the driven dyad (links 7 and 8 in Figure 3-29b or 3 and 4 in Figure P2-10) are a complicated arrangement of sun and planet gears with the planet axle in a circular track. These have the same effect as the simpler crank and connecting rod. Watt was forced to invent the sun and planet drive to get around James Pickard's 1780 patent on the crankshaft and connecting rod.

** View the video http://www.designofmachinery.com/DOM/Boot_Tester.mp4 to see an example of an application of the Chebyschev linkage.

3

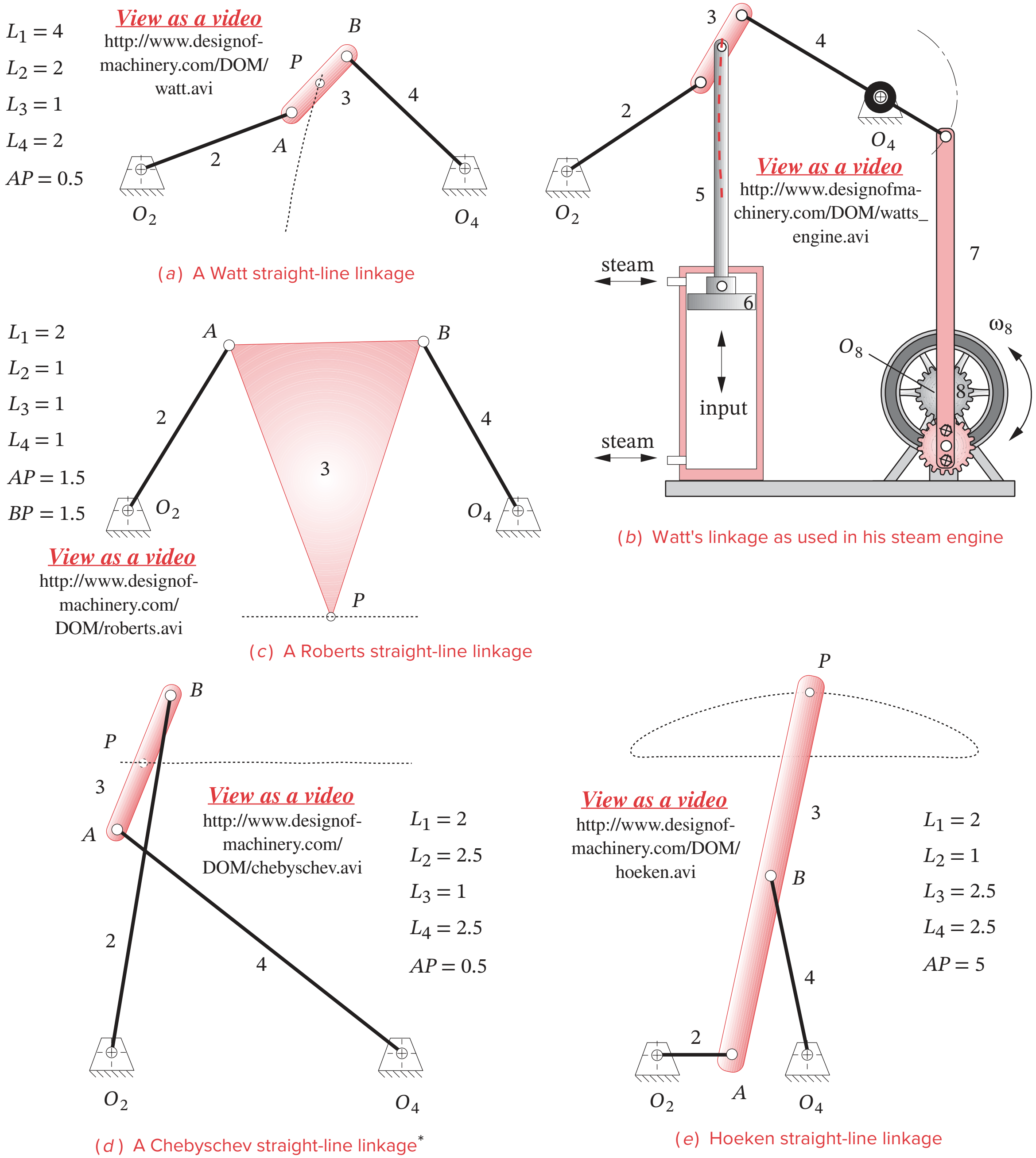
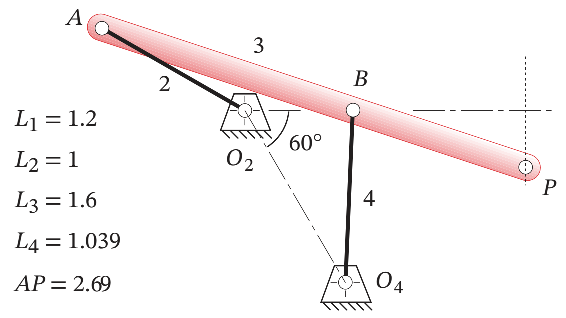


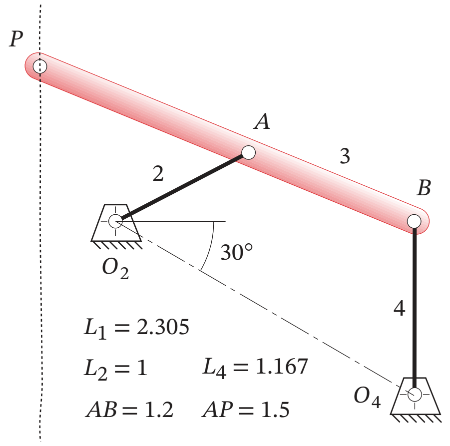
FIGURE 3-29 Part 1

Some common and classic approximate straight-line linkages

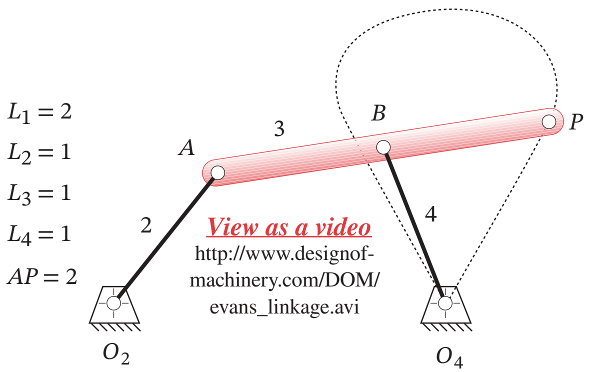
* The link ratios of the Chebyschev straight-line linkage shown have been reported differently by various authors. The ratios used here are those first reported (in English) by Kempe (1877). But Kennedy (1893) describes the same linkage, reportedly "as Chebyschev demonstrated it at the Vienna Exhibition of 1893" as having the link ratios 1, 3.25, 2.5, 3.25. We will assume the earliest reference by Kempe to be correct



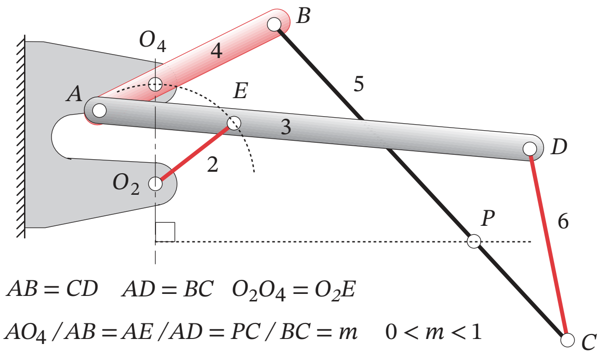
(*f*) Evans approx. straight-line linkage #1



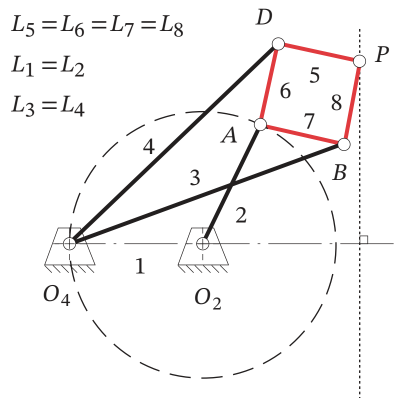
(*g*) Evans approx. straight-line linkage #2



(*h*) Evans approx. straight-line linkage #3



(*i*) Hart inversor exact straight-line linkage



(*j*) Peaucellier exact straight-line linkage

View as a video

http://www.designofmachinery.com/DOM/peaucellier.avi

FIGURE 3-29 Part 2

Approximate and exact straight-line linkages

The **Hoeken linkage**[16] in Figure 3-29e is a Grashof crank-rocker, which has a significant practical advantage. In addition, the Hoeken linkage has the feature of very *nearly constant velocity along the center portion of its straight-line motion*. It is interesting to note

3

that the **Hoeken** and **Chebyschev** linkages are cognates of one another.* The cognates shown in Figure 3-26 are the Chebyschev and Hoeken linkages.

Figure 3-29f shows one of **Evans'** many straight-line linkages. It is a triple rocker with a range of input link motion of about 27 to 333° between toggle positions. The portion of coupler curve shown is between 150° and 210° and has a very accurate straight line with a deviation of only 0.25% (0.0025 dec%) of the crank length.

Figure 3-29g shows a second **Evans straight-line linkag**e, also a triple rocker with a range of input link motion of about –81 to +81° between toggle positions. The portion of coupler curve shown is between –40 and 40° and has a long but less accurate straight line with a deviation of 1.5% (0.015 dec%) of the crank length.

Figure 3-29h shows a third **Evans straight-line linkag**e. It is a triple rocker with a range of input link motion of about –75 to +75° between toggle positions. The portion of coupler curve shown is all that is reachable between those limits and has two straight portions. The remainder of the coupler curve is a mirror image making a figure eight.

Some of these straight-line linkages are provided as built-in examples in program LINKAGES. AVI and Working Model files of many of them are also. Artobolevsky[20] shows seven Watt, seven Chebyschev, five Roberts, and sixteen Evans straight-line linkages in his Vol. I that include the ones shown here. A quick look in the downloadable Hrones and Nelson atlas of coupler curves will reveal a large number of coupler curves with **approximate straight-line** segments. They are quite common.

To generate an **exact straight line** with only pin joints requires more than four links. At least six links and seven pin joints are needed to generate an exact straight line with a pure revolute-jointed linkage, i.e., a Watt or Stephenson sixbar. Figure 3-29i shows the **Hart inversor exact straight-line sixbar mechanism**. A symmetrical geared fivebar mechanism (Figure 2-21), with a gear ratio of –1 and a phase angle of π radians, will also generate an exact straight line at the joint between links 3 and 4. But this linkage is merely a transformed Watt sixbar obtained by replacing one binary link with a higher joint in the form of a gear pair. This geared fivebar's straight-line motion can be seen by opening the file STRAIGHT.5BR in program LINKAGES, and animating the linkage.

Peaucellier† (1864) discovered an **exact straight-line** mechanism of eight bars and six pins, shown in Figure 3-29j.‡ Links 5, 6, 7, 8 form a rhombus of convenient size. Links 3 and 4 can be any convenient but equal lengths. When O_2O_4 exactly equals O_2A, point C generates an *arc of infinite radius*, i.e., **an exact straight line**. By moving the pivot O_2 left or right from the position shown, changing only the length of link 1, this mechanism *will generate true circle arcs with radii much larger than the link lengths*. Other exact straight-line linkages exist as well. See Artobolevsky.[20]

Designing Optimum Straight-Line Fourbar Linkages

Given the fact that an exact straight line can be generated with six or more links using only revolute joints, why use a fourbar approximate straight-line linkage at all? One reason is the desire for simplicity in machine design. The pin-jointed fourbar is the simplest possible 1-*DOF* mechanism. Another reason is that a very good approximation to a true straight line can be obtained with just four links, and this is often "good enough" for the needs of the machine being designed. Manufacturing tolerances will, after all, cause

* Hain[17] (1967) cites the Hoeken reference[16] (1926) for this linkage. Nolle [18] (1974) shows the Hoeken mechanism but refers to it as a Chebyschev crank-rocker without noting its cognate relationship to the Chebyschev double-rocker, which he also shows. It is certainly conceivable that Chebyschev, as one of the creators of the theorem of cognate linkages, would have discovered the "Hoeken" cognate of his own double-rocker. However, this author has been unable to find any mention of its genesis in the English literature other than the ones cited here.

† Peaucellier was a French army captain and military engineer who first proposed his "compas compose" or *compound compass* in 1864 but received no immediate recognition therefor. (He later received the "Prix Montyon" from the Institute of France.) The British-American mathematician James Sylvester reported on it to the *Atheneum Club* in London in 1874. He observed that "*The perfect parallel motion of Peaucellier looks so simple and moves so easily that people who see it at work almost universally express astonishment that it waited so long to be discovered*." A model of the Peaucellier linkage was passed around the table. The famous physicist Sir William Thomson (later Lord Kelvin) refused to relinquish it, declaring, "*No. I have not had nearly enough of it—it is the most beautiful*

(footnotes cont'd. opp. page)

any mechanism's performance to be less than ideal. As the number of links and joints increases, the probability that an exact straight-line mechanism will deliver its theoretical performance in practice is obviously reduced.

There is a real need for straight-line motions in machinery of all kinds, especially in automated production machinery. Many consumer products such as cameras, film, toiletries, razors, and bottles are manufactured, decorated, or assembled on sophisticated and complicated machines that contain a myriad of linkages and cam-follower systems. Traditionally, most of this kind of production equipment has been of the intermittent-motion variety. This means that the product is carried through the machine on a linear or rotary conveyor that stops for any operation to be done on the product, and then indexes the product to the next workstation where it again stops for another operation to be performed. The forces, torque, and power required to accelerate and decelerate the large mass of the conveyor (which is independent of, and typically larger than, the mass of the product) severely limit the speeds at which these machines can be run.

Economic considerations continually demand higher production rates, requiring higher speeds or additional, expensive machines. This economic pressure has caused many manufacturers to redesign their assembly equipment for continuous conveyor motion. When the product is in continuous motion in a straight line and at constant velocity, every workhead that operates on the product must be articulated to chase the product and match both its path and its constant velocity while performing the task. These factors have increased the need for straight-line mechanisms, including ones capable of near-constant velocity over the straight-line path.

A (near) perfect straight-line motion is easily obtained with a fourbar crank-slider mechanism. Ball-bushings (Figure 2-33) and ball-slides (Figure 2-38) are available commercially at moderate cost and make this a reasonable, low-friction solution to the straight-line path guidance problem. But, the cost and lubrication problems of a properly guided crank-slider mechanism are still greater than those of a pin-jointed fourbar linkage. Moreover, a crank-slider block has a velocity profile that is nearly sinusoidal (with some harmonic content) and is far from having constant velocity over any part of its motion. (See Section 3.10 for a modified crank-slider mechanism that has nearly constant slider velocity for part of its stroke.)

The Hoeken-type linkage offers an optimum combination of straightness and near constant velocity and is a crank-rocker, so it can be motor driven. Its geometry, dimensions, and coupler path are shown in Figure 3-30. This is a symmetrical fourbar linkage. Since the angle γ of line BP is specified and $L_3 = L_4 = BP$, only two link ratios are needed to define its geometry, say L_1 / L_2 and L_3 / L_2. If the crank L_2 is driven at constant angular velocity ω_2, the linear velocity V_x along the straight-line portion Δx of the coupler path will be very close to constant over a significant portion of crank rotation $\Delta\beta$.

A study was done to determine the errors in straightness and constant velocity of the Hoeken-type linkage over various fractions $\Delta\beta$ of the crank cycle as a function of the link ratios.[19] The structural error in position (i.e., straightness) ε_S and the structural error in velocity ε_V are defined using notation from Figure 3-30 as:

(continued from opp. page) *thing I have ever seen in my life*." Source: Strandh, S. (1979). *A History of the Machine.* A&W Publishers: New York, p. 67.

‡ This Peaucillier linkage figure is provided as animated AVI and Working Model files. Its filename is the same as the figure number.

View as a video
http://www.designofmachinery.com/DOM/hoeken.avi

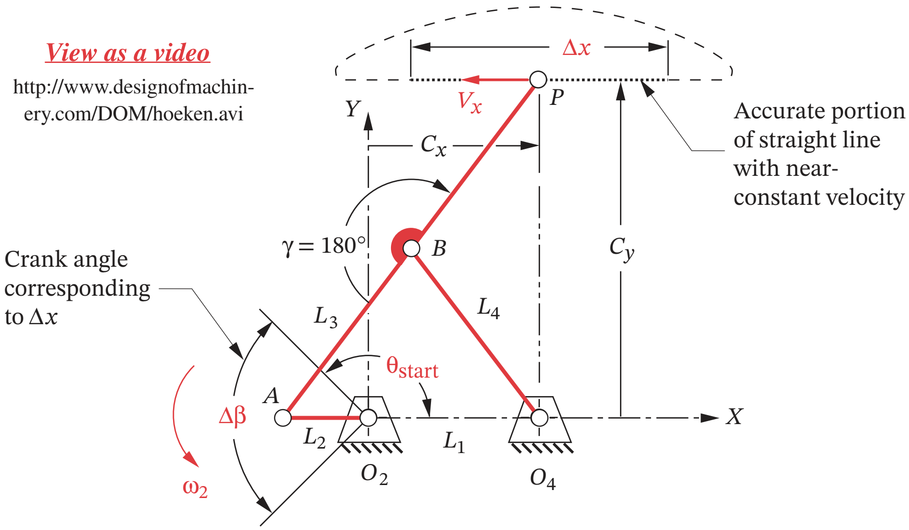


FIGURE 3-30

Hoekens linkage geometry: linkage shown with P at center of straight-line portion of path

$$\varepsilon_S = \frac{MAX_{i=1}^{n}\left(C_{y_i}\right) - MIN_{i=1}^{n}\left(C_{y_i}\right)}{\Delta x}$$

$$\varepsilon_V = \frac{MAX_{i=1}^{n}\left(V_{x_i}\right) - MIN_{i=1}^{n}\left(V_{x_i}\right)}{\bar{V}_x} \tag{3.5}$$

* See reference [19] for the derivation of equations 3.5.

TABLE 3-1 Link Ratios for Smallest Attainable Errors in Straightness and Velocity for Various Crank-Angle Ranges of a Hoeken-Type Fourbar Approximate Straight-Line Linkage [19]

Range of Motion			Optimized for Straightness						Optimized for Constant Velocity					
$\Delta\beta$ (deg)	θ (deg)	% of cycle	Maximum ΔC_y %	ΔV %	V_x / $(L_2\,\omega_2)$	Link Ratios L_1/L_2	L_3/L_2	$\Delta x/L_2$	Maximu m ΔV_x %	ΔC_y %	V_x / $(L_2\,\omega_2)$	Link Ratios L_1/L_2	L_3/L_2	$\Delta x/L_2$
20	170	5.6%	0.00001%	0.38%	1.725	2.975	3.963	0.601	0.006%	0.137%	1.374	2.075	2.613	0.480
40	160	11.1%	0.00004%	1.53%	1.717	2.950	3.925	1.193	0.038%	0.274%	1.361	2.050	2.575	0.950
60	150	16.7%	0.00027%	3.48%	1.702	2.900	3.850	1.763	0.106%	0.387%	1.347	2.025	2.538	1.411
80	140	22.2%	0.001%	6.27%	1.679	2.825	3.738	2.299	0.340%	0.503%	1.319	1.975	2.463	1.845
100	130	27.8%	0.004%	9.90%	1.646	2.725	3.588	2.790	0.910%	0.640%	1.275	1.900	2.350	2.237
120	120	33.3%	0.010%	14.68%	1.611	2.625	3.438	3.238	1.885%	0.752%	1.229	1.825	2.238	2.600
140	110	38.9%	0.023%	20.48%	1.565	2.500	3.250	3.623	3.327%	0.888%	1.178	1.750	2.125	2.932
160	100	44.4%	0.047%	27.15%	1.504	2.350	3.025	3.933	5.878%	1.067%	1.124	1.675	2.013	3.232
180	90	50.0%	0.096%	35.31%	1.436	2.200	2.800	4.181	9.299%	1.446%	1.045	1.575	1.863	3.456

The structural errors were computed separately for each of nine crank-angle ranges $\Delta\beta$ from 20° to 180°. Table 3-1 shows the link ratios that give the smallest possible structural error in either position or velocity over values of $\Delta\beta$ from 20° to 180°. Note that one cannot attain optimum straightness and minimum velocity error in the same linkage. However, reasonable compromises between the two criteria can be achieved, especially for smaller ranges of crank angle. The errors in both straightness and velocity increase as longer portions of the curve are used (larger $\Delta\beta$). The use of Table 3-1 to design a straight-line linkage will be shown with an example.

3

EXAMPLE 3-12

Designing a Hoeken-Type Straight-Line Linkage

Problem: A 100-mm-long straight-line motion is needed over 1/3 of the total cycle (120° of crank rotation). Determine the dimensions of a Hoeken-type linkage that will

(a) Provide minimum deviation from a straight line. Determine its maximum deviation from constant velocity.
(b) Provide minimum deviation from constant velocity. Determine its maximum deviation from a straight line.

Solution: (See Figure 3-30 and Table 3-1.)

1 Part (a) requires the most accurate straight line. Enter Table 3-1 at the 6th row which is for a crank-angle duration $\Delta\beta$ of the required 120°. The 4th column shows the minimum possible deviation from straight to be 0.01% of the length of the straight-line portion used. For a 100-mm length the absolute deviation will then be 0.01 mm (0.0004 in). The 5th column shows that its velocity error will be 14.68% of the average velocity over the 100-mm length. The absolute value of this velocity error of course depends on the speed of the crank.

2 The linkage dimensions for part (a) are found from the ratios in columns 7, 8, and 9. The crank length required to obtain the 100-mm length of straight line Δx is:

from Table 3-1: $$\frac{\Delta x}{L_2} = 3.238$$

$$L_2 = \frac{\Delta x}{3.238} = \frac{100 \text{ mm}}{3.23} = 30.88 \text{ mm} \tag{a}$$

The other link lengths are then:

from Table 3-1: $$\frac{L_1}{L_2} = 2.625$$

$$L_1 = 2.625 L_2 = 2.625(30.88 \text{ mm}) = 81.07 \text{ mm} \tag{b}$$

from Table 3-1: $$\frac{L_3}{L_2} = 3.438$$

$$L_3 = 3.438 L_2 = 3.438(30.88 \text{ mm}) = 106.18 \text{ mm} \tag{c}$$

The complete linkage is then $L_1 = 81.07$, $L_2 = 30.88$, $L_3 = L_4 = BP = 106.18$ mm. The nominal velocity V_x of the coupler point at the center of the straight line ($\theta_2 = 180°$) can be found from the factor in the 6th column which must be multiplied by the crank length L_2 and the crank angular velocity ω_2 in radians per second (rad/sec).

3 Part (b) requires the most accurate velocity. Again enter Table 3-1 at the 6th row which is for a crank angle duration $\Delta\beta$ of the required 120°. The 10th column shows the minimum possible deviation from constant velocity to be 1.885% of the average velocity V_x over the length of the straight-line portion used. The 11th column shows the deviation from straight to be 0.752% of the length of the straight-line portion used. For a 100-mm length the absolute deviation in straightness for this optimum constant velocity linkage will then be 0.75 mm (0.030 in).

4 Link lengths for this mechanism are found in the same way as was done in step 2 except that the link ratios 1.825, 2.238, and 2.600 from columns 13, 14, and 15 are used. The result is $L_1 = 70.19$, $L_2 = 38.46$, $L_3 = L_4 = BP = 86.08$ mm. The nominal velocity V_x of the coupler point at the center of the straight line ($\theta_2 = 180°$) can be found from the factor in the 12th column which must be multiplied by the crank length L_2 and the crank angular velocity ω_2 in rad/sec.

5 The first solution (step 2) gives an extremely accurate straight line over a significant part of the cycle. but its 15% deviation in velocity would probably be unacceptable if that factor were considered important. The second solution (step 3) gives less than 2% deviation from constant velocity, which may be viable for a design application. Its 3/4% deviation from straightness, while much greater than the first design, may be acceptable in some situations.

3.9 DWELL MECHANISMS *View the lecture video* (35:36)*

* http://www.designofmachinery.com/DOM/Dwell_Mechanisms.mp4

A common requirement in machine design problems is the need for a dwell in the output motion. A **dwell** is defined as *zero output motion for some nonzero input motion*. In other words, the motor keeps going, but the output link stops moving. Many production machines perform a series of operations which involve feeding a part or tool into a workspace, and then holding it there (in a dwell) while some task is performed. Then the part must be removed from the workspace, and perhaps held in a second dwell while the rest of the machine "catches up" by indexing or performing some other tasks. Cams and followers (Chapter 8) are often used for these tasks because it is trivially easy to create a dwell with a cam. But there is always a trade-off in engineering design, and cams have their problems of high cost and wear as described in Section 2.18.

It is also possible to obtain dwells with "pure" linkages of only links and pin joints, which have the advantage over cams of low cost and high reliability. Dwell linkages are more difficult to design than are cams with dwells. Linkages will usually yield only an approximate dwell but will be much cheaper to make and maintain than cams. Thus they may be worth the effort.

Single-Dwell Linkages

There are two usual approaches to designing single-dwell linkages. Both result in **sixbar mechanisms**, and both require first finding a fourbar with a suitable coupler curve. A **dyad** is then added to provide an output link with the desired dwell characteristic. The first approach to be discussed requires the design or definition of a fourbar with a coupler curve that contains an approximate circle arc portion, where the "arc" occupies the desired portion of the input link (crank) cycle designated as the dwell. An atlas of coupler curves is invaluable for this part of the task. Symmetrical coupler curves are also well suited to this task, and the information in Figure 3-21 can be used to find them.

3

EXAMPLE 3-13

Single-Dwell Mechanism with Only Revolute Joints

Problem: Design a sixbar linkage for 90° rocker motion over 300 crank degrees with dwell for the remaining 60°.

Solution: (See Figure 3-31.)

1 Search the H&N atlas for a fourbar linkage with a coupler curve having an approximate (pseudo) circle arc portion which occupies 60° of crank motion (12 dashes). The chosen fourbar is shown in Figure 3-31a.

2 Lay out this linkage to scale including the coupler curve and find the approximate center of the chosen coupler curve pseudo-arc using graphical geometric techniques. To do so, draw the chord of the arc and construct its perpendicular bisector as shown in Figure 3-31b. The center will lie on this bisector. Find it by striking arcs with your compass point on the bisector while adjusting the radius to get the best fit to the coupler curve. Label the arc center D.

3 Your compass should now be set to the approximate radius of the coupler arc. This will be the length of link 5 which is to be attached at the coupler point P.

4 Trace the coupler curve with the compass point, while keeping the compass pencil lead on the perpendicular bisector, and find the extreme location along the bisector that the compass lead will reach. Label this point E.

5 The line segment DE represents the maximum displacement that a link of length PD, attached at P, will reach along the bisector.

6 Construct a perpendicular bisector of the line segment DE, and extend it in a convenient direction.

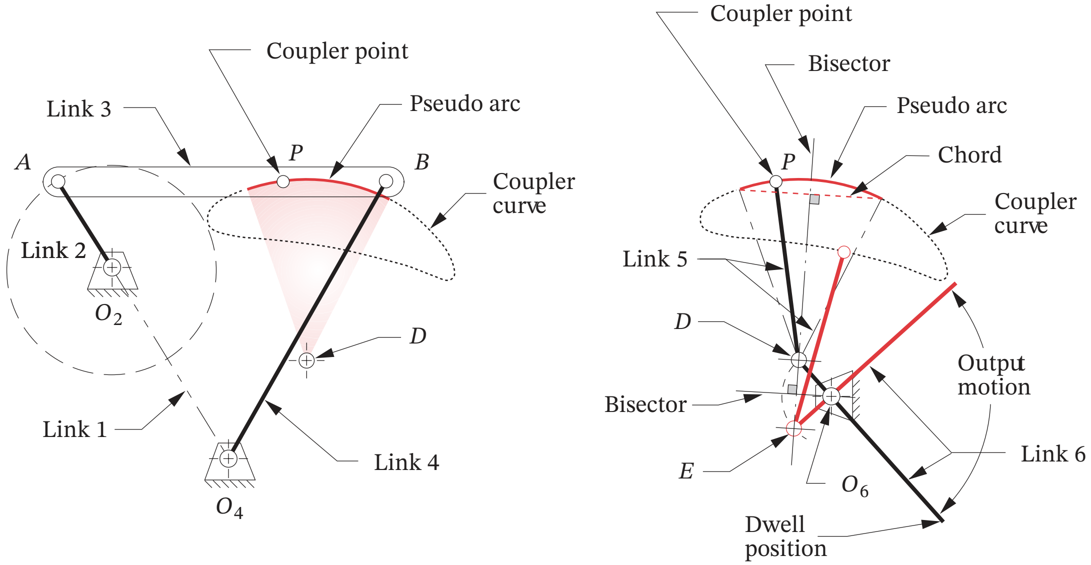
7 Locate fixed pivot O_6 on the bisector of DE such that lines O_6D and O_6E subtend the desired output angle, in this example, 90°.

8 Draw link 6 from D (or E) through O_6 and extend to any convenient length. This is the output link which will dwell for the specified portion of the crank cycle.

9 Check the transmission angles.

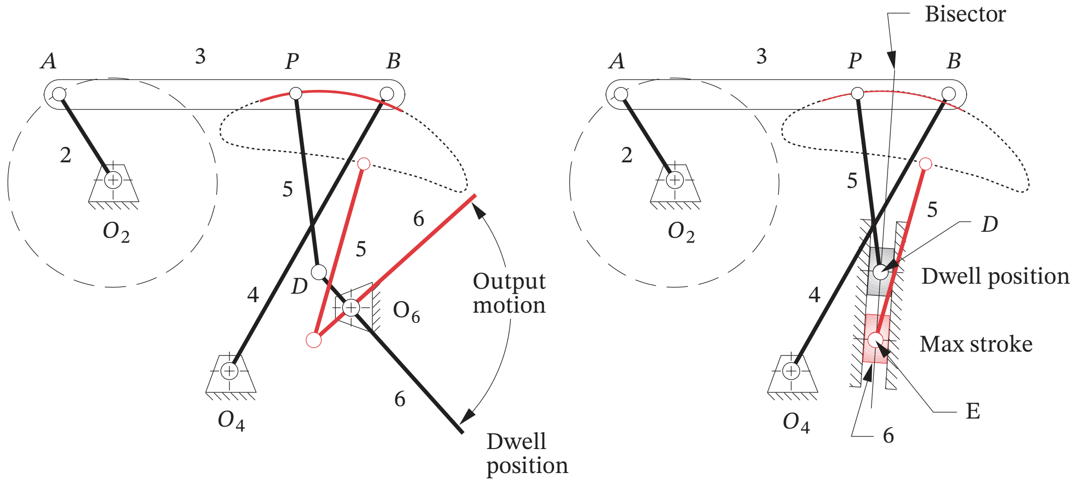
10 Make a model of the linkage and articulate it to check its function.

This linkage dwells because, during the time that the coupler point P is traversing the pseudo-arc portion of the coupler curve, the other end of link 5, attached to P and the same length as the arc radius, is essentially stationary at its other end, which is the arc center.



(*a*) Chosen fourbar crank-rocker with pseudo-arc section for 60° of link 2 rotation

(*b*) Construction of the output-dwell dyad



(*c*) Completed sixbar single-dwell linkage with rocker output option

(*d*) Completed sixbar single-dwell linkage with slider output option

View as a video
http://www.designof-machinery.com/DOM/dwell_revolute.avi

View as a video
http://www.designof-machinery.com/DOM/dwell_slider.avi

FIGURE 3-31

Design of a sixbar single-dwell mechanism with rocker output or slider output, using a pseudo-arc coupler curve

However the dwell at point *D* will have some "jitter" or oscillation, due to the fact that *D* is only an approximate center of the pseudo-arc on the sixth-degree coupler curve. When point *P* leaves the arc portion, it will smoothly drive link 5 from point *D* to point *E*, which will in turn rotate the output link 6 through its arc as shown in Figure 3-31c.* Note that we can have any angular displacement of link 6 we desire with the same links 2 to 5, as they alone completely define the dwell aspect. Moving pivot O_6 left and right along the bisector of line *DE* will change the angular displacement of link 6 but not its timing. In fact, a slider block could be substituted for link 6 as shown in Figure 3-31d,* and linear translation along line *DE* with the same timing and dwell at *D* will result. Input the file F03-31c.6br to program LINKAGES and animate to see the linkage of Example 3-13 in motion. The dwell in the motion of link 6 can be clearly seen in the animation, including the jitter due to its approximate nature.

* This figure is provided as animated AVI and Working Model files. Its filename is the same as the figure number.

Double-Dwell Linkages

It is also possible, using a fourbar coupler curve, to create a double-dwell output motion. One approach is the same as that used in the single-dwell of Example 3-13. Now a coupler curve is needed which has *two* approximate circle arcs of the same radius but with different centers, both convex or both concave. A link 5 of length equal to the radius of the two arcs will be added such that it and link 6 will remain nearly stationary at the center of each of the arcs, while the coupler point traverses the circular parts of its path. Significant motion of the output link 6 will occur only when the coupler point is between those arc portions. Higher-order linkages, such as the geared fivebar, can be used to create multiple-dwell outputs by a similar technique since they possess coupler curves with multiple, approximate circle arcs. See the built-in example double-dwell linkage in program LINKAGES for a demonstration of this approach.

A second approach uses a coupler curve with two approximate straight-line segments of appropriate duration. If a pivoted slider block (link 5) is attached to the coupler at this point, and link 6 is allowed to slide in link 5, it only remains to choose a pivot O_6 at the intersection of the straight-line segments extended. The result is shown in Figure 3-32. While block 5 is traversing the "straight-line" segments of the curve, it will not impart any angular motion to link 6. The approximate nature of the fourbar straight line causes some jitter in these dwells also.

EXAMPLE 3-14

Double-Dwell Mechanism.

Problem: Design a sixbar linkage for 80° rocker output motion over 20 crank degrees with dwell for 160°, return motion over 140° and second dwell for 40°.

Solution: (See Figure 3-32.)

1 Search the H&N atlas for a linkage with a coupler curve having two approximate straight-line portions. One should occupy 160° of crank motion (32 dashes), and the second 40° of crank motion (8 dashes). This is a wedge-shaped curve as shown in Figure 3-32a.

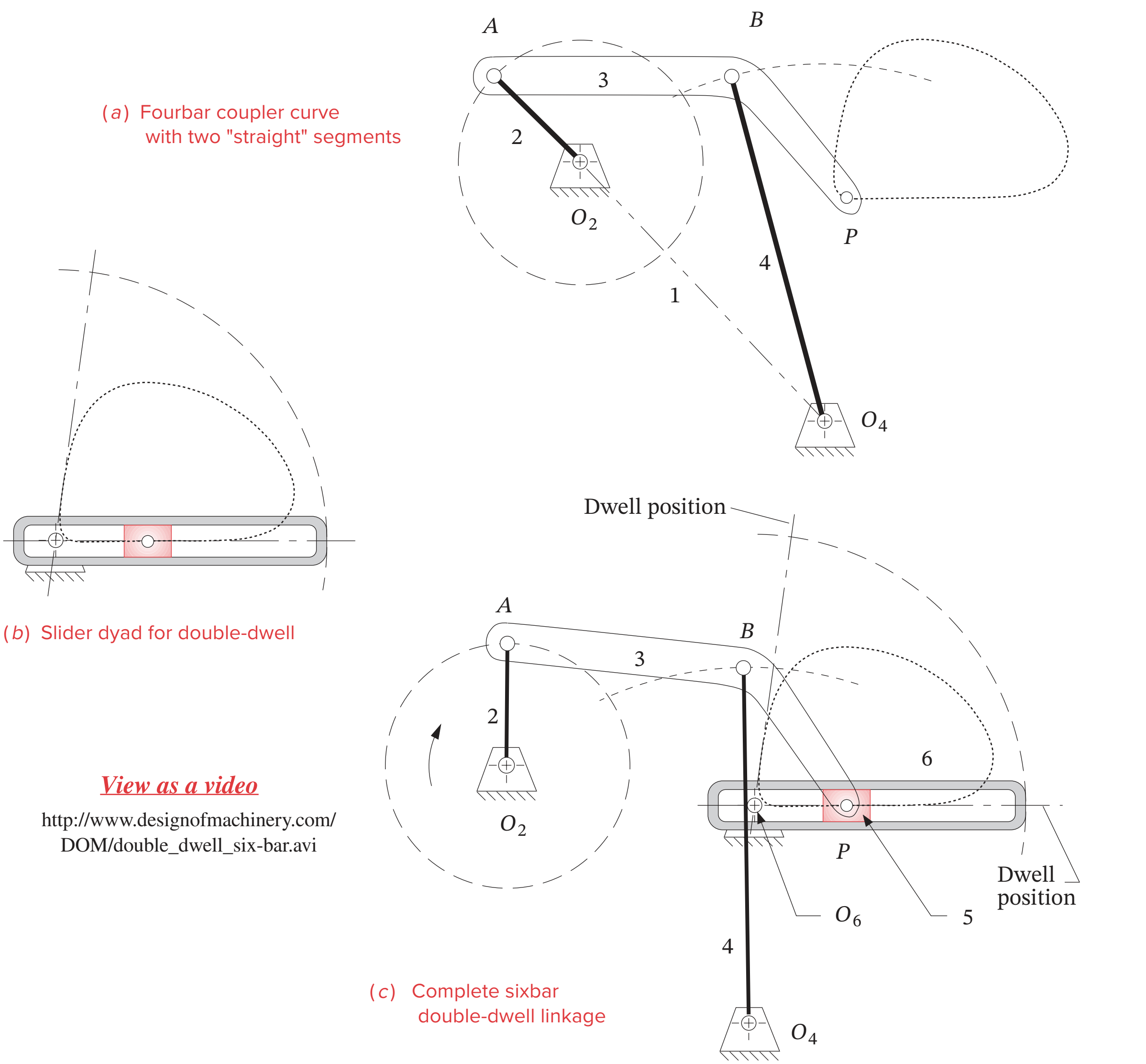
2 Lay out this linkage to scale including the coupler curve and find the intersection of two tangent lines colinear with the straight segments. Label this point O_6.

3 Design link 6 to lie along these straight tangents, pivoted at O_6. Provide a slot in link 6 to accommodate slider block 5 as shown in Figure 3-32b.

4 Connect slider block 5 to the coupler point P on link 3 with a pin joint. The finished sixbar is shown in Figure 3-32c.

5 Check the transmission angles.

It should be apparent that these linkage dwell mechanisms have some disadvantages. Besides being difficult to synthesize, they give only approximate dwells which have some



View as a video

http://www.designofmachinery.com/DOM/double_dwell_six-bar.avi

FIGURE 3-32

Double-dwell sixbar linkage

jitter on them. Also, they tend to be large for the output motions obtained, so do not package well. The acceleration of the output link can also be very high as in Figure 3-32, when block 5 is near pivot O_6. (Note the large angular displacement of link 6 resulting from a small motion of link 5.) Nevertheless they may be of value in situations where a completely stationary dwell is not required, and the low cost and high reliability of a linkage are important factors. Program LINKAGES has both single-dwell and double-dwell example linkages built in.

3.10 OTHER USEFUL LINKAGES

There are many practical machine design problems that can be solved with clever linkage design. One of the best references for these mechanisms is by Hain.[22] Another useful catalog of linkages is the four volumes of Artobolevsky.[20] We will present a few examples from these that we find useful. Some are fourbar linkages, others are Watt's or Stephenson's sixbars, or eightbar linkages. Artobolevsky provides link ratios, but Hain does not. Hain describes their graphical construction, so the dimensions of his linkages shown here are approximate, obtained by scaling his drawings.

Constant Velocity Piston Motion

The fourbar crank-slider linkage is probably the most frequently used linkage in machinery. Every internal combustion (IC) engine and reciprocating compressor has as many of them as it has cylinders. Manufacturing machinery uses them to obtain straight-line motions. In most cases this simple linkage is completely adequate for the application, converting continuous rotary input to oscillating straight-line output. One limitation is lack of control over the slider's velocity profile when the crank is driven with constant angular velocity. Altering the link ratios (crank vs. coupler) has a second-order effect on the shape of the slider's velocity and acceleration curves† but it will always be fundamentally a sinusoidal motion. In some cases, a constant or near constant velocity is needed on either the forward or backward stroke of the slider. An example is a piston pump for metering fluids whose flow rate needs to be constant during the delivery stroke. A direct solution is to use a cam to drive the piston with a constant velocity motion rather than use a crank-slider linkage. However, Hain[22] provides a pure linkage solution to this problem that adds a drag-link fourbar stage to the crank-slider with the drag-link geometry chosen to modulate the sinusoidal slider motion to be approximately constant velocity.

† This topic is addressed in depth in Chapter 13.

Figure 3-33 shows the result, which is effectively a Watt sixbar. Constant angular velocity is input to link 2 of the drag-link stage. It causes its "output" link 4 to have a non-constant angular velocity that repeats each cycle. This varying angular velocity becomes the "input" to the crank-slider stage 4-5-6 whose input link is now link 4. Thus, the drag link's velocity oscillation effectively "corrects" or modulates the slider velocity to be close to constant on the forward stroke as plotted in the figure. The deviation from constant velocity is $< 1\%$ over $240° < \theta_2 < 270°$ and $<= 4\%$ over $190° < \theta_2 < 316°$. Its velocity on the return stroke must therefore vary to a greater degree than in the unmodulated linkage. This is an example of the effect of cascading linkages. Each stage's output function becomes the input to the next, and the end result is their mathematical combination, analogous to adding terms to a series. See the *TKSolver* file Dragslider.tkw.

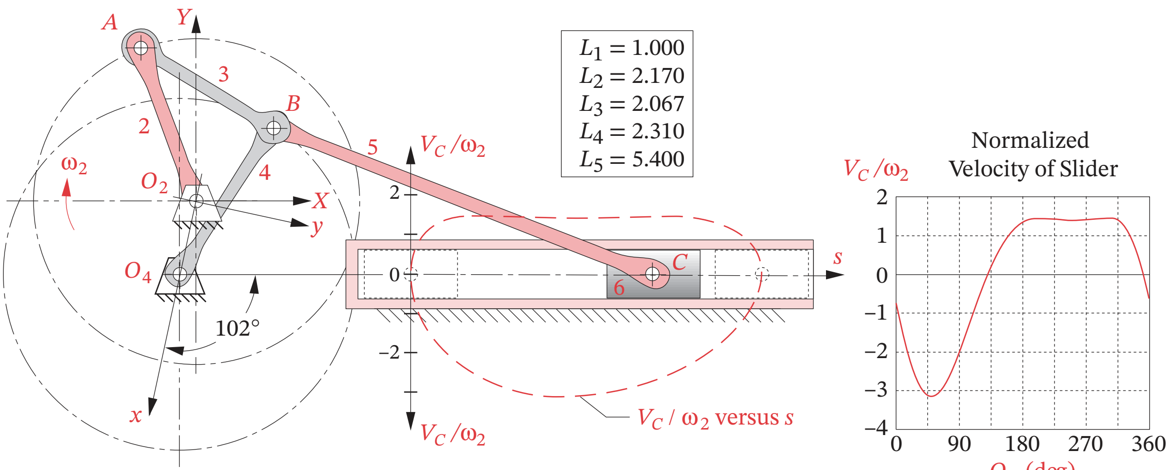


FIGURE 3-33

Approximate constant velocity, drag link driven slider-crank sixbar mechanism *Adapted from Hain* [22]

In addition to metering fluids, this linkage has application in situations where a part must be picked up from a nest on the stationary ground plane and transferred to a conveyor that is moving at constant velocity. The slider has points of zero velocity at each extreme of motion, exact straight line motion in both directions, and a long region of approximately constant velocity. Note however, that the Hoeken straight-line linkage of Section 3.8 gives a nearly exact straight line with close to constant velocity using only four links and four pin joints rather than the six links and slider track needed here. Hoeken's linkage is also useful for the pick-and-place-at-constant-velocity application.

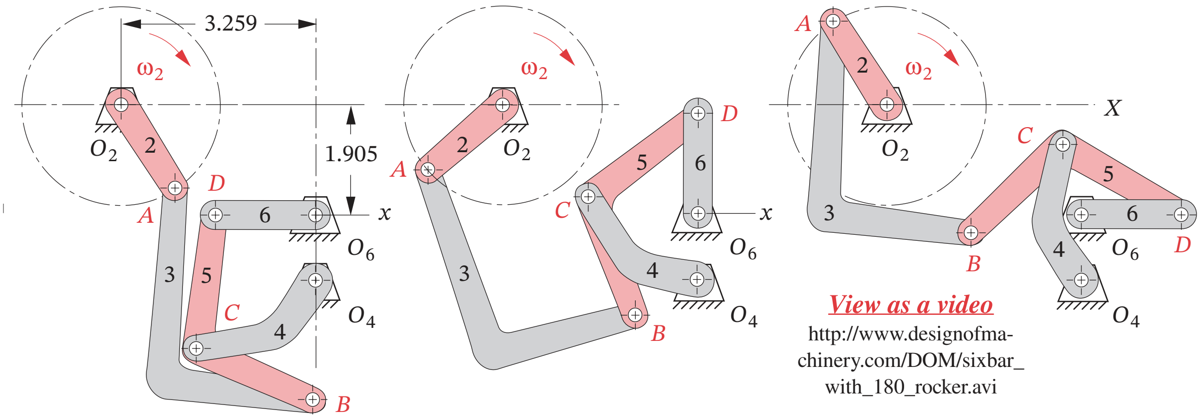
Large Angular Excursion Rocker Motion*

It is often desired to obtain a rocking motion through a large angle with continuous rotary input. A Grashof fourbar crank-rocker linkage is limited to about 120° of rocker oscillation if the transmission angles are to be kept above 30°. A rocker oscillation of 180° would obviously take the transmission angle to zero and also create a Barker Class III linkage with change points, an unacceptable solution. To get a larger oscillation than about 120° with good transmission angles requires six links. Hain[22] designed such a linkage (shown in Figure 3-34) as a Stephenson III sixbar that gives 180° of rocker motion with continuous rotation of the input crank. It is a non-quick-return linkage in which 180° of input crank rotation corresponds to the full oscillation of the output rocker.

An even larger rocker output of about 212° is obtained from the Watt II sixbar linkage shown in Figure 3-35. This mechanism is used to oscillate the agitator in some washing machines. The motor drives toothed crank 2 through a pinion *P*. Crank 2 oscillates rocker 4 through 102° via coupler 3. Rocker 4 serves as the input to rocker 6 through coupler 5. Rocker 6 is attached to the agitator in the washtub. The minimum transmission angles are 36° in stage 1 (links 2-3-4) and 23° in stage 2 (links 4-5-6).

Hain[22] also created a remarkable eightbar linkage that gives ±360° of oscillatory rocker motion from continuous unidirectional rotation of the input crank! This linkage,

* The linkages shown in Figures 3-34 and 3-35 can be animated in program LINKAGES by opening the files F03-34.6br and F03-35.6br, respectively.



(a) Extreme CCW position of link 6 (b) Mid position of link 6 (c) Extreme CW position of link 6

$O_4O_6 = 1.00$	$L_3 = AB = 4.248$	$L_6 = 1.542$	$DB = 3.274$
$L_2 = 1.556$	$L_4 = 2.125$	$CD = 2.158$	$\angle CDB = 36°$

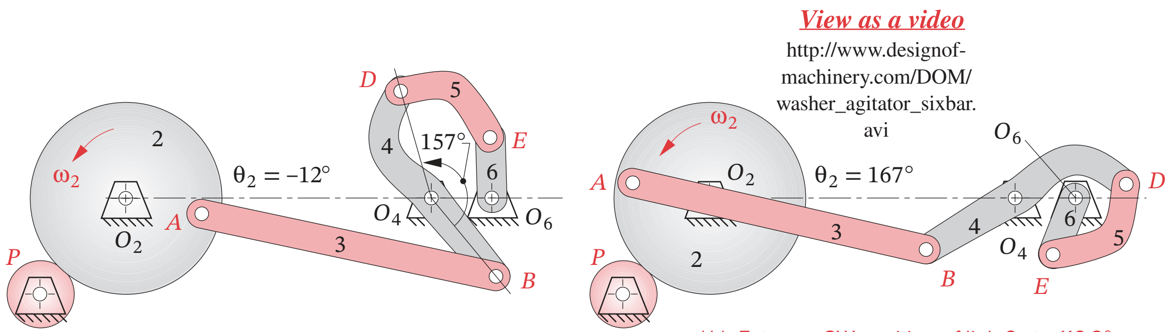
FIGURE 3-34

Stephenson's III sixbar with 180° oscillation of link 6 when crank 2 revolves fully *Adapted from Hain*[22] pp. 448-450

shown in Figure 3-36, has a minimum transmission angle of 30°. Slight changes to this linkage's geometry will give more or less than ±360° of output crank oscillation.

Remote Center Circular Motion

When a rotary motion is needed but the center of that rotation is not available to mount the pivot of a crank, a linkage can be used to describe an approximate or exact circular motion "in the air" remote from the fixed and moving pivots of the linkage. Artobolevsky[20]



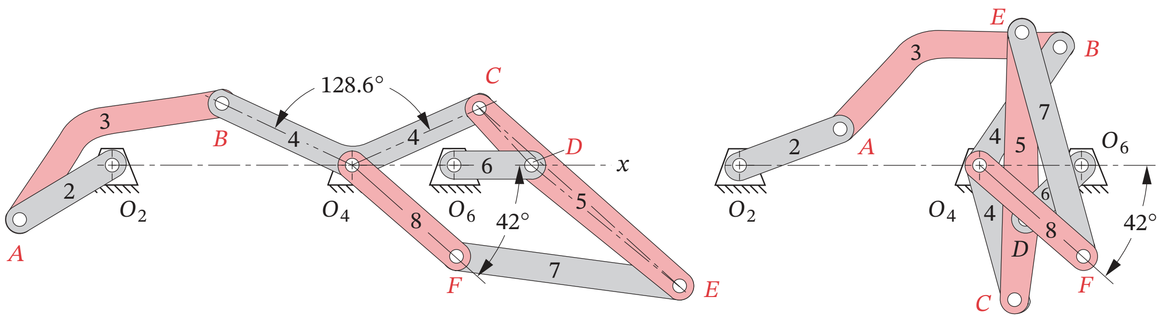
(a) Extreme CCW position of link 6 at +96.4° (b) Extreme CW position of link 6 at –116.2°

$L_2 = 1.000$	$L_3 = 3.800$	$L_5 = 1.286$	$L_6 = 0.771$	$O_4B = 1.286$	$O_4D = 1.429$	$O_2O_4 = 3.857$	$O_2O_6 = 4.643$

Link 6 rotates through 212.8° and back with every revolution of link 2

FIGURE 3-35

Washing machine agitator mechanism: constant speed motor drives link 2 and agitator is oscillated by link 6 at O_6



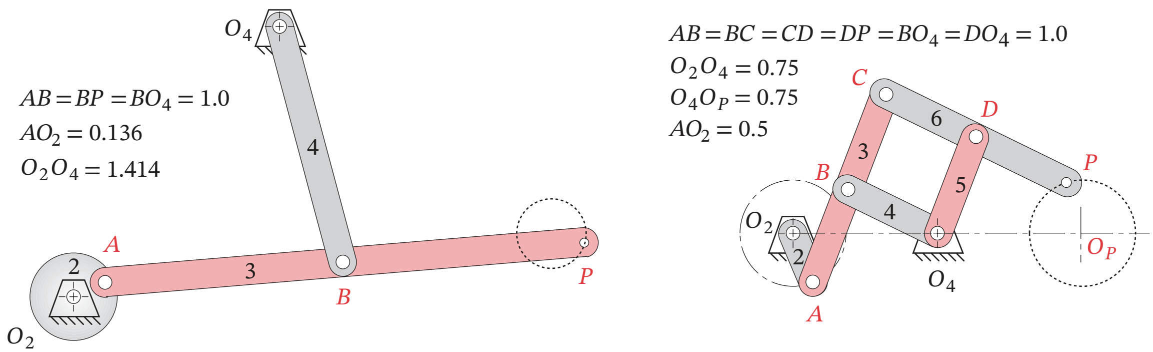
(a) First extreme position, $\theta_2 = 209°$ (b) Second extreme position, $\theta_2 = 19°$

$O_2O_4 = 1.00$ $L_2 = 0.450$ $L_3 = 0.990$ $L_6 = 0.325$ $L_7 = 0.938$ $L_8 = 0.572$ $CD = 0.325$ $CE = 1.145$
$DE = 0.823$ $O_4O_6 = 0.419$ $O_4B = O_4C = 0.590$ $\angle CDE = 173°$

FIGURE 3-36

Eightbar linkage with –360° oscillatory rotation of link 8 when crank 2 revolves fully *Adapted from Hain [22] pp. 368-370*

shows ten such mechanisms, two of which are reproduced in Figure 3-37. Figure 3-37a shows a Chebyschev fourbar approximate circle-tracing linkage. When the crank rotates CCW, point P traces a circle of the same diameter CW. Figure 3-37b shows a Delone exact circle-tracing sixbar linkage that contains a pantograph cell (B-C-D-O_4) that causes point P to mimic the motion of point A, giving an exact 1:1 replication of the circular motion of A about O_4, but rotating in the opposite direction. If a link were added between O_P and P, it would rotate at the same speed but in the opposite direction to link 2. Thus this linkage could be substituted for a pair of external gears (gearset) with a 1:1 ratio (see Chapter 9 for information on gears).



(a) Chebyschev approximate circle-tracing fourbar (b) Delone exact circle-tracing sixbar

FIGURE 3-37

Circle generating mechanisms *Adapted from Artobolevsky [20], Vol. 1, pp. 450-451*

3.11 REFERENCES

1 **Erdman, A. G., and J. E. Gustafson**. (1977). "LINCAGES: Linkage INteractive Computer Analysis and Graphically Enhanced Synthesis." ASME Paper: 77-DTC-5.

2 **Alt, H.** (1932). "Der Übertragungswinkel und seine Bedeutung für das Konstruieren periodischer Getriebe (The Transmission Angle and Its Importance for the Design of Periodic Mechanisms)." *Werkstattstechnik*, **26**, pp. 61-64.

3 **Hall, A. S.** (1961). *Kinematics and Linkage Design.* Waveland Press: Prospect Heights, IL, p. 146.

4 **Sandor, G. N., and A. G. Erdman**. (1984). *Advanced Mechanism Design: Analysis and Synthesis.* Vol. 2. Prentice-Hall: Upper Saddle River, NJ, pp. 177-187.

5 **Kaufman, R. E.** (1978). "Mechanism Design by Computer." *Machine Design*, October 26, 1978, pp. 94-100.

6 **Hall, A. S.** (1961). *Kinematics and Linkage Design.* Waveland Press: Prospect Heights, IL, pp. 33-34.

7a **Kempe, A. B.** (1876). "On a General Method of Describing Plane Curves of the *N*th Degree by Linkwork." *Proceedings London Mathematical Society*, **7**, pp. 213-216.

7b **Wunderlich, W.** (1963). "Höhere Koppelkurven." *Österreichisches Ingenieur Archiv*, **XVII**(3), pp. 162-165.

8a **Hrones, J. A., and G. L. Nelson**. (1951). *Analysis of the Fourbar Linkage.* MIT Technology Press: Cambridge, MA.

8b **Fichter, E. F., and K. H. Hunt**. (1979). "The Variety, Cognate Relationships, Class, and Degeneration of the Coupler Curves of the Planar 4R Linkage." *Proc. of 5th World Congress on Theory of Machines and Mechanisms*, Montreal, pp. 1028-1031.

9 **Kota, S.** (1992). "Automatic Selection of Mechanism Designs from a Three-Dimensional Design Map." *Journal of Mechanical Design*, **114**(3), pp. 359-367.

10 **Zhang, C., R. L. Norton, and T. Hammond**. (1984). "Optimization of Parameters for Specified Path Generation Using an Atlas of Coupler Curves of Geared Five-Bar Linkages." *Mechanism and Machine Theory*, **19**(6), pp. 459-466.

11 **Hartenberg, R. S., and J. Denavit**. (1959). "Cognate Linkages." *Machine Design*, April 16, 1959, pp. 149-152.

12 **Nolle, H.** (1974). "Linkage Coupler Curve Synthesis: A Historical Review - II. Developments after 1875." *Mechanism and Machine Theory*, **9**, 1974, pp. 325-348.

13 **Luck, K.** (1959). "Zur Erzeugung von Koppelkurven viergliedriger Getriebe." *Maschinenbautechnik (Getriebetechnik)*, **8**(2), pp. 97-104.

14 **Soni, A. H.** (1974). *Mechanism Synthesis and Analysis.* Scripta, McGraw-Hill: New York, pp. 381-382.

15 **Hall, A. S.** (1961). *Kinematics and Linkage Design.* Waveland Press: Prospect Heights, IL, p. 51.

16 **Hoeken, K.** (1926). "Steigerung der Wirtschaftlichkeit durch zweckmäßige." *Anwendung der Getriebelehre Werkstattstechnik*.

17 **Hain, K.** (1967). *Applied Kinematics.* D. P. Adams, translator. McGraw-Hill: New York, pp. 308-309.

18 **Nolle, H.** (1974). "Linkage Coupler Curve Synthesis: A Historical Review - I. Developments up to 1875." *Mechanism and Machine Theory*, **9**, pp. 147-168.

19 **Norton, R. L.** (1999). "In Search of the 'Perfect' Straight Line and Constant Velocity Too." *Proc. 6th Applied Mechanisms and Robotics Conference,* Cincinnati, OH.

20 **Artobolevsky, I. I.** (1975). *Mechanisms in Modern Engineering Design.* N. Weinstein, translator. Vols. I to IV. MIR Publishers: Moscow. pp. 431-447.

TABLE P3-0 Part 1
Topic/Problem Matrix

3.2 Type of Motion
3-1

3.3 Limiting Conditions
3-14, 3-15, 3-22, 3-23, 3-36, 3-39, 3-42, 3-93

3.4 Dimensional Synthesis
Two-Position
3-2, 3-3, 3-4, 3-20, 3-46, 3-47, 3-49, 3-50, 3-52, 3-53, 3-55, 3-56, 3-59, 3-60, 3-63, 3-64, 3-76, 3-77, 3-88
Three-Position with Specified Moving Pivots
3-5, 3-48, 3-51, 3-54, 3-57, 3-61, 3-65, 3-89
Three-Position with Specified Fixed Pivots
3-6, 3-58, 3-62, 3-66, 3-90

3.5 Quick-Return Mechanisms
Fourbar
3-7, 3-67, 3-68, 3-69, 3-91, 3-95
Sixbar
3-8, 3-9, 3-70, 3-71, 3-92, 3-96

3.6 Coupler Curves
3-15, 3-33, 3-34, 3-35, 3-78 to 3-83

3.7 Cognates
3-10, 3-16, 3-29, 3-30, 3-37, 3-40, 3-43
Parallel Motion
3-17, 3-18, 3-21, 3-28
Geared Fivebar Cognates of the Fourbar
3-11, 3-25, 3-38, 3-41, 3-44

3.8 Straight-Line Mechanisms
3-19, 3-31, 3-32, 3-76, 3-77, 3-94

21 **Thompson, T. J.** (2000). "A Web-Based Atlas of Coupler Curves and Centrodes of Fourbar Mechanisms." *Proc. of SCSI Conf. on Simulation and Multimedia in Eng. Education*, San Diego, pp. 209-213.

22 **Hain, K.** (1967). *Applied Kinematics.* D. P. Adams, translator. McGraw-Hill: New York.

23 **Roberts, S**. (1875). "Three-bar Motion in Plane Space." *Proc. Lond. Math. Soc.*, **7**, pp. 14-23.

24 **Cayley, A**. (1876). "On Three-bar Motion." *Proc. Lond. Math. Soc.*, **7**, pp. 136-166.

25 **Dijksman, E. A. and T. J. M. Smals**. (2000). "How to Exchange Centric Inverted Slider Cranks with λ-formed Fourbar Linkages." *Mechanism and Machine Theory*, **35**, pp. 305-327.

3.12 BIBLIOGRAPHY

*For additional information on **type synthesis**, the following are recommended:*

Artobolevsky, I. I. (1975). *Mechanisms in Modern Engineering Design.* N. Weinstein, translator. Vols. I to IV. MIR Publishers: Moscow.

Chironis, N. P., ed. (1965). *Mechanisms, Linkages, and Mechanical Controls*. McGraw-Hill: New York.

Chironis, N. P., ed. (1966). *Machine Devices and Instrumentation*. McGraw-Hill: New York.

Chironis, N. P., and N. Sclater, eds. (1996). *Mechanisms and Mechanical Devices Sourcebook*. McGraw-Hill: New York.

Jensen, P. W. (1991). *Classical and Modern Mechanisms for Engineers and Inventors.* Marcel Dekker: New York.

Jones, F., H. Horton, and J. Newell. (1967). *Ingenious Mechanisms for Engineers.* Vols. I to IV. Industrial Press: New York.

Olson, D. G., et al. (1985). "A Systematic Procedure for Type Synthesis of Mechanisms with Literature Review." *Mechanism and Machine Theory*, **20**(4), pp. 285-295.

Tuttle, S. B. (1967). *Mechanisms for Engineering Design.* John Wiley & Sons: New York.

*For additional information on **dimensional linkage synthesis**, the following are recommended:*

Djiksman, E. A. (1976). *Motion Geometry of Mechanisms.* Cambridge University Press: London.

Hain, K. (1967). *Applied Kinematics.* D. P. Adams, translator. McGraw-Hill: New York, p. 399.

Hall, A. S. (1961). *Kinematics and Linkage Design.* Waveland Press: Prospect Heights, IL.

Hartenberg, R. S., and J. Denavit. (1964). *Kinematic Synthesis of Linkages.* McGraw-Hill: New York.

McCarthy, J. M. (2000). *Geometric Design of Linkages*. Springer-Verlag: New York.

Molian, S. (1982). *Mechanism Design: An Introductory Text.* Cambridge University Press: Cambridge.

Sandor, G. N., and A. G. Erdman. (1984). *Advanced Mechanism Design: Analysis and Synthesis.* Vol. 2. Prentice-Hall: Upper Saddle River, NJ.

Tao, D. C. (1964). *Applied Linkage Synthesis.* Addison-Wesley: Reading, MA.

*For information on **spatial linkages**, the following are recommended:*

Haug, E. J. (1989). *Computer Aided Kinematics and Dynamics of Mechanical Systems.* Allyn and Bacon: Boston.

Nikravesh, P. E. (1988). *Computer Aided Analysis of Mechanical Systems.* Prentice-Hall: Upper Saddle River, NJ.

Suh, C. H., and C. W. Radcliffe. (1978). *Kinematics and Mechanism Design.* John Wiley & Sons: New York.

3.13 PROBLEMS†

*3-1 Define the following examples as path, motion, or function generation cases.

a. A telescope aiming (star tracking) mechanism
b. A backhoe bucket control mechanism
c. A thermostat adjusting mechanism
d. A computer printer head moving mechanism
e. An XY plotter pen control mechanism

3-2 Design a fourbar Grashof crank-rocker for 90° of output rocker motion with no quick return. (See Example 3-1.) Build a model and determine the toggle positions and the minimum transmission angle from the model.

*3-3 Design a fourbar mechanism to give the two positions shown in Figure P3-1 of output rocker motion with no quick return. (See Example 3-2.) Build a model and determine the toggle positions and the minimum transmission angle from the model.

3-4 Design a fourbar mechanism to give the two positions shown in Figure P3-1 of coupler motion. (See Example 3-3.) Build a model and determine the toggle positions and the minimum transmission angle from the model. Add a driver dyad. (See Example 3-4.)

*3-5 Design a fourbar mechanism to give the three positions of coupler motion with no quick return shown in Figure P3-2. (See also Example 3-5.) Ignore the points O_2 and O_4 shown. Build a model and determine the toggle positions and the minimum transmission angle from the model. Add a driver dyad. (See Example 3-4.)

*3-6 Design a fourbar mechanism to give the three positions shown in Figure P3-2 using the fixed pivots O_2 and O_4 shown. Build a model and determine the toggle positions and the minimum transmission angle from the model. Add a driver dyad.

3-7 Repeat Problem 3-2 with a quick-return time ratio of 1:1.4. (See Example 3-9.)

*3-8 Design a sixbar drag link quick-return linkage for a time ratio of 1:2 and output rocker motion of 60°.

3-9 Design a crank-shaper quick-return mechanism for a time ratio of 1:3 (see Figure 3-14).

*3-10 Find the two cognates of the linkage in Figure 3-17. Draw the Cayley and Roberts diagrams. Check your results with program LINKAGES.

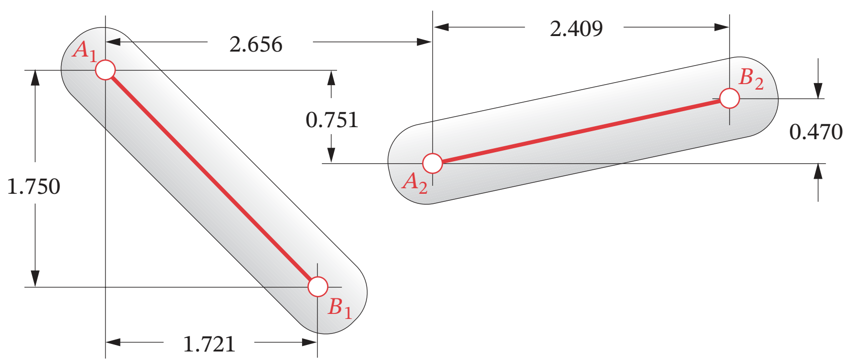


FIGURE P3-1
Problems 3-3 to 3-4

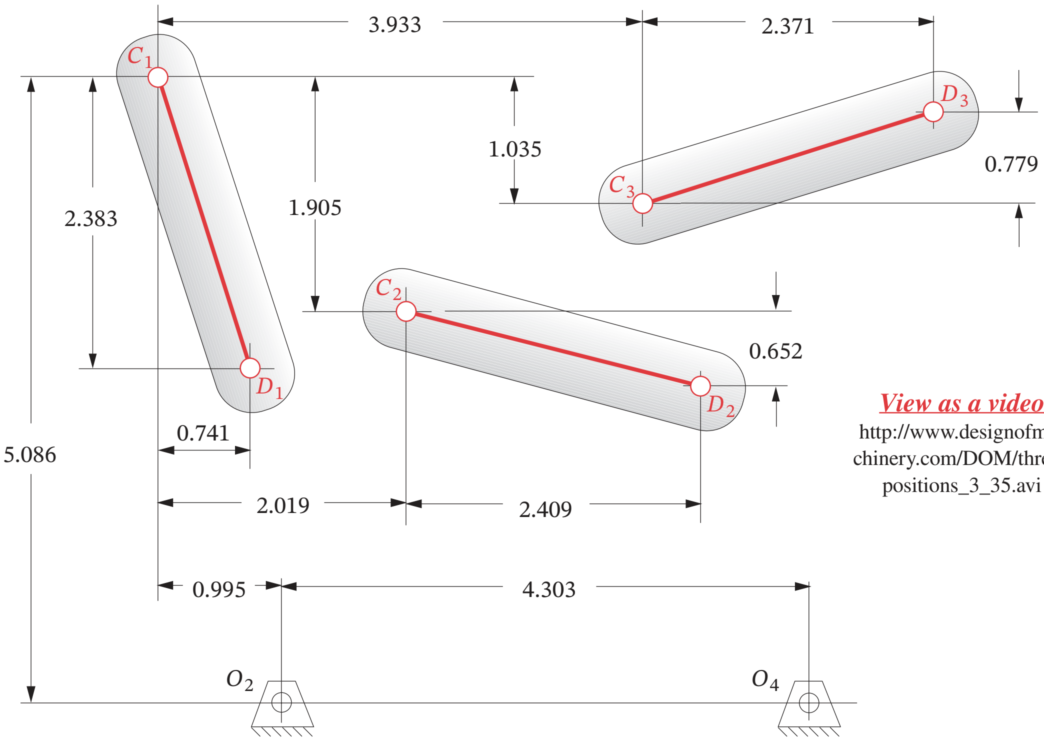
TABLE P3-0 Part 2
Topic/Problem Matrix

3.9 Dwell Mechanisms
Single Dwell
3-12, 3-72, 3-73, 3-74
Double Dwell
3-13, 3-26, 3-27

3

† All problem figures are provided as PDF files, and some are also provided as animated AVI and Working Model files. PDF filenames are the same as the figure number. Run the file *Animations.html* to access and run the animations.

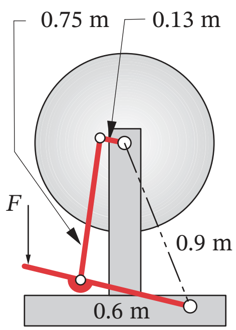
* Answers in Appendix F.



View as a video
http://www.designofmachinery.com/DOM/three_positions_3_35.avi

FIGURE P3-2

Problems 3-5 to 3-6



View as a video
http://www.designofmachinery.com/DOM/treadle_wheel.avi

FIGURE P3-3

Problem 3-14 Treadle-operated grinding wheel

3-11 Find the three equivalent geared fivebar linkages for the three fourbar cognates in Figure 3-25a. Check your results by comparing the coupler curves with program LINKAGES.

3-12 Design a sixbar single-dwell linkage for a dwell of 90° of crank motion, with an output rocker motion of 45°.

3-13 Design a sixbar double-dwell linkage for a dwell of 90° of crank motion, with an output rocker motion of 60°, followed by a second dwell of about 60° of crank motion.

3-14 Figure P3-3 shows a treadle-operated grinding wheel driven by a fourbar linkage. Make a model of the linkage to any convenient scale. Find its minimum transmission angles from the model. Comment on its operation. Will it work? If so, explain how it does.

3-15 Figure P3-4 shows a non-Grashof fourbar linkage that is driven from link O_2A. All dimensions are in centimeters (cm).

a. Find the transmission angle at the position shown.
b. Find the toggle positions in terms of angle AO_2O_4.
c. Find the maximum and minimum transmission angles over its range of motion by graphical techniques.
d. Draw the coupler curve of point P over its range of motion.

3-16 Draw the Roberts diagram for the linkage in Figure P3-4 and find its two cognates. Are they Grashof or non-Grashof?

3-17 Design a Watt I sixbar to give parallel motion that follows the coupler path of point *P* of the linkage in Figure P3-4.

3-18 Add a driver dyad to the solution of Problem 3-17 to drive it over its possible range of motion with no quick return. (The result will be an eightbar linkage.)

3-19 Design a pin-jointed linkage that will guide the forks of the fork lift truck in Figure P3-5 up and down in an approximate straight line over the range of motion shown. Arrange the fixed pivots so they are close to some part of the existing frame or body of the truck.

3-20 Figure P3-6 shows a "V-link" off-loading mechanism for a paper roll conveyor. Design a pin-jointed linkage to replace the air cylinder driver that will rotate the rocker arm and V-link through the 90° motion shown. Keep the fixed pivots as close to the existing frame as possible. Your fourbar linkage should be Grashof and be in toggle at each extreme position of the rocker arm.

3-21 Figure P3-7 shows a walking-beam transport mechanism that uses a fourbar coupler curve, replicated with a parallelogram linkage for parallel motion. Note the duplicate crank and coupler shown ghosted in the right half of the mechanism—they are redundant and have been removed from the duplicate fourbar linkage. Using the same fourbar driving stage (links L_1, L_2, L_3, L_4 with coupler point *P*), design a Watt I sixbar linkage that will drive link 8 in the same parallel motion using two fewer links.

*3-22 Find the maximum and minimum transmission angles of the fourbar driving stage (links L_1, L_2, L_3, L_4) in Figure P3-7 to graphical accuracy.

*3-23 Figure P3-8 shows a fourbar linkage used in a power loom to drive a comblike reed against the thread, "beating it up" into the cloth. Determine its Grashof condition and its minimum and maximum transmission angles to graphical accuracy.

3-24 Draw the Roberts diagram and find the cognates of the linkage in Figure P3-9.

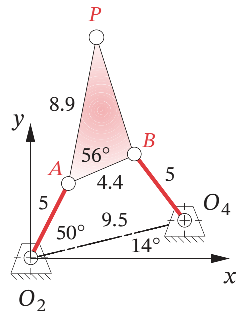


FIGURE P3-4

Problems 3-15 to 3-18

* Answers in Appendix F.

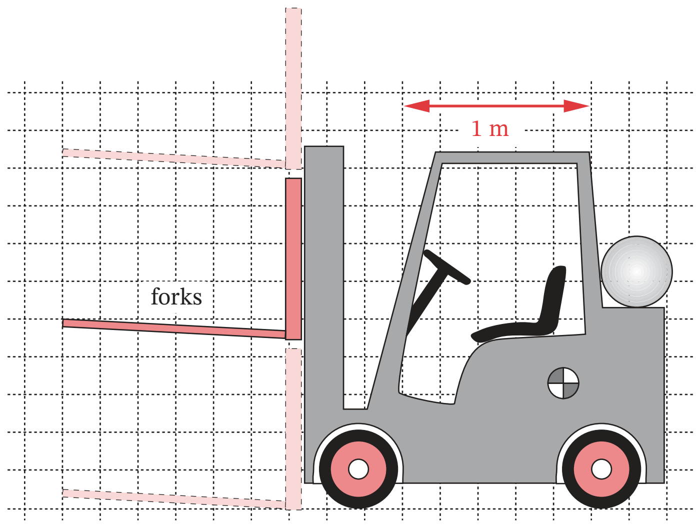


FIGURE P3-5

Problem 3-19

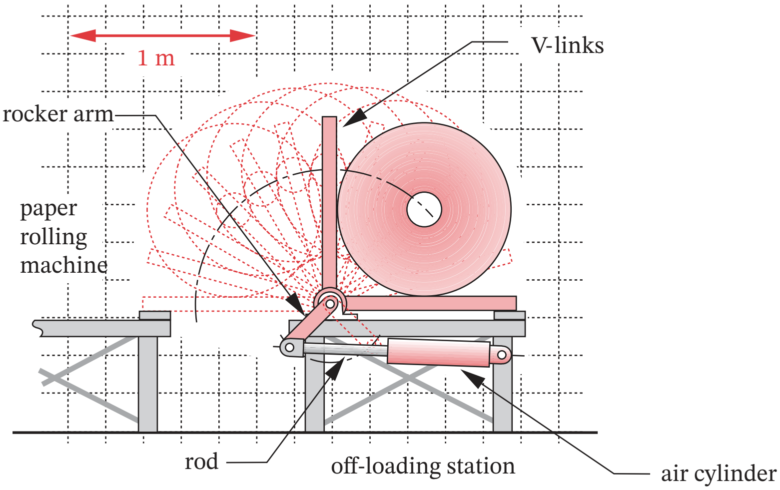
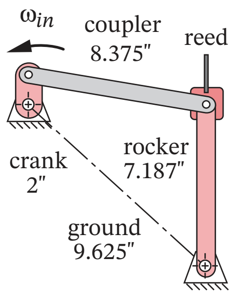


FIGURE P3-6

Problem 3-20



View as a video
http://www.designofmachinery.com/DOM/loom_laybar_drive.avi

FIGURE P3-8

Problem 3-23
Loom laybar drive

3-25 Find the equivalent geared fivebar mechanism cognate of the linkage in Figure P3-9.

3-26 Use the linkage in Figure P3-9 to design an eightbar double-dwell mechanism that has a rocker output through 45°.

3-27 Use the linkage in Figure P3-9 to design an eightbar double-dwell mechanism that has a slider output stroke of 5 crank units.

3-28 Use two of the cognates in Figure 3-26b to design a Watt I sixbar parallel motion mechanism that carries a link through the same coupler curve at all points. Comment on its similarities to the original Roberts diagram.

3-29 Find the cognates of the Watt straight-line mechanism in Figure 3-29a.

3-30 Find the cognates of the Roberts straight-line mechanism in Figure 3-29c.

View as a video
http://www.designofmachinery.com/DOM/walking_beam_eight-bar.avi

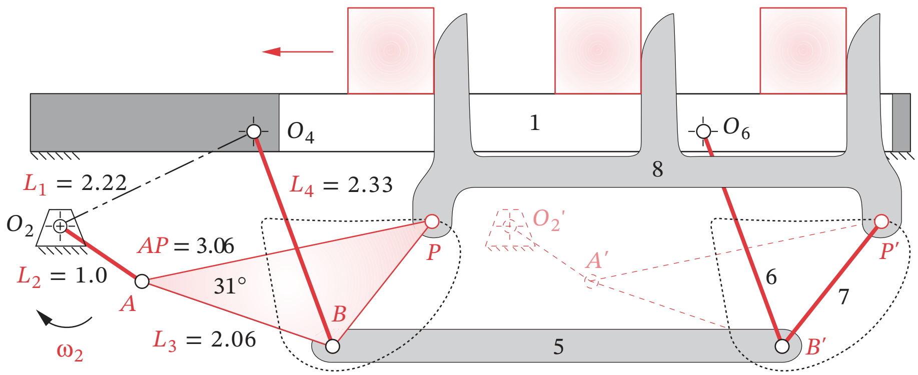


FIGURE P3-7

Problems 3-21 to 3-22 Straight-line walking-beam eightbar transport mechanism

View as a video
http://www.designofmachinery.com/DOM/cognates_hw_3_24.avi

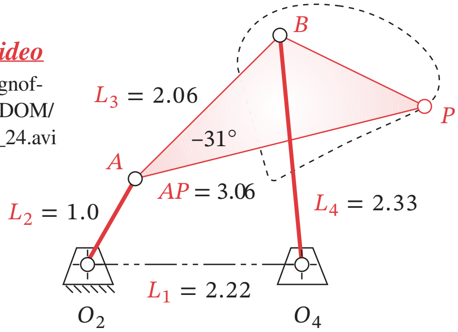


FIGURE P3-9

Problems 3-24 to 3-27

*3-31 Design a Hoeken straight-line linkage to give minimum error in velocity over 22% of the cycle for a 15-cm-long straight-line motion. Specify all linkage parameters.

3-32 Design a Hoeken straight-line linkage to give minimum error in straightness over 39% of the cycle for a 20-cm-long straight-line motion. Specify all linkage parameters.

3-33 Design a linkage that will give a symmetrical "kidney bean" shaped coupler curve as shown in Figure 3-16. Use the data in Figure 3-21 to determine the required link ratios and generate the coupler curve with program LINKAGES.

3-34 Repeat Problem 3-33 for a "double straight" coupler curve.

3-35 Repeat problem 3-33 for a "scimitar" coupler curve with two distinct cusps. Show that there are (or are not) true cusps on the curve by using program LINKAGES. (*Hint*: Think about the definition of a cusp and how you can use the program's data to show it.)

*3-36 Find the Grashof condition, inversion, any limit positions, and the extreme values of the transmission angle (to graphical accuracy) of the linkage in Figure P3-10.

3-37 Draw the Roberts diagram and find the cognates of the linkage in Figure P3-10.

3-38 Find the three geared fivebar cognates of the linkage in Figure P3-10.

* Answers in Appendix F.

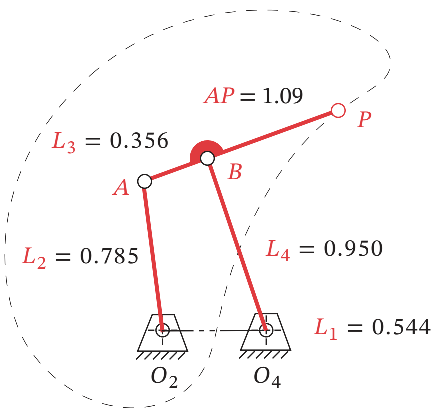


FIGURE P3-10

Problems 3-36 to 3-38

3

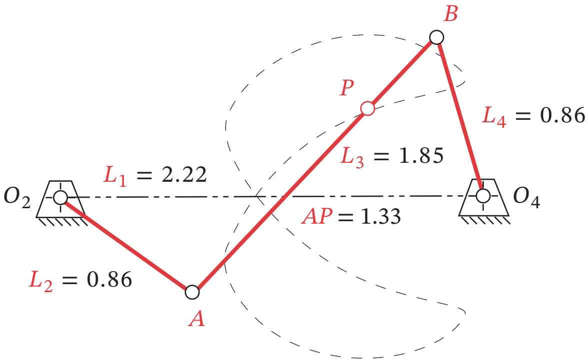


FIGURE P3-11

Problems 3-39 to 3-41

* Answers in Appendix F.

*3-39 Find the Grashof condition, any limit positions, and the extreme values of the transmission angle (to graphical accuracy) of the linkage in Figure P3-11.

3-40 Draw the Roberts diagram and find the cognates of the linkage in Figure P3-11.

3-41 Find the three geared fivebar cognates of the linkage in Figure P3-11.

*3-42 Find the Grashof condition, any limit positions, and the extreme values of the transmission angle (to graphical accuracy) of the linkage in Figure P3-12.

3-43 Draw the Roberts diagram and find the cognates of the linkage in Figure P3-12.

3-44 Find the three geared fivebar cognates of the linkage in Figure P3-12.

3-45 Prove that the relationships between the angular velocities of various links in the Roberts diagram as shown in Figure 3-25 are true.

3-46 Design a fourbar linkage to move the object in Figure P3-13 from position 1 to 2 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions designed making it a sixbar. All fixed pivots should be on the base.

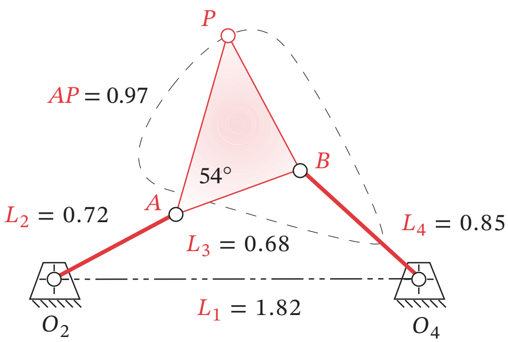


FIGURE P3-12

Problems 3-42 to 3-44

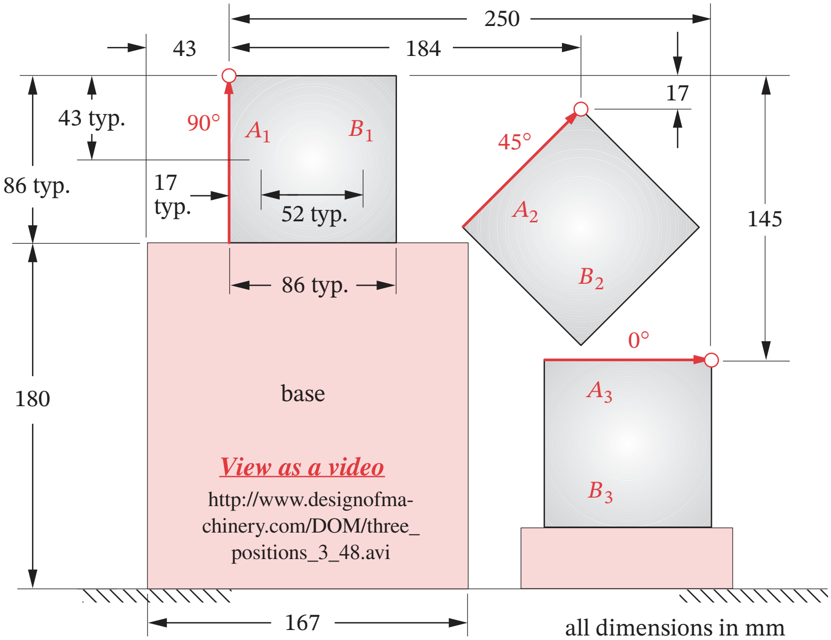


FIGURE P3-13

Problems 3-46 to 3-48

3-47 Design a fourbar linkage to move the object in Figure P3-13 from position 2 to 3 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions designed making it a sixbar. All fixed pivots should be on the base.

3-48 Design a fourbar linkage to move the object in Figure P3-13 through the three positions shown using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions designed, making it a sixbar. All fixed pivots should be on the base.

3-49 Design a fourbar linkage to move the object in Figure P3-14 from position 1 to 2 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions designed making it a sixbar. All fixed pivots should be on the base.

3-50 Design a fourbar linkage to move the object in Figure P3-14 from position 2 to 3 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions designed making it a sixbar. All fixed pivots should be on the base.

3-51 Design a fourbar linkage to move the object in Figure P3-14 through the three positions shown using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions designed, making it a sixbar. All fixed pivots should be on the base.

3-52 Design a fourbar linkage to move the object in Figure P3-15 from position 1 to 2 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions designed, making it a sixbar. All fixed pivots should be on the base.

3-53 Design a fourbar linkage to move the object in Figure P3-15 from position 2 to 3 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions designed, making it a sixbar. All fixed pivots should be on the base.

3

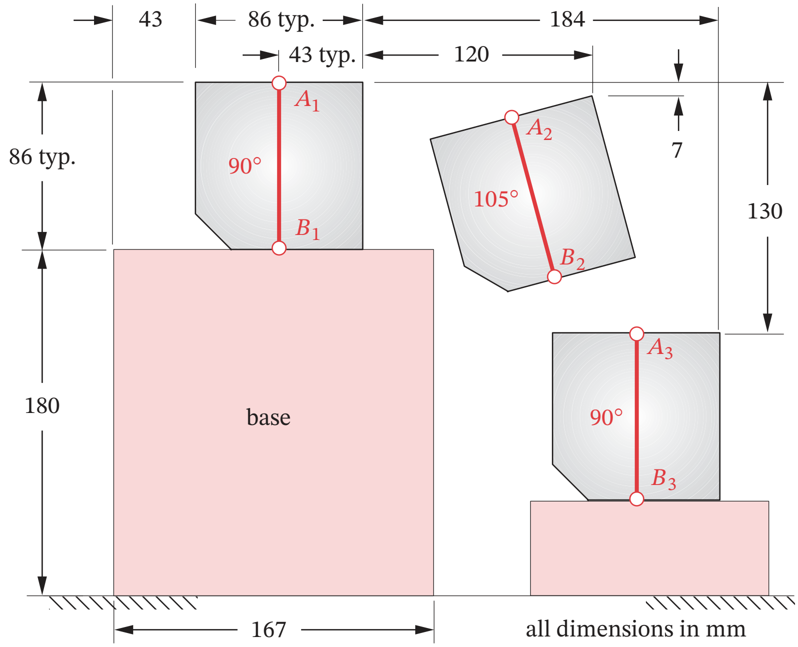


FIGURE P3-14

Problems 3-49 to 3-51

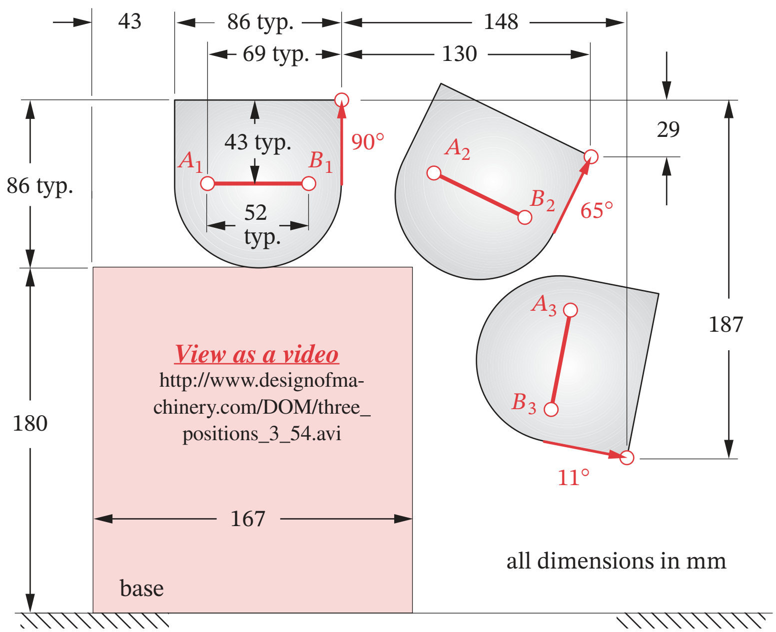


FIGURE P3-15

Problems 3-52 to 3-54

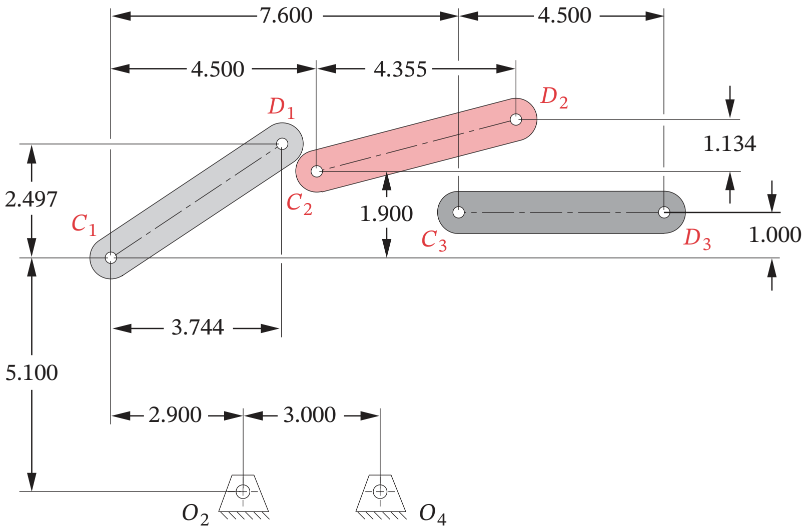


FIGURE P3-16

Problems 3-55 to 3-58

3-54 Design a fourbar linkage to move the object in Figure P3-15 through the three positions shown using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions designed, making it a sixbar. All fixed pivots should be on the base.

3-55 Design a fourbar mechanism to move the link shown in Figure P3-16 from position 1 to position 2. Ignore the third position and the fixed pivots O_2 and O_4 shown. Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-56 Design a fourbar mechanism to move the link shown in Figure P3-16 from position 2 to position 3. Ignore the first position and the fixed pivots O_2 and O_4 shown. Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-57 Design a fourbar mechanism to give the three positions shown in Figure P3-16. Ignore the fixed pivots O_2 and O_4 shown. Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-58 Design a fourbar mechanism to give the three positions shown in Figure P3-16 using the fixed pivots O_2 and O_4 shown. (See Example 3-7.) Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-59 Design a fourbar mechanism to move the link shown in Figure P3-17 from position 1 to position 2. Ignore the third position and the fixed pivots O_2 and O_4 shown. Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-60 Design a fourbar mechanism to move the link shown in Figure P3-17 from position 2 to position 3. Ignore the first position and the fixed pivots O_2 and O_4 shown. Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3

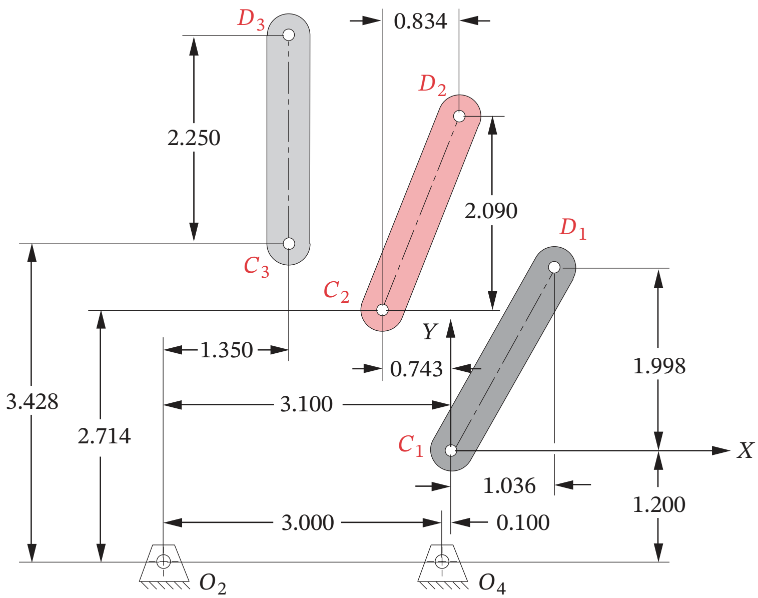


FIGURE P3-17

Problems 3-59 to 3-62

3-61 Design a fourbar mechanism to give the three positions shown in Figure P3-17. Ignore the fixed pivots O_2 and O_4 shown. Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-62 Design a fourbar mechanism to give the three positions shown in Figure P3-17 using the fixed pivots O_2 and O_4 shown. (See Example 3-7.) Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-63 Design a fourbar mechanism to move the link shown in Figure P3-18 from position 1 to position 2. Ignore the third position and the fixed pivots O_2 and O_4 shown. Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-64 Design a fourbar mechanism to move the link shown in Figure P3-18 from position 2 to position 3. Ignore the first position and the fixed pivots O_2 and O_4 shown. Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-65 Design a fourbar mechanism to give the three positions shown in Figure P3-18. Ignore the fixed pivots O_2 and O_4 shown. Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-66 Design a fourbar mechanism to give the three positions shown in Figure P3-18 using the fixed pivots O_2 and O_4 shown. (See Example 3-7.) Build a model and add a driver dyad to limit its motion to the range of positions designed, making it a sixbar.

3-67 Design a fourbar Grashof crank-rocker for 120° of output rocker motion with a quick-return time ratio of 1:1.2. (See Example 3-9.)

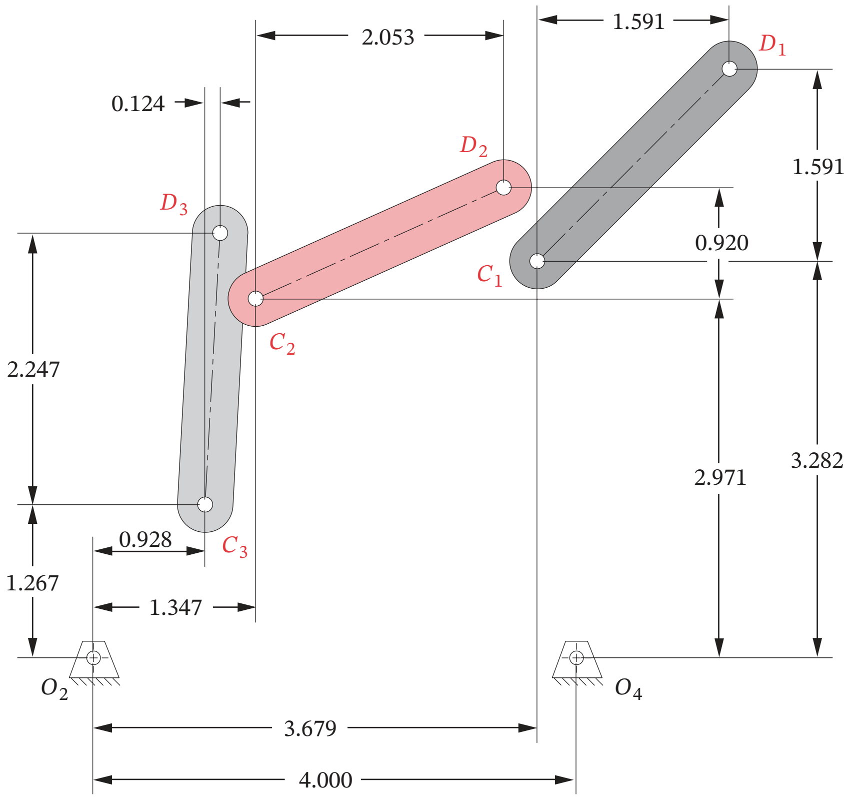


FIGURE P3-18

Problems 3-63 to 3-66

3-68 Design a fourbar Grashof crank-rocker for 100° of output rocker motion with a quick-return time ratio of 1:1.5. (See Example 3-9.)

3-69 Design a fourbar Grashof crank-rocker for 80° of output rocker motion with a quick-return time ratio of 1:1.33. (See Example 3-9.)

3-70 Design a sixbar drag link quick-return linkage for a time ratio of 1:4 and output rocker motion of 50°.

3-71 Design a crank shaper quick-return mechanism for a time ratio of 1:2.5 (See Figure 3-14).

3-72 Design a sixbar, single-dwell linkage for a dwell of 70° of crank motion, with an output rocker motion of 30° using a symmetrical fourbar linkage having the following parameters: ground link ratio = 2.0, common link ratio = 2.0, and coupler angle $\gamma = 40°$. (See Example 3-13.)

3-73 Design a sixbar, single-dwell linkage for a dwell of 100° of crank motion, with an output rocker motion of 50° using a symmetrical fourbar linkage having the following parameters: ground link ratio = 2.0, common link ratio = 2.5, and coupler angle $\gamma = 60°$. (See Example 3-13.)

3-74 Design a sixbar, single-dwell linkage for a dwell of 80° of crank motion, with an output rocker motion of 45° using a symmetrical fourbar linkage having the following parameters: ground link ratio = 2.0, common link ratio = 1.75, and coupler angle $\gamma = 70°$. (See Example 3-13.)

3-75 Using the method of Example 3-11, show that the sixbar Chebyschev straight-line linkage of Figure P2-5 is a combination of the fourbar Chebyschev straight-line linkage of Figure 3-29d and its Hoeken cognate of Figure 3-29e. See also Figure 3-26 for additional information useful to this solution. Graphically construct the Chebyschev sixbar parallel motion linkage of Figure P2-5a from its two fourbar linkage constituents and build a physical or computer model of the result.

3-76 Design a driver dyad to drive link 2 of the Evans straight-line linkage in Figure 3-29f from 150° to 210°. Make a model of the resulting sixbar linkage and trace the coupler curve.

3-77 Design a driver dyad to drive link 2 of the Evans straight-line linkage in Figure 3-29g from –40° to 40°. Make a model of the resulting sixbar linkage and trace the coupler curve.

3-78 Figure 6 on page *ix* of the Hrones and Nelson atlas of fourbar coupler curves shows a 50-point coupler that was used to generate the curves in the atlas. Using the definition of the vector **R** given in Figure 3-17b of the text, determine the 10 possible pairs of values of ϕ and R for the first row of points above the horizontal axis if the grid point spacing is one-half the length of the unit crank.

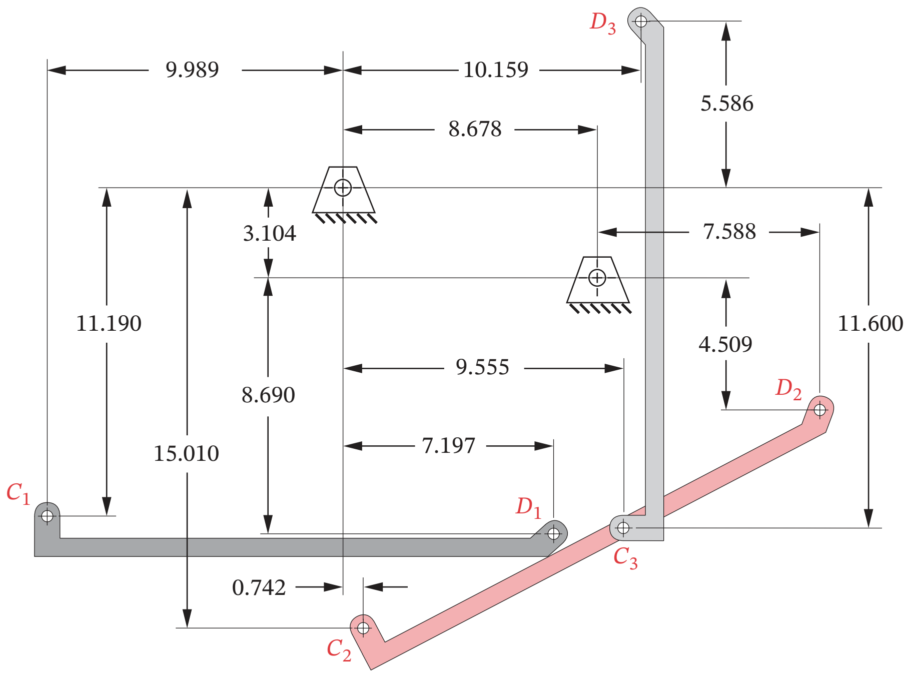


FIGURE P3-19

Problems 3-84 to 3-87

*3-79 The first set of 10 coupler curves on page 1 of the Hrones and Nelson atlas of fourbar coupler curves has $A = B = C = 1.5$. Model this linkage with program LINKAGES using the coupler point farthest to the left in the row shown on page 1 and plot the resulting coupler curve. Note that you will have to orient link 1 at the proper angle in LINKAGES to get the curve as shown in the atlas.

3-80 Repeat problem 3-79 for the set of coupler curves on page 17 of the Hrones and Nelson atlas (see page 32 of the PDF file) which has $A = 1.5$, $B = C = 3.0$, using the coupler point farthest to the right in the row shown.

3-81 Repeat problem 3-79 for the set of coupler curves on page 21 of the Hrones and Nelson atlas (see page 36 of the PDF file) which has $A = 1.5$, $B = C = 3.5$, using the second coupler point from the right end in the row shown.

3-82 Repeat problem 3-79 for the set of coupler curves on page 34 of the Hrones and Nelson atlas (see page 49 of the PDF file) which has $A = 2.0$, $B = 1.5$, $C = 2.0$, using the coupler point farthest to the right in the row shown.

3-83 Repeat problem 3-79 for the set of coupler curves on page 115 of the Hrones and Nelson atlas (see page 130 of the PDF file) which has $A = 2.5$, $B = 1.5$, $C = 2.5$, using the fifth coupler point from the left end in the row shown.

* Answers in Appendix F.

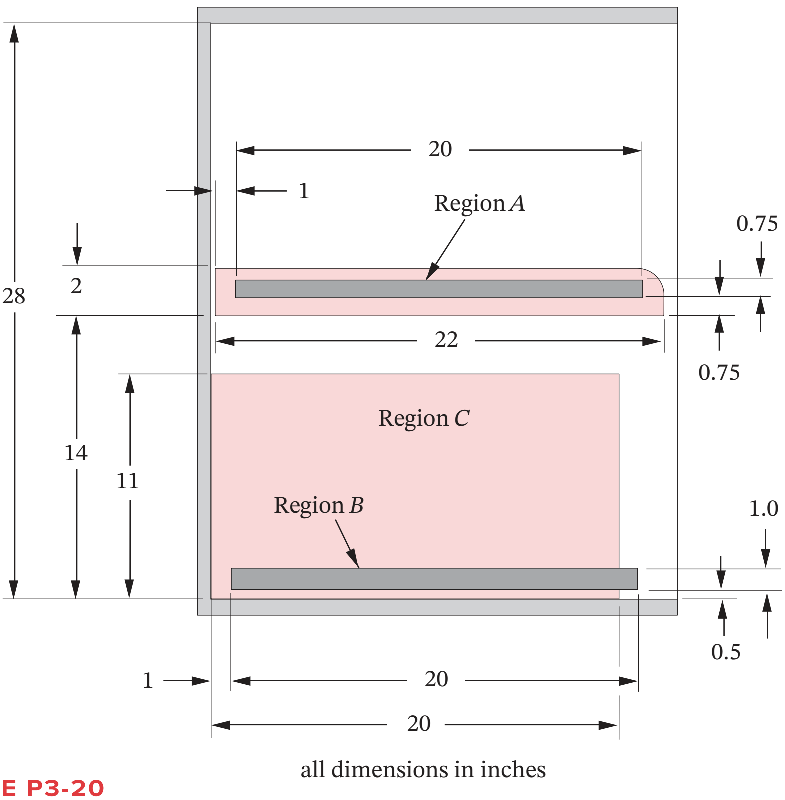


FIGURE P3-20

Problem 3-88

3-84 Design a fourbar mechanism to move the link shown in Figure P3-19 from position 1 to position 2. Ignore the third position and the fixed pivots O_2 and O_4 shown. Build a cardboard model that demonstrates the required movement.

3-85 Design a fourbar mechanism to move the link shown in Figure P3-19 from position 2 to position 3. Ignore the first position and the fixed pivots O_2 and O_4 shown. Build a cardboard model that demonstrates the required movement.

3-86 Design a fourbar mechanism to give the three positions shown in Figure P3-19. Ignore the points O_2 and O_4 shown. Build a cardboard model that has stops to limit its motion to the range of positions designed.

3-87 Design a fourbar mechanism to give the three positions shown in Figure P3-19 using the fixed pivots O_2 and O_4 shown. See Example 3-7. Build a cardboard model that has stops to limit its motion to the range of positions designed.

3-88 The side view of the upper section of a kitchen-pantry cabinet is shown in Figure P3-20. It has a removable shelf 14 in above the bottom of the section but it is too high off the floor to be useful. Design a fourbar linkage to move the shelf from the position shown in the figure to a lower position keeping it horizontal throughout. The moving pivots should be in Region *A* and the fixed pivots should be in Region *B*. Additionally, the shelf should not intrude into Region *C* and it should be stable when in the fully raised position.

3-89 Design a fourbar mechanism to give the three positions of coupler motion shown in Figure P3-21. (See also Example 3-5.) Ignore the points O_2 and O_4 shown.

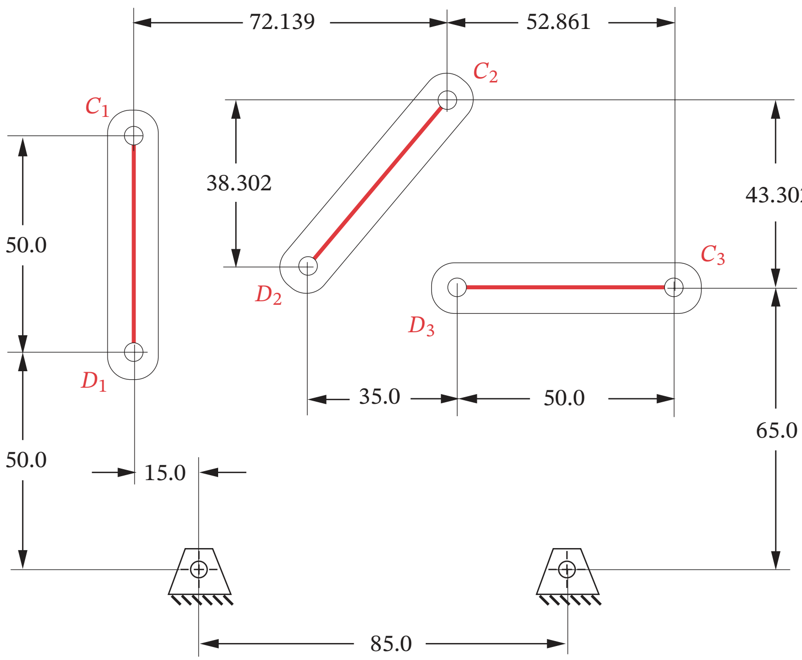


FIGURE P3-21

Problems 3-89 and 3-90

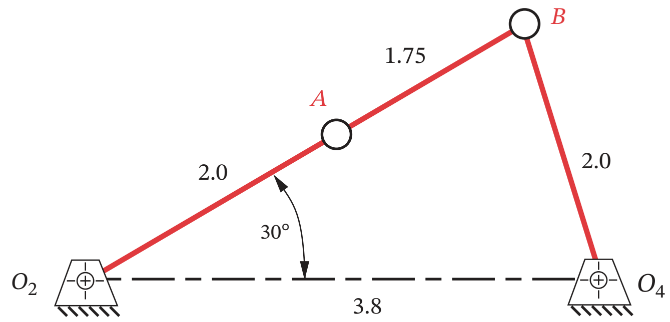


FIGURE P3-22

Problem 3-93

3-90 Design a fourbar mechanism to give the three positions shown in Figure P3-21 using the fixed pivots O_2 and O_4 shown.

3-91 Design a fourbar Grashof crank-rocker for 60 degrees of output rocker motion with a quick-return time ratio of 1:1.25.

3-92 Design a crank-shaper quick-return mechanism for a time ratio of 1:4 (Figure 3-14).

3-93 Figure P3-22 shows a non-Grashof fourbar linkage that is driven from link O_2A. All dimensions are in inches (in).

(a) Find the transmission angle at the position shown.
(b) Find the toggle positions in terms of angle AO_2O_4.

3-94 The Peaucellier straight line linkage shown in Figure 3-29(j) will generate true circle arcs if the fixed pivot O_2 is moved to the left or right with only the length of the ground link being changed. Determine, graphically, the radius of the circular arc traced by point P over the range of $0° \leq \theta_2 \leq 60° \leq$ if the links have the following lengths: $L_1 = 12$, $L_2 = 10$, $L_3 = L_4 = 22$, and $L_5 = L_6 = L_7 = L_8 = 6.5$.

3-95 Design a fourbar Grashof crank-rocker for 80 degrees of output rocker motion with a quick-return time ratio of 1:1.333.

3-96 Design a sixbar drag link quick-return linkage for a time ratio of 1 : 2.6, and output rocker motion of 70 degrees. (See Example 3-10.)

3.14 PROJECTS

These larger-scale project statements deliberately lack detail and structure and are loosely defined. Thus, they are similar to the kind of "identification of need" problem statement commonly encountered in engineering practice. It is left to the student to structure the problem through background research and to create a clear goal statement and set of performance specifications before attempting to design a solution. This design process is spelled out in Chapter 1 and should be followed in all of these examples. These projects can be done as an exercise in mechanism synthesis alone or can be revisited and thoroughly analyzed by the methods presented in later chapters as well. All results should be documented in a professional engineering report.

P3-1 The tennis coach needs a better tennis ball server for practice. This device must fire a sequence of standard tennis balls from one side of a standard tennis court over the net such that they land and bounce within each of the three court areas defined by the court's white lines. The order and frequency of a ball's landing in any one of the three court areas must be random. The device should operate automatically and unattended except for the refill of balls. It should fire 50 balls between reloads. The timing of ball releases should vary. For simplicity, a motor driven pin-jointed linkage design is preferred.

P3-2 A quadriplegic patient has lost all motion except that of her head. She can only move a small "mouth stick" to effect a switch closure. She was an avid reader before her injury and would like again to be able to read standard hardcover books without the need of a person to turn pages for her. Thus, a reliable, simple, and inexpensive automatic page turner is needed. The book may be placed in the device by an assistant. It should accommodate as wide a range of book sizes as possible. Book damage is to be avoided and safety of the user is paramount.

P3-3 Grandma's off her rocker again! Junior's run down to the Bingo parlor to fetch her, but we've got to do something about her rocking chair before she gets back. She's been complaining that her arthritis makes it too painful to push the rocker. So, for her 100th birthday in 2 weeks, we're going to surprise her with a new, automated, motorized rocking chair. The only constraints placed on the problem are that the device must be safe and must provide interesting and pleasant motions, similar to those of her present *Boston* rocker, to all parts of the occupant's body. Since simplicity is the mark of good design, a linkage solution with only full pin joints is preferred.

P3-4 The local amusement park's business is suffering as a result of the proliferation of computer game parlors. They need a new and more exciting ride which will attract new customers. The only constraints are that it must be safe, provide excitement, and not subject the occupants to excessive accelerations or velocities. Also it must be as compact as possible, since space is limited. Continuous rotary input and full pin joints are preferred.

P3-5 The student section of ASME is sponsoring a spring fling on campus. They need a mechanism for their "Dunk the Professor" booth which will carry the unfortunate (untenured) volunteer into and out of the water tub. The contestants will provide the inputs to a multiple-*DOF* mechanism. If they know their kinematics, they can provide a combination of inputs which will dunk the victim.

P3-6 The National House of Flapjacks wants to automate its flapjack production. It needs a mechanism that will automatically flip the flapjacks "on the fly" as they travel through the griddle on a continuously moving conveyor. This mechanism must track the constant velocity of the conveyor, pick up a pancake, flip it over, and place it back onto the conveyor.

P3-7 Many varieties and shapes of computer video monitors now exist. Their long-term use leads to eyestrain and body fatigue. There is a need for an adjustable stand which will hold the video monitor and the separate keyboard at any position the user deems comfortable. The computer's central processor unit (CPU) can be remotely located. This device should be freestanding to allow use with a comfortable chair, couch, or lounge of the user's choice. It should not require the user to assume the conventional "seated at a desk" posture to use the computer. It must be stable in all positions and safely support the equipment's weight.

P3-8 Most small boat trailers must be submerged in the water to launch or retrieve the boat. This greatly reduces the life of the trailer, especially in salt water. A need exists for a trailer that will remain on dry land while it launches or retrieves the boat. No part of the trailer should get wet. User safety is of greatest concern, as is protection of the boat from damage.

P3-9 The "Save the Skeet" foundation has requested a more humane skeet launcher be designed. While they have not yet succeeded in passing legislation to prevent the wholesale slaughter of these little devils, they are concerned about the inhumane aspects of the large accelerations imparted to the skeet as it is launched into the sky for the sportsman to shoot it down. The need is for a skeet launcher that will smoothly accelerate the clay pigeon onto its desired trajectory.

P3-10 The coin-operated "kid bouncer" machines found outside supermarkets typically provide a very unimaginative rocking motion to the occupant. There is a need for a superior "bouncer" which will give more interesting motions while remaining safe for small children.

P3-11 Horseback riding is a very expensive hobby or sport. There is a need for a horseback riding simulator to train prospective riders sans the expensive horse. This device should provide similar motions to the occupant as she would feel in the saddle under various gaits such as a walk, canter, or gallop. A more advanced version might contain jumping motions as well. User safety is most important.

P3-12 The nation is on a fitness craze. Many exercise machines have been devised. There is still room for improvement to these devices. They are typically designed for the young, strong athlete. There is also a need for an ergonomically optimum exercise machine for the older person who needs gentler exercise.

P3-13 A paraplegic patient needs a device to get himself from his wheelchair into the Jacuzzi with no assistance. He has good upper body and arm strength. Safety is paramount.

P3-14 The army has requested a mechanical walking device to test army boots for durability. It should mimic a person's walking motion and provide forces similar to an average soldier's foot.

P3-15 NASA wants a zero-g machine for astronaut training. It must carry one person and provide a negative 1-g acceleration for as long as possible.

P3-16 The Amusement Machine Co. Inc. wants a portable "whip" ride that will give two or four passengers a thrilling but safe ride, and which can be pulled behind a pickup truck from one location to another.

P3-17 The Air Force has requested a pilot training simulator which will give potential pilots exposure to g forces similar to those they will experience in dogfight maneuvers.

P3-18 Cheers needs a better "mechanical bull" simulator for its "yuppie" bar in Boston. It must give a thrilling "bucking bronco" ride but be safe.

P3-19 Despite the improvements in handicap access, many curbs block wheelchairs from public places. Design an attachment for a conventional wheelchair which will allow it to get up over a curb.

P3-20 A carpenter needs a dumping attachment to fit in her pickup truck so she can dump building materials. She can't afford to buy a dump truck.

P3-21 The carpenter in Project P3-20 wants an inexpensive lift gate designed to fit her full-sized pickup truck in order to lift and lower heavy cargo to the truck bed.

P3-22 The carpenter in Project P3-20 is very demanding (and lazy). She also wants a device to lift sheet rock and blueboard into place on ceilings or walls to hold it while she screws it on.

P3-23 Click and Clack, the tappet brothers, need a better transmission jack for their Good News Garage. This device should position a transmission under a car (on a lift) and allow it to be maneuvered into place safely and quickly.

P3-24 A paraplegic with good upper body strength, who was an avid golfer before his injury, wants a mechanism to allow him to stand up in his wheelchair in order to once again play golf. It must not interfere with normal wheelchair use, though it could be removed from the chair when he is not golfing.

P3-25 A wheelchair lift is needed to raise the wheelchair and person 3 ft from the garage floor to the level of the first floor of the house. Safety, reliability, and cost are of major concern.

P3-26 A paraplegic needs a mechanism that can be installed on a full-size 3-door pickup truck that will lift the wheelchair into the area behind the driver's seat. This person has excellent upper body strength and, with the aid of specially installed handles on the truck, can get into the cab from the chair. The truck can be modified as necessary to accommodate this task. For example, attachment points can be added to its structure and the back seat of the truck can be removed if necessary.

P3-27 There is demand for a better *baby transport device*. Many such devices are on the market. Some are called carriages, others strollers. Some are convertible to multiple uses. Our marketing survey data so far seems to indicate that the customers want portability (i.e., foldability), light weight, one-handed operation, and large wheels. Some of these features are present in existing devices. We need a better design that more completely meets the needs of the customer. The device must be stable, effective, and safe for the baby and the operator. Full joints are preferred to half joints and simplicity is the mark of good design. A linkage solution with manual input is desired.

P3-28 A boat owner has requested that we design a lift mechanism to automatically move a 1000-lb, 15-ft boat from a cradle on land to the water. A seawall protects the owner's yard, and the boat cradle sits above the seawall. The tidal variation is 4 ft and the seawall is 3 ft above the high tide mark. Your mechanism will be attached to land and move the boat from its stored position on the cradle to the water and return it to the cradle. The device must be safe and easy to use and not overly expensive.

P3-29 The landfills are full! We're about to be up to our ears in trash! The world needs a better trash compactor. It should be simple, inexpensive, quiet, compact, and safe. It can either be manually powered or motorized, but manual operation is preferred to keep the cost down. The device must be stable, effective, and safe for the operator.

P3-30 A small contractor needs a mini-dumpster attachment for his pickup truck. He has made several trash containers which are 4 ft x 4 ft x 3.5 ft high. The empty container weighs 150 lb. He needs a mechanism which he can attach to his fleet of standard, full-size pickup trucks (Chevrolet, Ford, or Dodge). This mechanism should be able to pick up the full trash container from the ground, lift it over the closed tailgate of the truck, dump its contents into the truck bed, and then return it empty to the ground. He would like not to tip his truck over in the process. The mechanism should store permanently on the truck in such a manner as to allow the normal use of the pickup truck at all other times. You may specify any means of attachment of your mechanism to the container and to the truck.

P3-31 As a feast day approaches, the task of inserting the leaves in the dining room table presents itself. Typically, the table leaves are stored in some forgotten location, and when found have to be carried to the table and manually placed in it. Wouldn't it be better if the leaves (leaf) were stored within the table itself and were automatically inserted into place when the table was opened? The only constraints imposed on the problem are that the device must be simple to use, preferably using the action of opening the table halves as the actuating motion. That is, as you pull the table open, the stored leaf should be carried by the mechanism of your design into its proper place in order to extend the dining surface. Thus, a linkage solution with manual input is desired and full joints are preferred to half joints, though either may be used.

P3-32 Small sailboats often are not "self-bailing," meaning that they accumulate rainwater and can sink at the mooring if not manually bailed (emptied of water) after a rainstorm. These boats usually do not have a power source such as a battery aboard. Design a mechanism that can be quickly attached to, detached from, and stored in a 20-foot-long daysailer, and that will use wave action (boat rocking) as the input to a bilge pump to automatically keep the boat dry and afloat when left at a mooring.

P3-33 A machine uses several 200 kg servomotors that bolt underneath the machine's bedplate, which is 0.75 m above the floor. The machine frame has a 400 mm square front opening through which the motor can be inserted. It must be extended 0.5 m horizontally to its installed location. Design a mechanism to transport the motor from the stockroom to the machine, insert it under the machine and raise it 200 mm into position. Your mechanism also should be capable of removing a motor from the machine.

P3-34 A paraplegic client has requested that we design a mechanism to attach to his wheelchair that will store his backpack behind his seatback. This mechanism must also bring the backpack around toward the front of the chair so that he can access its contents. He has some use of his upper body and so can do something to cause your mechanism to move. It should safely lock itself in place when stowed behind the seatback. It should not upset the chair's stabilty or limit its mobility.

P3-35 In an effort to reduce chronic back injury among janitorial staff, our client, Ready Refuse, has requested that we design a mechanism to safely lift an office size rectangular trash or recycling container and dump it into a large rolling trash barrel. The mechanism needs to be motorized to dump the smaller container automatically. To operate the system, the user will roll the large trash barrel up to the rectangular container, which is sitting on the floor, and press a button that will cause the mechanism move through the required motion and dump the contents of the container into the large barrel. The grip design between your mechanism and the top lip of the rectangular trash container is being designed by another team at Widgets Perfected Inc.; assume that it works. Your task is to design the motorized mechanism that dumps the container without spilling the contents outside of the large trash barrel.